MODEL G0872
60W 17" X 23" CNC LASER CUTTER/ENGRAVER

OWNER'S MANUAL

(For models manufactured since 01/19)

Meets FDA Safety Standards

#CRKS20917 PRINTED IN CHINA
V1.03.21

WARNING!

This manual provides critical safety instructions on the proper setup, operation, maintenance, and service of this machine/tool. Save this document, refer to it often, and use it to instruct other operators.

Failure to read, understand and follow the instructions in this manual may result in fire or serious personal injury—including amputation, electrocution, or death.

The owner of this machine/tool is solely responsible for its safe use. This responsibility includes but is not limited to proper installation in a safe environment, personnel training and usage authorization, proper inspection and maintenance, manual availability and comprehension, application of safety devices, cutting/sanding/grinding tool integrity, and the usage of personal protective equipment.

The manufacturer will not be held liable for injury or property damage from negligence, improper training, machine modifications or misuse.

WARNING!

Some dust created by power sanding, sawing, grinding, drilling, and other construction activities contains chemicals known to the State of California to cause cancer, birth defects or other reproductive harm. Some examples of these chemicals are:

- **Lead from lead-based paints.**
- **Crystalline silica from bricks, cement and other masonry products.**
- **Arsenic and chromium from chemically-treated lumber.**

Your risk from these exposures varies, depending on how often you do this type of work. To reduce your exposure to these chemicals: Work in a well ventilated area, and work with approved safety equipment, such as those dust masks that are specially designed to filter out microscopic particles.

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INTRODUCTION

Contact Info

We stand behind our machines! If you have questions or need help, contact us with the information below. Before contacting, make sure you get the **serial number** and **manufacture date** from the machine ID label. This will help us help you faster.

Grizzly Technical Support
1815 W. Battlefield
Springfield, MO 65807
Phone: (570) 546-9663
Email: techsupport@grizzly.com

We want your feedback on this manual. What did you like about it? Where could it be improved? Please take a few minutes to give us feedback.

Grizzly Documentation Manager
P.O. Box 2069
Bellingham, WA 98227-2069
Email: manuals@grizzly.com

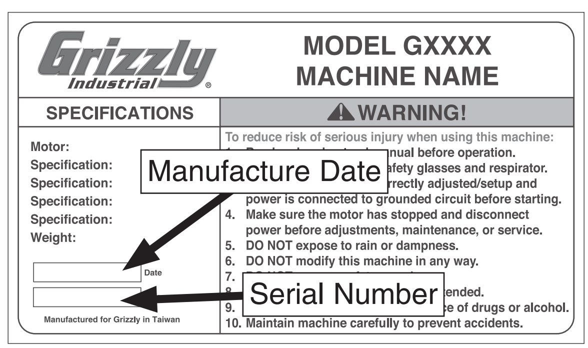
Manual Accuracy

We are proud to provide a high-quality owner's manual with your new machine!

We made every effort to be exact with the instructions, specifications, drawings, and photographs in this manual. Sometimes we make mistakes, but our policy of continuous improvement also means that **sometimes the machine you receive is slightly different than shown in the manual**.

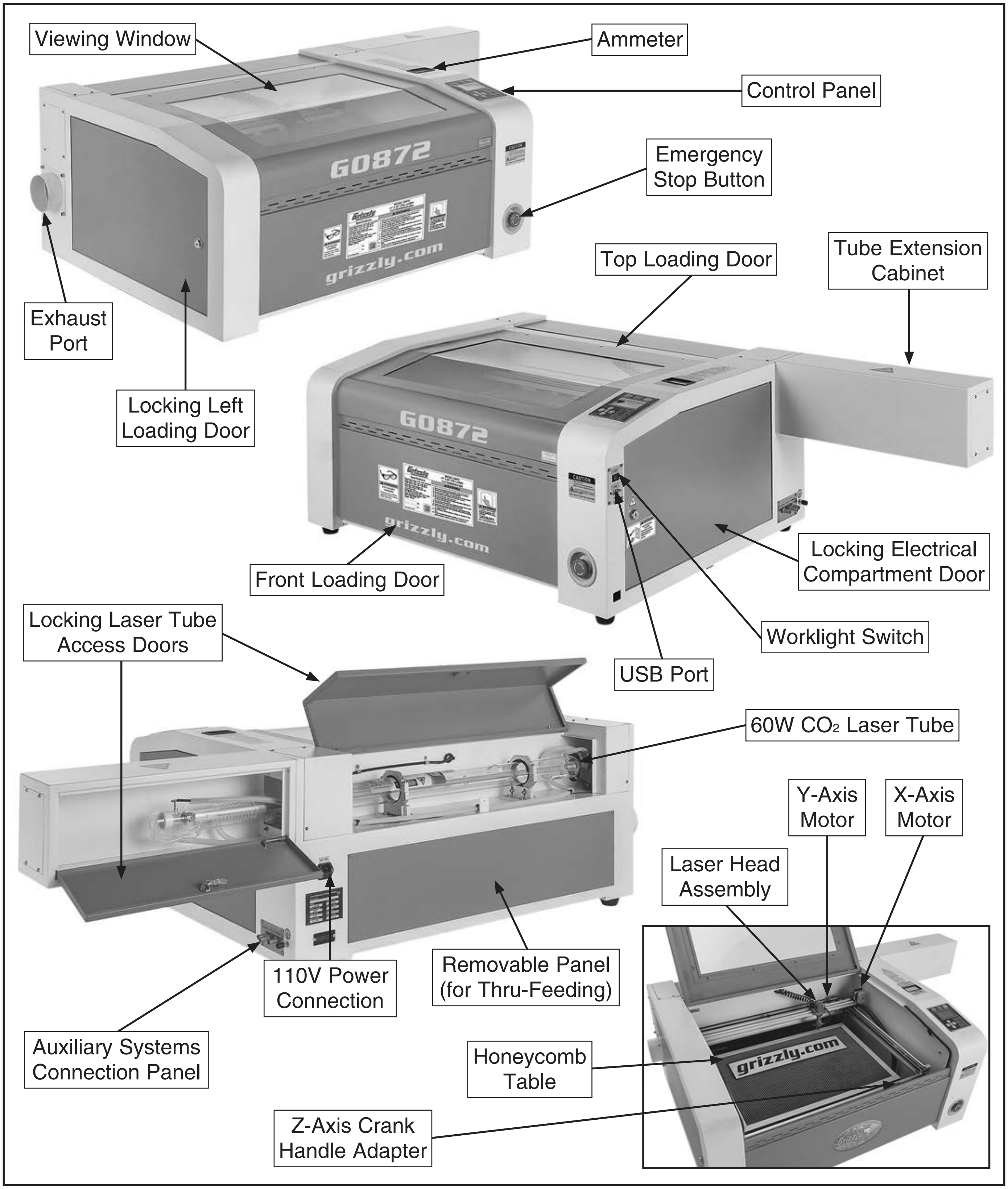
If you find this to be the case, and the difference between the manual and machine leaves you confused or unsure about something, check our website for an updated version. We post current manuals and manual updates for free on our website at **www.grizzly.com**.

Alternatively, you can call our Technical Support for help. Before calling, make sure you write down the **Manufacture Date** and **Serial Number** from the machine ID label (see below). This information is required for us to provide proper tech support, and it helps us determine if updated documentation is available for your machine.

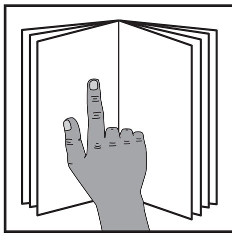


Identification

Become familiar with the names and locations of the controls and features shown below to better understand the instructions in this manual.



Controls & Components



⚠WARNING

To reduce your risk of serious injury, read this entire manual BEFORE using machine.

Refer to the following figures and descriptions to become familiar with the basic controls and components of this machine. Understanding these items and how they work will help you understand the rest of the manual and minimize your risk of injury when operating this machine.

Control Panel

The control panel (see **Figure 1**) is used for controlling machine operations and allows access to direct commands and selectable menu functions. Refer to **Command Tree** on **Page 73** for an extended map of menu features.

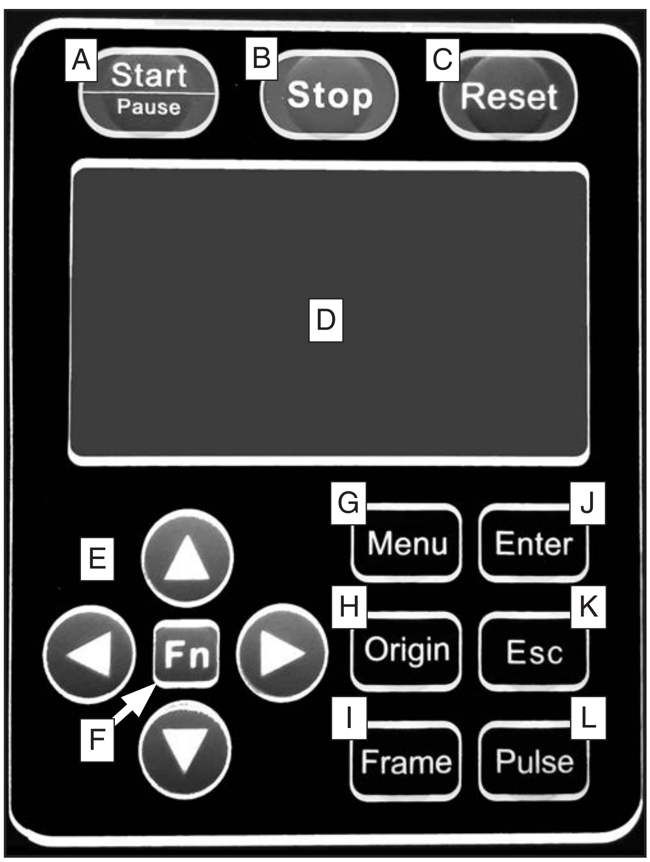


Figure 1. Control panel.

A. Start/Pause Button: Runs or pauses a toolpath operation.

B. Stop Button: Stops a running program or cancels a command, depending on function selected.

C. Reset Button: Resets a value or command, depending on menu item selected.

D. Screen Display: Visual interface between user and current machine state.

E. Arrow Nav Buttons: Navigates menu items or manually moves laser head.

F. Fn Button: Accesses root setup screen for advanced adjustments.

G. Menu Button: Accesses main menu screen.

H. Origin Button: Designates current physical location of laser head assembly as origin.

I. Frame Button: Instructs laser head to trace working envelope of image to be cut.

J. Enter Button: Opens currently highlighted command.

K. Esc Button: Exits selected command or returns to previous screen.

L. Pulse Button: Powers laser for a fraction of a second to assist with laser alignment or general troubleshooting purposes.

Additional Components

M. Ammeter: Located above control panel, shows amount of current being used by laser tube in milliamperes (mA).

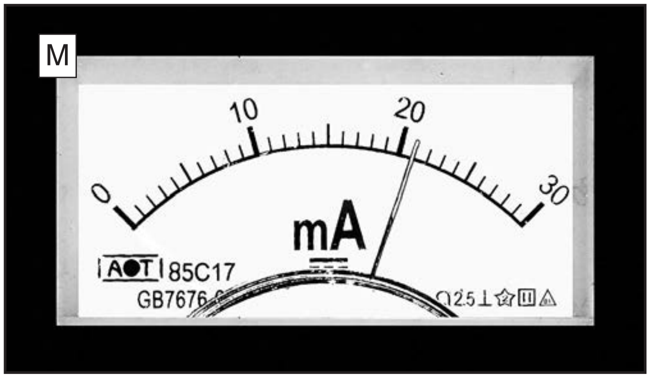


Figure 2. Ammeter indicating 21 mA.

N. Worklight Switch: Turns worklight under laser head assembly ***ON*** or ***OFF***.

O. USB 2.0 Port: Dedicated USB port for transferring files to machine.

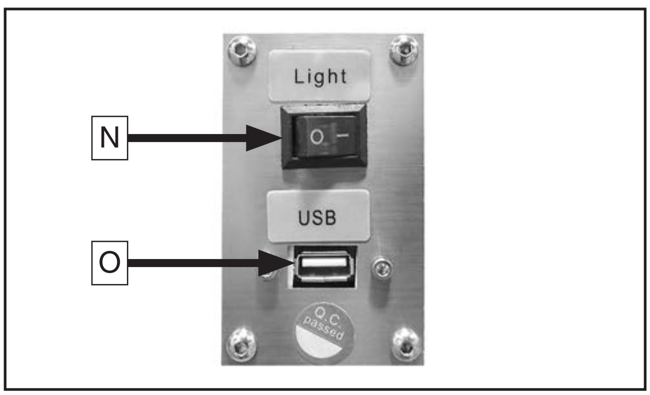


Figure 3. Worklight switch and USB Port.

P. Emergency Stop Button: Disables power to machine. To reset, twist button clockwise until it pops out.

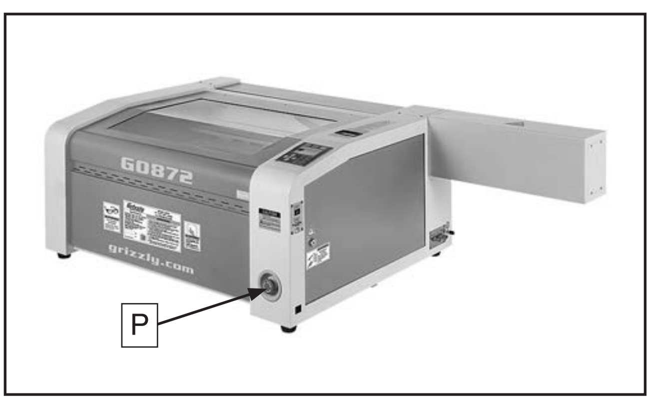


Figure 4. Emergency stop button location.

Q. Z-Axis Crank Handle: Removable hand crank that raises or lowers table height when rotated. Crank can be left in place during laser operations.

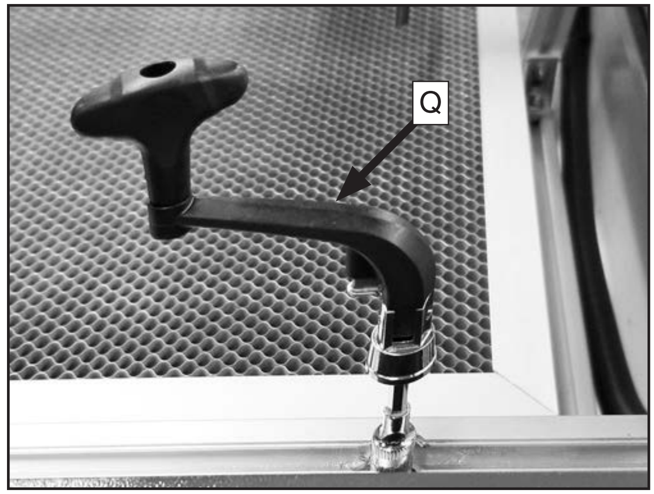


Figure 5. Z-Axis crank handle attached to table.

R. 60W CO_2 Laser Tube: CO_2 gas-filled laser tube rated for 60W output. Supported by two soft-mount saddles and straps.

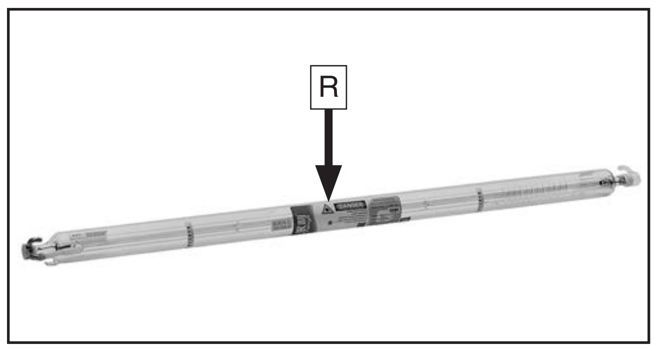


Figure 6. Laser tube (removed from machine for clarity).

S. 110V Power Connection: Connects machine to power supply with included power cord.

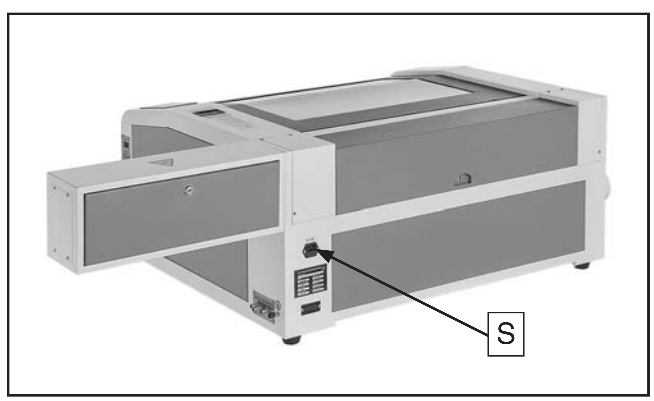


Figure 7. Power connection location.

Auxiliary Systems Connection Panel

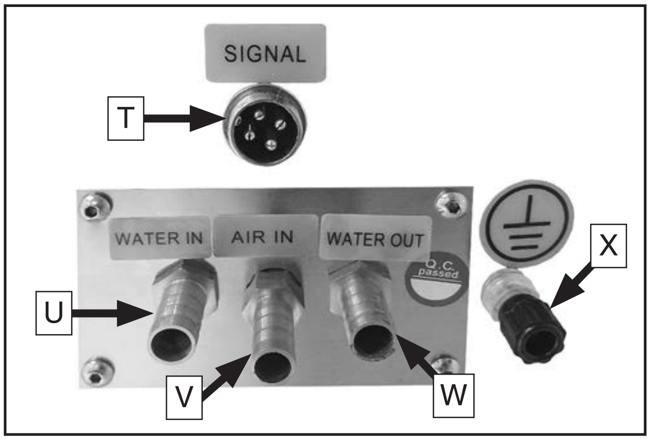


Figure 8. Auxiliary systems connection panel components.

T. Water Chiller Signal Connector: Provides water flow and high-temperature alarms when connected to water chiller.

U. Water Inlet Fitting: Supply connection for water chiller system.

V. Air Inlet Fitting: Supply connection for air pump.

W. Water Outlet Fitting: Return connection for water chiller system.

X. Earth Ground Post: Allows dissipation of static electricity generated during operation.

Auxiliary Systems

All three auxiliary systems are mandatory for laser operation: water chiller, air pump, and exhaust fan.

Y. Water Chiller: Cools laser tube by continuously circulating water through an internal loop of supply and return hoses.

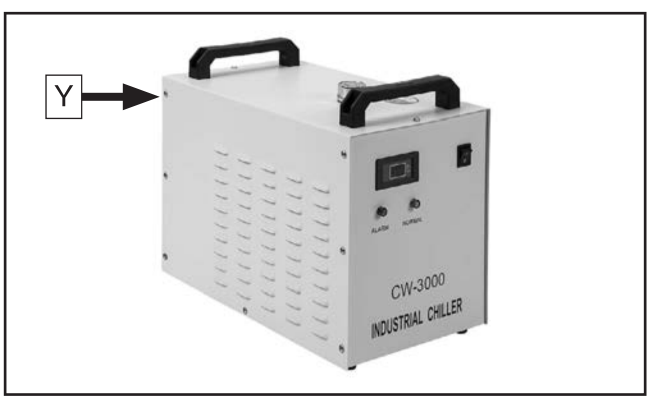


Figure 9. Water chiller system.

Z. Air Pump: Blows debris and gases away from point of laser burn using a jet of air produced by a diaphragm-style pump.

Figure 10. Diaphragm-style air pump.

AA. Exhaust Fan: Vents debris and gases created during laser operations out of machine using a centrifugal-style exhaust fan.

Figure 11. Centrifugal-style exhaust fan.

Glossary of Terms

The following is a list of common definitions, terms, and phrases used throughout this manual as they relate to this CNC laser and laser cutting in general. Become familiar with these terms for assembling, adjusting, or operating this machine. Your safety is **VERY** important to us at Grizzly!

Anode: The positive (+) terminal, electrode, or element of an electron tube or electrolytic cell.

Axis: Direction of movement. On a typical three-axis machine, axes are X (left to right), Y (front to back) & Z (up and down). Axis directions are described as positive or negative. On this machine, negative movement is defined as movement towards the front (Y), left (X), and downward (Z) portion of the working envelope.

CAD (Computer Aided Design): CAD software is used to create a digital model of a project.

CAM (Computer Aided Manufacturing): CAM software converts CAD models into a toolpath defined by code that CNC machines interpret.

Cathode: The negative (-) terminal, electrode, or element of an electron tube or electrolytic cell.

CNC (Computer Numerical Control): Automated operation of a machine by a computer program via written instructions.

ESD (Electrostatic Discharge): A sudden flow of electricity between two electrically charged objects caused by contact, an electrical short, or dielectric breakdown.

Focal Length: The distance from the focal point of a lens or mirror to the corresponding principal plane.

Frame Slop: Machine error code that indicates travel exceeds working envelope of X- and Y-axes.

Home Position: Machine designated zero point on all axes.

Name Over Lap: Machine error code which indicates file with same name is detected in destination memory location.

Origin: User designated zero point for a workpiece from which laser will reference positioning of all cutting/engraving.

Profile Toolpath: A toolpath that cuts around or along the profile of a set of vectors. Typically used to cut out the shape of a design.

Right-Hand Rule: A rule that uses the shape of the right hand to establish the standard orientation of vector quantities normal to a plane.

Soft Limits: Axis limits imposed by workspace boundaries based on controller settings and the location of home.

Stepper Motor: An electric motor (typically DC) that moves in precise steps when pulses are received. Has very accurate positioning and speed control.

Thermal Lens Effect: When energy from a laser beam passing through a sample is absorbed, causing heating of the sample along the beam path.

Toolpath: User-defined route that the laser follows to cut or engrave a workpiece.

TrackFrame: Machine command which verifies toolpath boundaries of queued image do not exceed available working envelope of workpiece.

Working Envelope: Total area that laser tip can travel within that does not exceed physical machine boundaries.

XSlop Over: Machine error code that indicates travel exceeds working envelope of X-axis.

YSlop Over: Machine error code that indicates travel exceeds working envelope of Y-axis.

MACHINE DATA SHEET

Grizzly Industrial, Inc.®

Customer Service #: (570) 546-9663 · To Order Call: (800) 523-4777 · Fax #: (800) 438-5901

MODEL G0872
60W 17" X 23" CNC LASER CUTTER/ENGRAVER

Product Dimensions:

- Weight .... 220 lbs.
- Width (side-to-side) x Depth (front-to-back) x Height .... 59 x 32 x 18 in.
- Footprint (Length/Width) .... 29-1/2 x 37-1/2 in.

Shipping Dimensions:

- Type .... Wood Crate
- Content .... Machine
- Weight .... 287 lbs.
- Length x Width x Height .... 54 x 38 x 26 in.
- Must Ship Upright .... Yes

Electrical:

- Power Requirement .... 110V, Single-Phase, 60 Hz
- Full-Load Current Rating 9.1A
- Minimum Circuit Size 15A
- Connection Type .... Cord & Plug
- Power Cord Included .... Yes
- Power Cord Length .... 6 ft.
- Power Cord Gauge .... 16 AWG
- Plug Included .... Yes
- Included Plug Type 5-15
- Switch Type .... Control Panel

Auxiliary Systems:

Water Chiller

- Power Requirement .... 110V, Single-Phase, 60 Hz
- Amps 1A

Exhaust Fan

- Power Requirement .... 110V, Single-Phase, 60 Hz
- Amps 4.9A

Air Pump

- Power Requirement .... 110V, Single-Phase, 60 Hz
- Amps 1A

Motor:

X-Axis Motor

- Type .... NEMA 17
- Amps 1.1A

Y-Axis Motor

- Type .... NEMA 17
- Amps 1.1A

Main Specifications:

Laser Information

Type........ Sealed CO_2 Laser Tube
Wattage........ 60W
Wavelength........ 1064nm
Class........ 4
Focus........ Manual
Cooling System........ Water

Cutting Information

Cutting Area........ 17 x 23 in.
Cutting Speed........ 0–1900 mm/min.
Minimum Shaping Character Size........ 0.04 x 0.04 in.
Resetting Position Accuracy........ 0.002 in.
Maximum Workpiece Height........ 3-1/2 in.
Cutting Accuracy........ 0.05mm

Table Information

Table Length........ 23-1/2 in.
Table Width........ 17-3/4 in.
Table Adjustment........ Manual

Construction Materials

Table........ Aluminum Lattice
Cabinet........ Steel
Paint Type/Finish........ Enamel

Other Related Information

Number of Exhaust Ports........ 1
Exhaust Port Size........ 3-3/4 in.
Design Software........ RDCam

Other Specifications:

Country of Origin........ China
Warranty........ 1 Year
Approximate Assembly & Setup Time........ 2 Hours
Serial Number Location........ Machine ID Label
ISO 9001 Factory........ Yes

Features:

LCD Control Panel
Manual Z-Axis Adjustment
Circular Bearing Rails
USB 2.0 Port
RDCam Laser Cutting/Engraving Software

Accessories:

Water Chiller System for Cooling Laser Tube
Exhaust Fan for Fume Extraction from Laser Compartment
Air Pump for Removing Debris During Operations
(2) Exhaust Ducts 4" x 60"

SECTION 1: SAFETY

For Your Own Safety, Read Instruction Manual Before Operating This Machine

The purpose of safety symbols is to attract your attention to possible hazardous conditions. This manual uses a series of symbols and signal words intended to convey the level of importance of the safety messages. The progression of symbols is described below. Remember that safety messages by themselves do not eliminate danger and are not a substitute for proper accident prevention measures. Always use common sense and good judgment.

DANGER — **Indicates an imminently hazardous situation which, if not avoided, WILL result in death or serious injury.**

WARNING — **Indicates a potentially hazardous situation which, if not avoided, COULD result in death or serious injury.**

CAUTION — **Indicates a potentially hazardous situation which, if not avoided, MAY result in minor or moderate injury. It may also be used to alert against unsafe practices.**

NOTICE — **Alerts the user to useful information about proper operation of the machine to avoid machine damage.**

Safety Instructions for Machinery

WARNING

OWNER'S MANUAL. Read and understand this owner's manual BEFORE using machine.

TRAINED OPERATORS ONLY. Untrained operators have a higher risk of being hurt or killed. Only allow trained/supervised people to use this machine. When machine is not being used, disconnect power, remove switch keys, or lock-out machine to prevent unauthorized use—especially around children. Make your workshop kid proof!

DANGEROUS ENVIRONMENTS. Do not use machinery in areas that are wet, cluttered, or have poor lighting. Operating machinery in these areas greatly increases the risk of accidents and injury.

MENTAL ALERTNESS REQUIRED. Full mental alertness is required for safe operation of machinery. Never operate under the influence of drugs or alcohol, when tired, or when distracted.

ELECTRICAL EQUIPMENT INJURY RISKS. You can be shocked, burned, or killed by touching live electrical components or improperly grounded machinery. To reduce this risk, only allow qualified service personnel to do electrical installation or repair work, and always disconnect power before accessing or exposing electrical equipment.

DISCONNECT POWER FIRST. Always disconnect machine from power supply BEFORE making adjustments, changing tooling, or servicing machine. This prevents an injury risk from unintended startup or contact with live electrical components.

EYE PROTECTION. Always wear ANSI-approved safety glasses or a face shield when operating or observing machinery to reduce the risk of eye injury or blindness from flying particles. Everyday eyeglasses are NOT approved safety glasses.

⚠WARNING

WEARING PROPER APPAREL. Do not wear clothing, apparel or jewelry that can become entangled in moving parts. Always tie back or cover long hair. Wear non-slip footwear to reduce risk of slipping and losing control or accidentally contacting cutting tool or moving parts.

HAZARDOUS DUST. Dust created by machinery operations may cause cancer, birth defects, or long-term respiratory damage. Be aware of dust hazards associated with each workpiece material. Always wear a NIOSH-approved respirator to reduce your risk.

HEARING PROTECTION. Always wear hearing protection when operating or observing loud machinery. Extended exposure to this noise without hearing protection can cause permanent hearing loss.

REMOVE ADJUSTING TOOLS. Tools left on machinery can become dangerous projectiles upon startup. Never leave chuck keys, wrenches, or any other tools on machine. Always verify removal before starting!

USE CORRECT TOOL FOR THE JOB. Only use this tool for its intended purpose—do not force it or an attachment to do a job for which it was not designed. Never make unapproved modifications—modifying tool or using it differently than intended may result in malfunction or mechanical failure that can lead to personal injury or death!

AWKWARD POSITIONS. Keep proper footing and balance at all times when operating machine. Do not overreach! Avoid awkward hand positions that make workpiece control difficult or increase the risk of accidental injury.

CHILDREN & BYSTANDERS. Keep children and bystanders at a safe distance from the work area. Stop using machine if they become a distraction.

GUARDS & COVERS. Guards and covers reduce accidental contact with moving parts or flying debris. Make sure they are properly installed, undamaged, and working correctly BEFORE operating machine.

FORCING MACHINERY. Do not force machine. It will do the job safer and better at the rate for which it was designed.

NEVER STAND ON MACHINE. Serious injury may occur if machine is tipped or if the cutting tool is unintentionally contacted.

STABLE MACHINE. Unexpected movement during operation greatly increases risk of injury or loss of control. Before starting, verify machine is stable and mobile base (if used) is locked.

USE RECOMMENDED ACCESSORIES. Consult this owner's manual or the manufacturer for recommended accessories. Using improper accessories will increase the risk of serious injury.

UNATTENDED OPERATION. To reduce the risk of accidental injury, turn machine ***OFF*** and ensure all moving parts completely stop before walking away. Never leave machine running while unattended.

MAINTAIN WITH CARE. Follow all maintenance instructions and lubrication schedules to keep machine in good working condition. A machine that is improperly maintained could malfunction, leading to serious personal injury or death.

DAMAGED PARTS. Regularly inspect machine for damaged, loose, or mis-adjusted parts—or any condition that could affect safe operation. Immediately repair/replace BEFORE operating machine. For your own safety, DO NOT operate machine with damaged parts!

MAINTAIN POWER CORDS. When disconnecting cord-connected machines from power, grab and pull the plug—NOT the cord. Pulling the cord may damage the wires inside. Do not handle cord/plug with wet hands. Avoid cord damage by keeping it away from heated surfaces, high traffic areas, harsh chemicals, and wet/damp locations.

EXPERIENCING DIFFICULTIES. If at any time you experience difficulties performing the intended operation, stop using the machine! Contact our Technical Support at (570) 546-9663.

Additional Safety for CNC Laser Cutters/Engravers

⚠WARNING

Severe eye injury or blindness can occur from looking directly into laser beam, or staring at laser contact point for more than a few seconds. Touching hot machine parts and workpieces can cause serious skin burns. To reduce these risks, operator and bystanders MUST completely heed the warnings below.

EYE INJURIES. Operator and bystanders MUST wear ANSI-approved safety glasses rated for use with a Class 4 laser when machine is operating. DO NOT look directly into laser beam, or stare at laser contact point for more than a few seconds, or severe eye injury or blindness may result.

AVOID SKIN BURNS. NEVER put hands in or near path of laser. Material cut by a laser can be hot. ALWAYS wear leather gloves when handling processed material. Allow machine to cool before starting any adjustment or service/maintenance procedure.

FIRE HAZARD. Laser beam produces extremely high temperatures and significant amounts of heat as material is cut. DO NOT process materials that are highly flammable or explosive. Keep flammable materials well away from machine during operations. If materials do catch fire during operations, extinguish immediately. ALWAYS keep a properly maintained fire extinguisher nearby.

REFLECTIVE MATERIALS. DO NOT process materials with reflective surfaces. These materials will redirect the laser beam, exposing operator and bystanders to serious injury, and causing damage to mechanical and electrical components inside machine.

UNATTENDED MACHINE. DO NOT leave machine unattended during operation. Materials may catch fire during operation. Fires MUST be extinguished immediately to prevent personal injury or property damage.

SAFETY DEVICES. DO NOT modify or disable safety devices on machine. Laser is designed to shut off if cover is opened. Severe injury may occur if operator or bystanders come into contact with laser beam during operation.

SAFE OPERATING LOCATION. DO NOT place machine where it can be exposed to rain or moisture. Exposure to water creates a shock hazard and will reduce life of machine.

POWER DISCONNECT. To reduce risk of electrocution or injury from unexpected startup, make sure machine is turned OFF and disconnected from power before starting any inspection, adjustment, or service/maintenance procedure.

PROPERLY MAINTAIN MACHINE. Keep machine in proper working condition to help ensure all safety components function as intended. Perform routine inspections and all necessary maintenance indicated in owner's manual. Never operate machine with damaged or worn parts.

⚠WARNING

Like all machinery there is potential danger when operating this machine. Accidents are frequently caused by lack of familiarity or failure to pay attention. Use this machine with respect and caution to decrease the risk of operator injury. If normal safety precautions are overlooked or ignored, serious personal injury may occur.

⚠CAUTION

No list of safety guidelines can be complete. Every shop environment is different. Always consider safety first, as it applies to your individual working conditions. Use this and other machinery with caution and respect. Failure to do so could result in serious personal injury, damage to equipment, or poor work results.

Additional Safety for Toxic Fumes Generated by Laser Cutting

⚠WARNING

Long-term respiratory damage, toxicity, cancer, or birth defects can occur from inhaling fumes, vapors, and particulates generated while cutting substrates without adequate ventilation, exhaust, and air filtration. All users must be properly trained on the potential hazards, control measures, manufacturer's operating procedures, use of personal protective equipment (PPE), emergency procedures, and safety precautions for CNC laser operations. To reduce these risks, operator and bystanders MUST completely heed the warnings below.

TOXIC MATERIALS. Exposure to certain types of fumes can result in serious, potentially deadly health effects. To reduce this risk, research toxicity of material types you work with, and always seek to minimize/eliminate exposure to yourself and others. Obtain the Safety Data Sheet (SDS) from material manufacturer BEFORE operations, and never knowingly engrave or cut a workpiece that has been treated with or contains material which releases toxic byproducts when heated.

TOXIC FUMES. Cutting or engraving metals and plastics give off highly toxic fumes, vapors, and air particulates containing zinc, lead, beryllium, cadmium, mercury, fluorine, hexavalent chromium, chlorine gas, and many others. These fumes and air contaminants can damage the machine and harm your health. If the air filtration system or exhaust system is malfunctioning, immediately stop operations and correct the issue.

ADEQUATE VENTILATION. Only use CNC lasers in spaces with adequate ventilation. Some materials can produce vapors and fumes that may irritate the nose, throat, and respiratory tract, or cause suffocation. Only operate CNC lasers with a fully-functioning exhaust and air filtration system. Utilize additional personnel to monitor operator from outside the operating area in the event of equipment failure.

EXHAUST SYSTEM. Only operate CNC laser with a properly installed and functioning exhaust system. To effectively exhaust fumes and particulates from the machine during operations, use an exhaust system rated for a minimum of **200 CFM** (Cubic Feet per Minute) at 6" static pressure.

AIR FILTRATION SYSTEM. CNC lasers must be equipped with an air filtration system that utilizes MERV 15+ or HEPA filters. NEVER modify air filtration systems or bypass safety features. Only operate air filtration systems with all filters and covers in place during operation. If any filter is missing or has been replaced with a non-specification filter, the filtration system will not properly filter contaminated air and will be unsafe to use.

INSPECTIONS/MAINTENANCE. Always inspect exhaust and air filtration systems for leaking gaskets or seals prior to machine operations. Repair or replace defective components before starting.

FILTER CLEANING/DISPOSAL. Filters must be changed regularly according to the frequency of use, or as specified by the manufacturer. When servicing filters, make sure operator and any bystanders are wearing Personal Protective Equipment (PPE). When vacuuming filters and cabinet, only use a shop vacuum that is equipped with a MERV 15+ or HEPA filter, or dangerous particulates may be spread throughout the area contaminating the air. Wrap all waste filters in air-tight plastic bags, then mark and dispose of according to current laws and regulations.

EXPERIENCING DIFFICULTIES. Keep in mind that CNC laser hazards are intensified in a confined space. If you are experiencing difficulties performing the intended operation, stop using the equipment, and contact the Occupational Safety and Health Administration (OSHA) at (800) 321-6742, or online at **www.osha.gov** to find out how to design and maintain the best overall CNC laser toxic fume extraction system for your needs.

SECTION 2: POWER SUPPLY

Availability

Before installing the machine, consider the availability and proximity of the required power supply circuit. If an existing circuit does not meet the requirements for this machine, a new circuit must be installed. To minimize the risk of electrocution, fire, or equipment damage, installation work and electrical wiring must be done by an electrician or qualified service personnel in accordance with all applicable codes and standards.

⚠WARNING

Electrocution, fire, shock, or equipment damage may occur if machine is not properly grounded and connected to power supply.

Full-Load Current Rating

The full-load current rating is the amperage a machine draws at 100% of the rated output power. On machines with multiple motors, this is the amperage drawn by the largest motor or sum of all motors and electrical devices that might operate at one time during normal operations.

Full-Load Current Rating at 110V..... 9.1 Amps

The full-load current is not the maximum amount of amps that the machine will draw. If the machine is overloaded, it will draw additional amps beyond the full-load rating.

If the machine is overloaded for a sufficient length of time, damage, overheating, or fire may result—especially if connected to an undersized circuit. To reduce the risk of these hazards, avoid overloading the machine during operation and make sure it is connected to a power supply circuit that meets the specified circuit requirements.

Circuit Information

A power supply circuit includes all electrical equipment between the breaker box or fuse panel in the building and the machine. The power supply circuit used for this machine must be sized to safely handle the full-load current drawn from the machine for an extended period of time. (If this machine is connected to a circuit protected by fuses, use a time delay fuse marked D.)

⚠CAUTION

For your own safety and protection of property, consult an electrician if you are unsure about wiring practices or electrical codes in your area.

Note: *Circuit requirements in this manual apply to a dedicated circuit—where only one machine will be running on the circuit at a time. If machine will be connected to a shared circuit where multiple machines may be running at the same time, consult an electrician or qualified service personnel to ensure circuit is properly sized for safe operation.*

Circuit Requirements

This machine is prewired to operate on a power supply circuit that has a verified ground and meets the following requirements:

Nominal Voltage 110V, 115V, 120V
Cycle ....60 Hz
Phase .... Single-Phase
Power Supply Circuit .... 15 Amps
Plug/Receptacle ....5-15 (Included)

Grounding Requirements

This machine MUST be grounded. In the event of certain malfunctions or breakdowns, grounding reduces the risk of electric shock by providing a path of least resistance for electric current.

This machine is equipped with a power cord that has an equipment-grounding wire and a grounding plug. Only insert plug into a matching receptacle (outlet) that is properly installed and grounded in accordance with all local codes and ordinances. DO NOT modify the provided plug!

Laser machines generate strong electrical fields that can charge the machine housing with micro voltages. These voltages must be allowed to dissipate through a separate, physical earth ground connection (see **Figure 12**) than what is provided by the power cord.

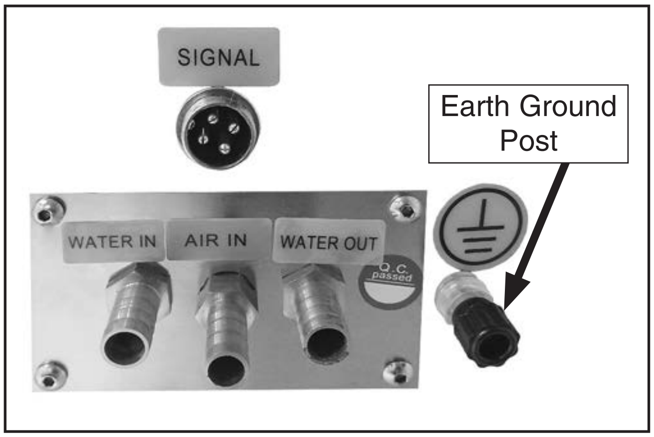


Figure 12. Earth ground post location.

Resistance to ground of a single-made electrode is **5 Ohms or less**. Refer to an electrician for guidance and testing, if required.

NOTICE

Choose a location where the physical earth ground can be made with an anti-static grounding rod for communication and digital equipment. This is especially true for areas prone to thunderstorms, dry weather, and electrostatic discharge.

⚠WARNING

Serious injury could occur if you connect machine to power before completing setup process. DO NOT connect to power until instructed later in this manual.

Improper connection of the equipment-grounding wire can result in a risk of electric shock. The wire with green insulation (with or without yellow stripes) is the equipment-grounding wire. If repair or replacement of the power cord or plug is necessary, do not connect the equipment-grounding wire to a live (current carrying) terminal.

Check with a qualified electrician or service personnel if you do not understand these grounding requirements, or if you are in doubt about whether the tool is properly grounded. If you ever notice that a cord or plug is damaged or worn, disconnect it from power, and immediately replace it with a new one.

Extension Cords

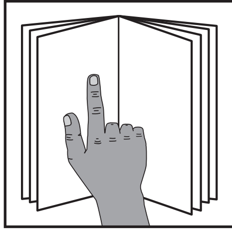
We do not recommend using an extension cord with this machine. If you must use an extension cord, only use it if absolutely necessary and only on a temporary basis.

Extension cords cause voltage drop, which can damage electrical components and shorten motor life. Voltage drop increases as the extension cord size gets longer and the gauge size gets smaller (higher gauge numbers indicate smaller sizes).

Any extension cord used with this machine must be in good condition and contain a ground wire and matching plug/receptacle. Additionally, it must meet the following size requirements:

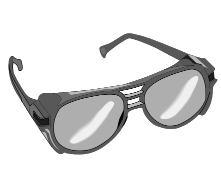
Minimum Gauge Size .... 14 AWG
Maximum Length (Shorter is Better) .... 50 ft.

SECTION 3: SETUP



⚠WARNING

To reduce your risk of serious injury, read this entire manual BEFORE using machine.



⚠WARNING

Eye injury hazard! Always wear safety glasses when using this machine.

⚠WARNING

This machine and its components are very heavy. Get lifting help or use lifting equipment such as a pallet jack to move heavy items.

Needed for Setup

The following items are needed, but not included, for the setup/assembly of this machine.

Description	**Qty**
• Additional Person for Lifting	1
• Safety Glasses (for each person)	1
• Pallet Jack (rated for 500 lbs.)	1
• Power Drill w/Phillips Bit #2	1
• Crowbar	1
• Level	1
• Utility Knife	1
• Distilled Water	4 Gal.

Unpacking

This machine was carefully packaged for safe transport. When unpacking, separate all enclosed items from packaging materials and inspect them for shipping damage. ***If items are damaged, please call us immediately at (570) 546-9663.***

IMPORTANT: Save all packaging materials until you are completely satisfied with the machine and have resolved any issues between Grizzly or the shipping agent. *You MUST have the original packaging to file a freight claim. It is also extremely helpful if you need to return your machine later.*

To prevent machine damage to sensitive parts and ensure your safety: prior to assembly, it is mandatory that the installer read, understand, and apply best practices and safeguards. It is also highly recommended that the operator is included in the assembly process because some day-to-day operational adjustments are linked to areas being assembled.

- DO NOT make adjustments until instructed.
- Use a flashlight to inspect hidden machine areas for any stray packing materials or foreign objects that could catch fire or jam mechanisms during operation.
- DO NOT touch or wipe mirrors. If a mirror needs to be cleaned, refer to **Cleaning Laser Optics** on **Page 45**.
- While assembling machine, familiarize yourself with machine component locations and recognize their purpose.
- DO NOT use any type of power tool for assembly to avoid overtightening fasteners.
- DO NOT overtighten factory-installed fasteners and components that are associated with laser system, or misalignment and component damage may occur.

Inventory

The following is a list of items shipped with your machine. Before beginning setup, lay these items out and inventory them.

If any non-proprietary parts are missing (e.g. a nut or a washer), we will gladly replace them; or for the sake of expediency, replacements can be obtained at your local hardware store.

NOTICE

If you cannot find an item on this list, carefully check around/inside the machine and packaging materials. Often, these items get lost in packaging materials while unpacking or they are pre-installed at the factory.

Main Components (Figures 13–16)	**Qty**
A. CNC Laser Machine w/Laser Tube	1
B. Honeycomb Table 17" x 23"	1
C. Water Chiller w/Tubing	1
D. Water Chiller Power Cord	1
E. Water Chiller Signal Cord	1
F. Air Pump w/Tubing	1
G. Air Pump Barbed Fitting	1
H. Exhaust Fan	1
I. Exhaust Ducting 4" x 60"	2

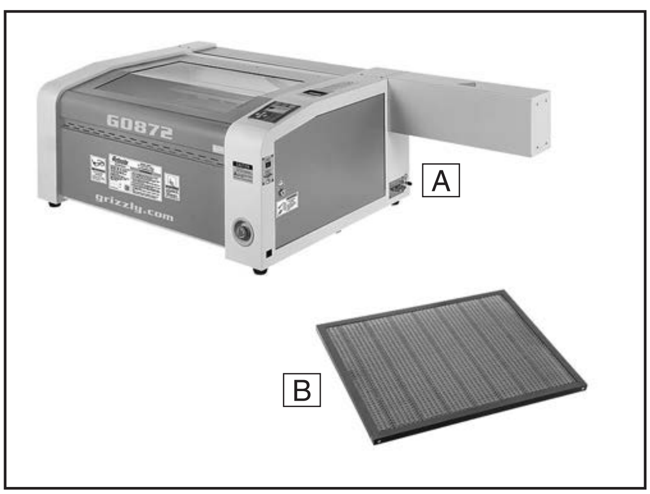


Figure 13. CNC laser machine inventory.

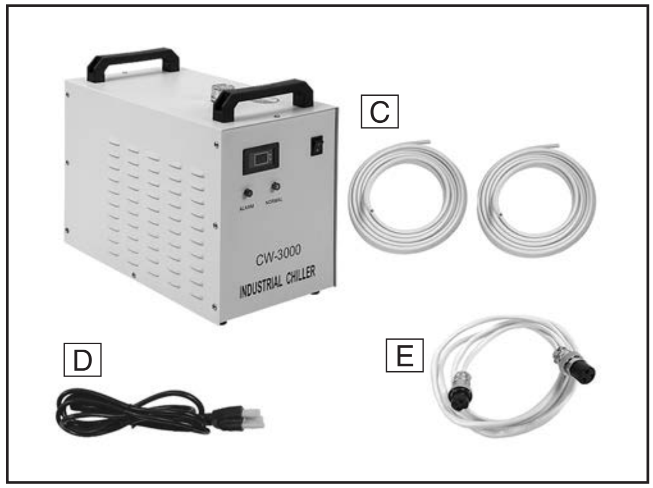


Figure 14. Water chiller inventory.

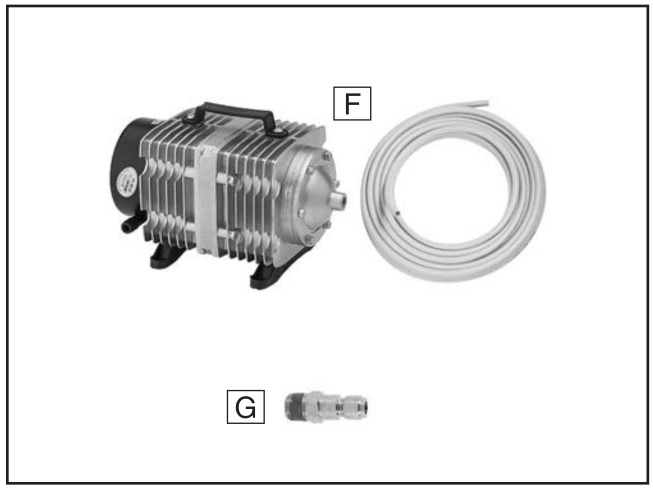


Figure 15. Air pump inventory.

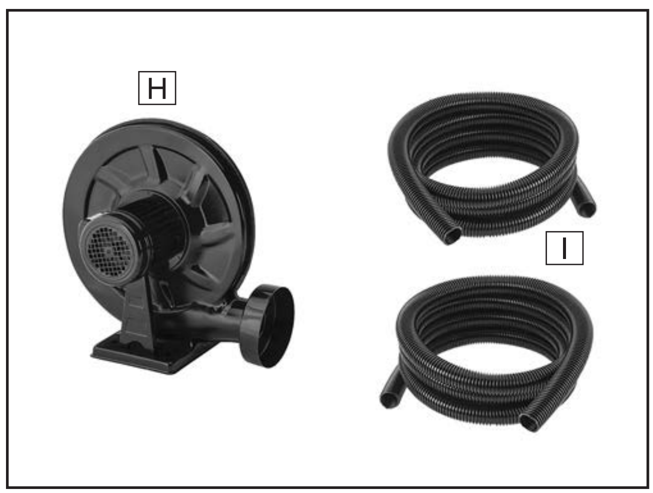


Figure 16. Exhaust system inventory.

Loose Parts (Figure 17)	Qty
J. Toolbox	1
K. Z-Axis Crank Handle	1
L. Optics Disassembly Tool	1
M. Laser Beam Alignment Gauge	1
N. Mirror Alignment Gauge	1
O. Focus Gauge	1
P. Laser Cutting Depth Gauge	1
Q. USB Flash Drive	1
R. Software Installation Discs	2
S. Absorbent Cotton Bags	2
T. Exhaust Port 3¾"	1
U. Hose Clamps 4"	3
V. Hex Wrenches 2, 2.5, 3, 5mm	1 Ea.
W. Locking Door Keys	4
X. Limit Switch (Spare Parts)	1
Y. Light Emitting Diodes (Spare Parts)	2
Z. Power Cord	1
AA. Flat Head Screwdriver ¼"	1

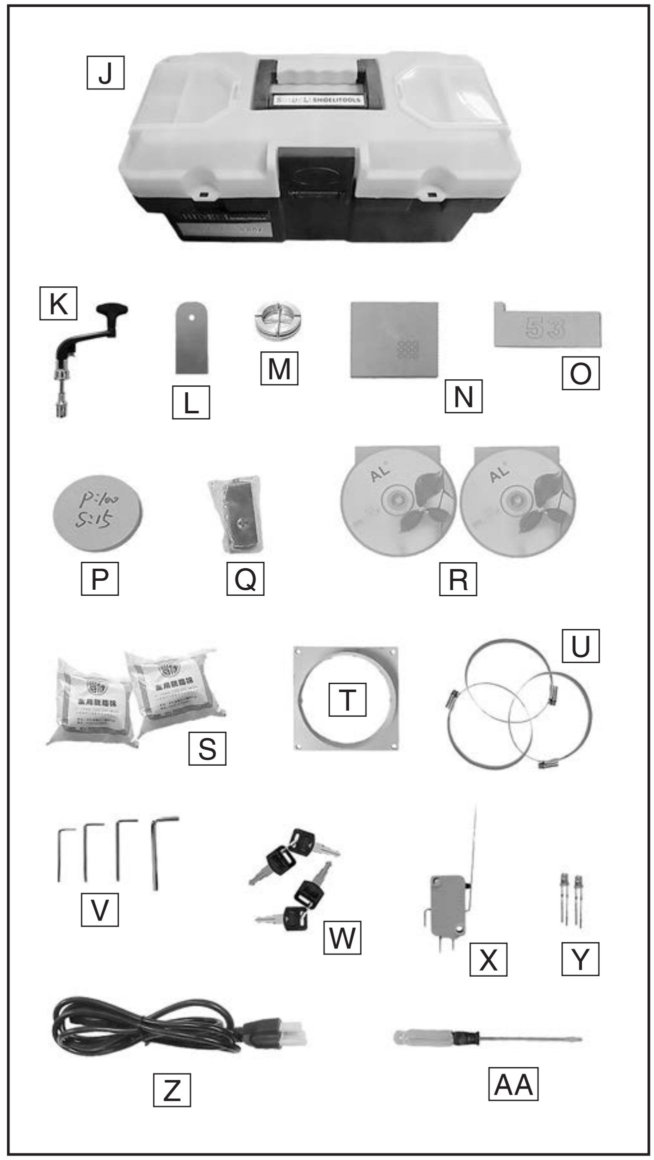


Figure 17. Loose parts inventory.

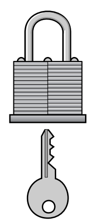
Site Considerations

Weight Load

Refer to the **Machine Data Sheet** for the weight of your machine. Make sure that the surface upon which the machine is placed will bear the weight of the machine, additional equipment that may be installed on the machine, and the heaviest workpiece that will be used. Additionally, consider the weight of the operator and any dynamic loading that may occur when operating the machine.

Space Allocation

Consider the largest size of workpiece that will be processed through this machine and provide enough space around the machine for adequate operator material handling or the installation of auxiliary equipment. With permanent installations, leave enough space around the machine to open or remove doors/covers as required by the maintenance and service described in this manual. **See below for required space allocation.**



⚠CAUTION

Children or untrained people may be seriously injured by this machine. Only install in an access restricted location.

Physical Environment

The physical environment where the machine is operated is important for safe operation and longevity of machine components. For best results, operate this machine in a dry environment that is free from excessive moisture, hazardous chemicals, airborne abrasives, or extreme conditions. Extreme conditions for this type of machinery are generally those where the ambient temperature range exceeds 41°–104°F; the relative humidity range exceeds 20%–95% (non-condensing); or the environment is subject to vibration, shocks, or bumps.

Electrical Installation

Place this machine near an existing power source. Make sure all power cords are protected from traffic, material handling, moisture, chemicals, or other hazards. Make sure to leave enough space around machine to disconnect power supply or apply a lockout/tagout device, if required.

Lighting

Lighting around the machine must be adequate enough that operations can be performed safely. Shadows, glare, or strobe effects that may distract or impede the operator must be eliminated.

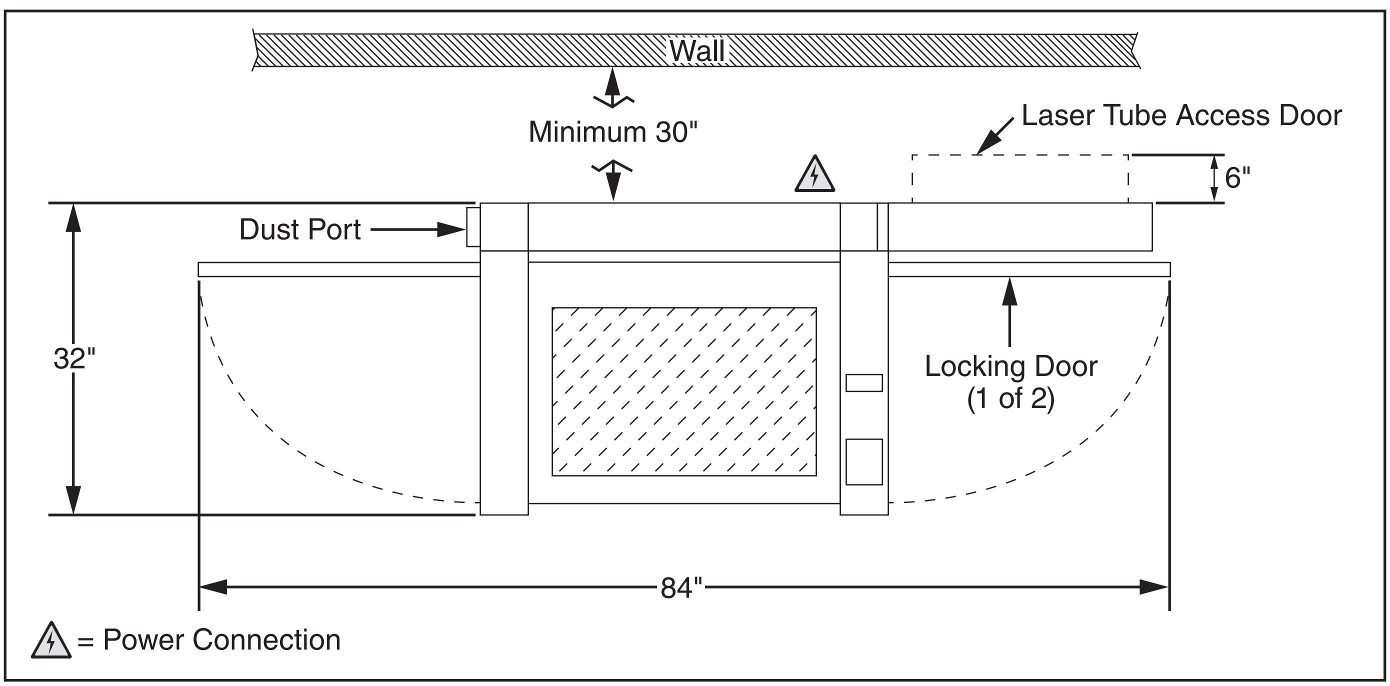


Figure 18. Model G0872 minimum working clearances.

Lifting & Placing

⚠WARNING

This machine and its components are very heavy. Get lifting help or use lifting equipment such as a pallet jack to move heavy items.

DO NOT attempt to lift or move this machine without using the proper lifting equipment (such as a pallet jack) or the necessary assistance from other people. Each piece of lifting equipment must be rated for **at least 500 lbs.** to support dynamic loads that may be applied while lifting. Refer to **Needed for Setup** on **Page 16** for complete list of needed equipment for setup and installation.

Verify location meets the following conditions:

- Adequate ventilation so machine does not fill an enclosed area with toxic fumes from cutting certain types of materials.

- Immediate access to auxiliary systems for verifying operation and ease of maintenance.

To lift and place machine:

1. Using a pallet jack, move crate to machine work site location.

2. Remove crate top and sides, components inside crate, and blocks around machine base.

3. To reduce weight, remove honeycomb table and loose parts toolbox from laser machine interior.

Note: *Honeycomb table is not permanently mounted and requires no tools to remove.*

4. With assistance from an additional person, lift machine and place it in desired location.

Leveling

Leveling machinery helps precision components remain straight and flat during the lifespan of the machine. The bed of a machine may slowly twist over time.

After placing machine on a flat, even surface, use a level to check each axis. Repeat as needed.

Installing Water Chiller System

The water chiller system should be located away from any area where freezing temperatures may occur. This includes locations where winter power outages allow work areas to drop below freezing.

In desert locations where ambient temperatures can rise over 100°F (37°C), you may be required to purchase a refrigeration-style water chiller, have bags of ice readily available and replaced, or incorporate additional water chilling equipment in the same loop. The water temperature MUST be kept below 122°F (50°C) for proper laser operation and maximum tube life.

The work area must be properly cleaned to prevent contaminants from being drawn into the water chiller and restricting airflow to the radiator. Keep the water chiller a minimum of 12" away from all obstructions.

IMPORTANT: WATER IN and WATER OUT connections on the auxiliary systems connection panel are NOT reversible. Correct hose orientation is required (see **Figure 19** for typical water chiller system setup).

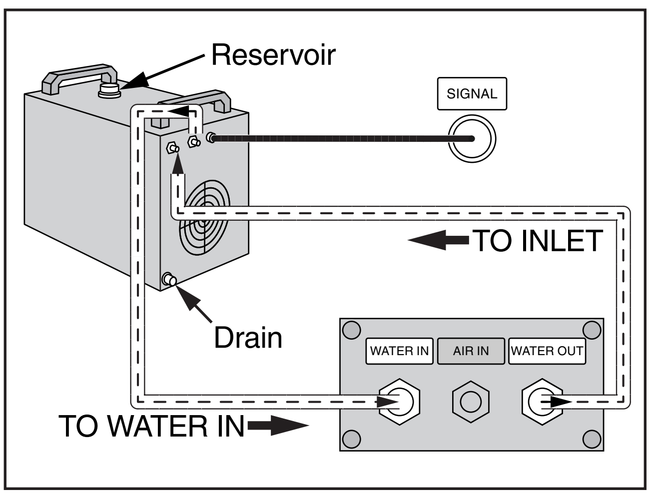


Figure 19. Typical water chiller system setup.

To install water chiller system:

1. Remove any dust caps or plugs (if equipped) from lines and fittings on laser machine and water chiller.

2. On water chiller (see **Figure 20**), install water chiller tubing onto INLET and OUTLET fittings, then connect signal cord to ALARM OUTPUT receptacle.

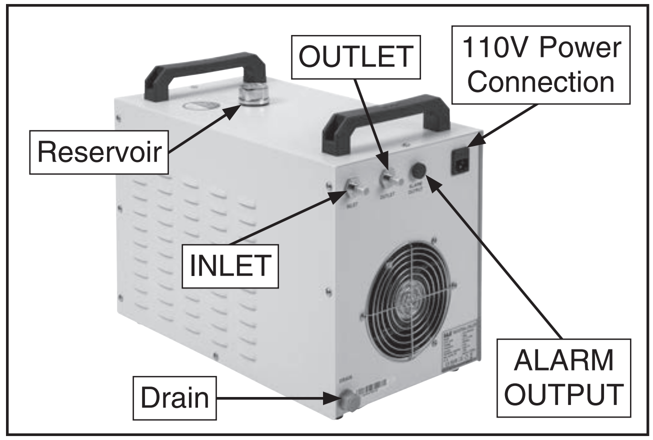


Figure 20. Water chiller connections.

3. Connect opposite end of OUTLET hose to auxiliary systems connection panel WATER IN fitting (see **Figure 21**).

4. Connect opposite end of INLET hose to auxiliary systems connection panel WATER OUT fitting (see **Figure 21**).

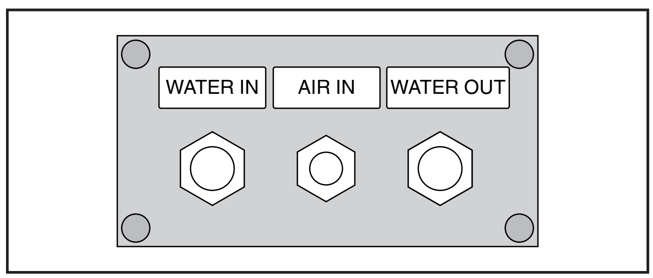


Figure 21. Water and air connection points on auxiliary systems connection panel.

5. Connect signal cord to SIGNAL receptacle on machine, and fill water chiller reservoir with 3 gallons of distilled water.

6. Connect water chiller system to power and turn ***ON***. Allow water to cycle for one minute and verify air bubbles have released from laser tube.

   Note: *Loosen laser tube saddle straps then slowly rotate laser tube while slightly raising tube at cathode-end to release air bubbles.*

7. Remove reservoir cap and verify water level is 3"–6" from reservoir opening.

Installing Air Pump

The air pump is designed to blow air directly on the laser focal point. Place the air pump away from areas where dust can be drawn in through the inlet port.

Note: *A barbed fitting on the inlet port can be fitted with a hose to draw air from an alternate source, if required.*

The pump has cooling fins to release heat from operation and should NEVER be located inside of an unventilated compartment, or any area where temperature will increase from pump operation.

To install air pump:

1. Thread barbed fitting into air pump outlet port (see **Figure 22**) and tighten.

Figure 22. Air pump components.

2. Install air pump tubing onto outlet port barbed fitting, and connect to auxiliary systems connection panel AIR IN fitting (see **Figure 21** on **Page 21**).

3. Connect air pump to an available 110V grounded power source.

4. Place air pump, electrical cord, and hoses in a location that prevents tripping hazards, hose kinks, and abrasive damage.

   Note: *The air pump generates vibrations and should be secured to prevent pump from "walking" during operation.*

Installing Exhaust Fan

The exhaust fan vents debris and gases created during laser operations using a centrifugal-style exhaust fan. The exhaust fan should be mounted in a fixed location that reduces vibration, and the inlet and outlet ducting should be secured at each end to prevent separation. Extending ducting over six feet, or adding multiple elbow connections is NOT recommended due to reduced exhaust efficiency.

The exhaust fan *does not* remove residual odors from the machine when not in use. If materials have an unpleasant odor while being cut, consider re-locating the machine to an area with greater ambient ventilation.

If mandated by local fire codes, a spark arrester may have to be incorporated into the exhaust system. One example of this is when the outlet duct is connected to a vented container partially filled with water. Inside the vented container, one or more baffles are used to direct smoke and embers toward the water, which will then naturally rise up and outward from a vent on top of the container.

IMPORTANT: The machine and exhaust fan are only to be used for laser cutting operations. DO NOT use machine as a downdraft table for sanding or other operations.

⚠WARNING

Gases and fumes generated by laser machines are hazardous to your health. It is your responsibility to install a dedicated and rated fume extractor if toxic gases or fumes are produced during laser cutting operations.

⚠WARNING

Dust and embers from laser machines present a fire hazard. DO NOT direct exhaust ports anywhere combustible materials exist, or combine laser exhaust fan with woodworking dust collection systems.

To install exhaust fan:

1. Remove lower end cover and (4) cap screws from left side of machine (see **Figure 23**).

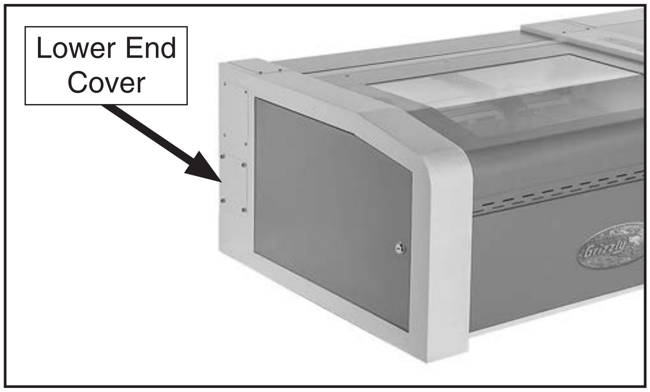


Figure 23. Location of end cover and securing cap screws.

2. Install exhaust port (see **Figure 24**) using (4) cap screws previously removed in **Step 1**.

   Note: *Save lower end cap removed during* ***Step 1*** *in the event machine must be prepared for long-term storage.*

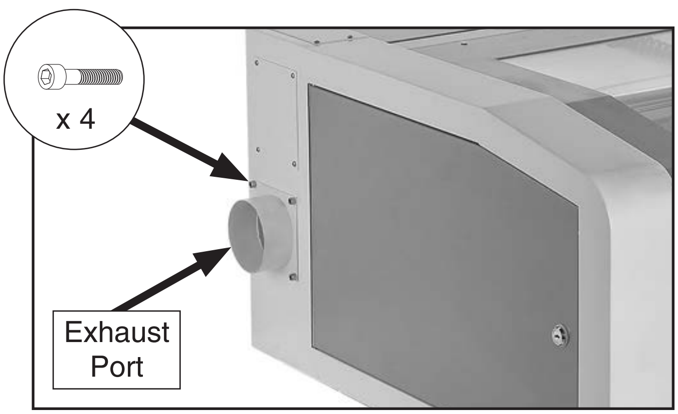


Figure 24. Exhaust port installed.

3. Place exhaust fan, electrical cord, and ducting in a location that prevents tripping hazards, duct kinks, and abrasive damage.

   Note: *Ensure ducting is kept as straight as possible for maximum efficiency.*

4. Use included hose clamps to connect and secure one end of ducting to exhaust port on machine, then connect and secure other end of ducting to inlet port of exhaust fan, as shown in **Figure 25**.

Minimum CFM at Exhaust Port: 200 CFM
DO NOT confuse this CFM recommendation with the rating of the fume extractor. To determine the CFM at the exhaust port, you must consider these variables: (1) CFM rating of the fume extractor, (2) hose type and length between air filters (MERV 15+, HEPA, Carbon, etc.) and the machine, (3) number of branches or wyes, and (4) amount of other open lines throughout the system. Explaining how to calculate these variables is beyond the scope of this manual. Consult an expert or purchase a dedicated CNC laser "how-to" book.

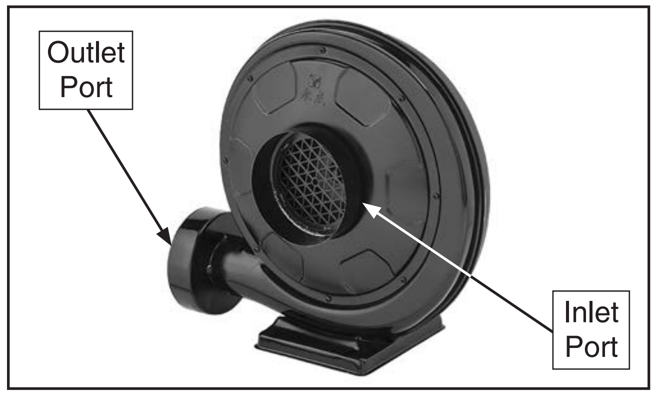


Figure 25. Exhaust fan inlet and outlet ports.

5. Connect and secure remaining ducting to exhaust fan outlet port (see **Figure 25**).

6. Route and connect ducting to an air filtration system that utilizes a MERV 15+ or HEPA filter.

7. Secure exhaust fan in an accessible location where ducting can be removed and cleaned during maintenance.

8. Connect exhaust fan to an available 110V grounded power source.

   Note: *If exhaust outlet ducting is routed outside, install a metal screen on duct opening to prevent invasive animals from nesting in exhaust system.*

Test Run

Once assembly is complete, test run the machine to ensure it is properly connected to power and safety components are functioning correctly.

If you find an unusual problem during the test run, immediately stop the machine, disconnect it from power, and fix the problem BEFORE operating the machine again. The **Troubleshooting** table in the **SERVICE** section of this manual can help.

> **⚠WARNING**
>
> **Serious injury or death can result from using this machine BEFORE understanding its controls and related safety information. DO NOT operate, or allow others to operate, machine until the information is understood.**

> **⚠WARNING**
>
> **DO NOT start machine until all preceding setup instructions have been performed. Operating an improperly set up machine may result in malfunction or unexpected results that can lead to serious injury, death, or machine/property damage.**

The Test Run consists of verifying the following: 1) Auxiliary systems power up and run properly, 2) stepper motors run correctly and machine properly homes, 3) limit switches function correctly, 4) top loading door interlock switch operates properly, and 5) Emergency Stop button functions properly.

To test run machine:

1. Clear all setup tools away from machine.

2. Turn auxiliary systems ***ON*** and verify correct operation according to **SECTION 3: SETUP** on **Page 21**.

3. Press Emergency Stop button in.

4. Connect machine to power by inserting power cord plug into a matching receptacle.

5. Twist Emergency Stop button clockwise until it springs out (see **Figure 26**). This resets the switch and turns machine ***ON***.

Figure 26. Resetting Emergency Stop button.

6. Verify machine starts up and runs smoothly without any unusual problems or noises.

   — Screen display will show system status information (see **Figure 27**), and after one audible beep, laser head assembly will home to upper right corner of table before moving to origin (or position of last cut).

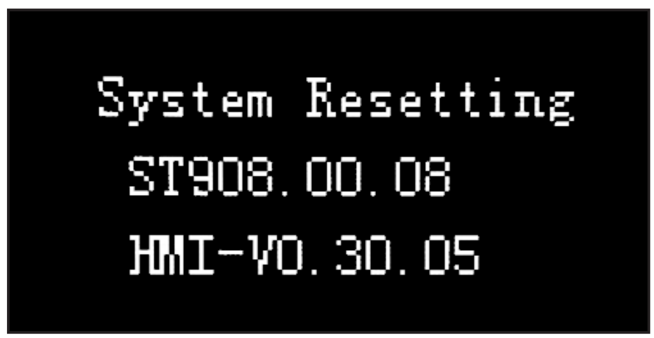


Figure 27. Status screen display during start-up.

7. Open top loading door and locate x- and y-axis limit switches, and top loading door interlock switch (see **Figure 28**).

Figure 28. Location of safety switch components.

8. Verify x-axis limit switch is functioning correctly by pushing in switch lever (see **Figure 29**) until an audible click is heard.

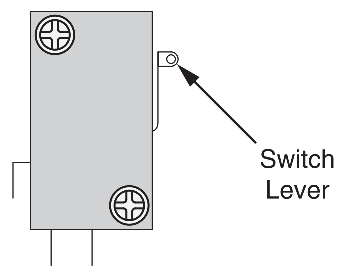


Figure 29. Example of typical limit switch lever.

9. With limit switch lever held in place, press arrow nav buttons on control panel to test movement of laser head assembly. The laser head assembly should *not* move.

   — If the laser head assembly *does not* move, the limit switch is functioning correctly.

   — If the laser head assembly *does* move, disconnect machine from power. The limit switch safety feature is NOT working properly and must be replaced before operating the machine.

10. Repeat **Steps 8–9** on y-axis limit switch to verify proper operation.

11. Put on Class 4 laser safety glasses (see **Figure 30**).

Figure 30. Example of typical Class 4 laser safety glasses.

DANGER

Operator and bystanders MUST wear ANSI-approved safety glasses rated for use with a CLASS 4 laser when machine is operating. NEVER look directly into laser beam, or stare at laser contact point for more than a few seconds, or severe eye injury or blindness will occur.

12. With top loading door still open, press Pulse button on control panel to attempt a test fire of the laser. The laser should *not* fire.

    — If the laser *does not* audibly beep and briefly pulse ***ON***, the top loading door interlock switch is functioning correctly.

    — If the laser *does* audibly beep and briefly pulse ***ON***, disconnect machine from power. The top loading door interlock switch safety feature is NOT working properly and must be replaced before operating the machine.

13. Press Emergency Stop button to turn machine ***OFF***.

    — If the machine *does* turn ***OFF***, Emergency Stop button is functioning correctly. Congratulations! Test Run is complete.

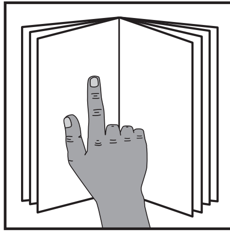
    — If the machine *does not* turn ***OFF***, immediately disconnect machine from power. The safety feature of the Emergency Stop button is NOT working properly and must be replaced before operating the machine.

SECTION 4: OPERATIONS

Operation Overview

The purpose of this overview is to provide the novice machine operator with a basic understanding of how the machine is used during operation, so the machine controls/components discussed later in this manual are easier to understand.

Due to the generic nature of this overview, it is **not** intended to be an instructional guide. To learn more about specific operations, read this entire manual, seek additional training from experienced machine operators, and do additional research outside of this manual by reading "how-to" books, trade magazines, or websites.



WARNING

To reduce your risk of serious injury, read this entire manual BEFORE using machine.

WARNING

DO NOT start machine until all preceding setup instructions have been performed. Operating an improperly set up machine may result in malfunction or unexpected results that can lead to serious injury, death, or machine/property damage.

NOTICE

If you are not experienced with this type of machine, WE STRONGLY RECOMMEND that you seek additional training outside of this manual. Read books/magazines or get formal training before beginning any projects. Regardless of the content in this section, Grizzly Industrial will not be held liable for accidents caused by lack of training.

To complete a typical operation, the operator performs the following:

1. Creates artwork using desired design software, prepares artwork for cutting/engraving using RDCam software, and exports .RD file for upload (**Page 27**).
2. Inspects auxiliary systems (water chiller, air pump, and exhaust fan) before every use.
3. Turns machine, worklight, and auxiliary systems ***ON***.
4. Uploads design to machine and queues file for cutting/engraving (**Page 31**).
5. Selects and installs appropriate table for operation.
6. Verifies workpiece is suitable for cutting/engraving.
7. Places workpiece between table and laser head assembly.
8. Sets laser focal length (**Page 36**).
9. Sets origin by pushing Origin button on control panel.
10. Verifies total operation time by performing Worktime preview (**Page 37**).
11. Verifies working envelope by performing TrackFrame operation (**Page 37**).
12. Puts on Class 4 laser safety glasses and wears them while operating machine.
13. Begins laser operations (**Page 33**).
14. Removes workpiece and scrap material from cabinet once operations are completed.
15. Turns machine and auxiliary systems ***OFF***.
16. Cleans and prepares machine for additional operations.

Preparing Artwork in RDCam Software

The Model G0872 uses RDCam software to format artwork into numerical code that the laser uses to cut or engrave a design. The following procedure will guide you through basic RDCAM controls, importing artwork into RDCam, and exporting it to an .RD format file for transferring to the machine.

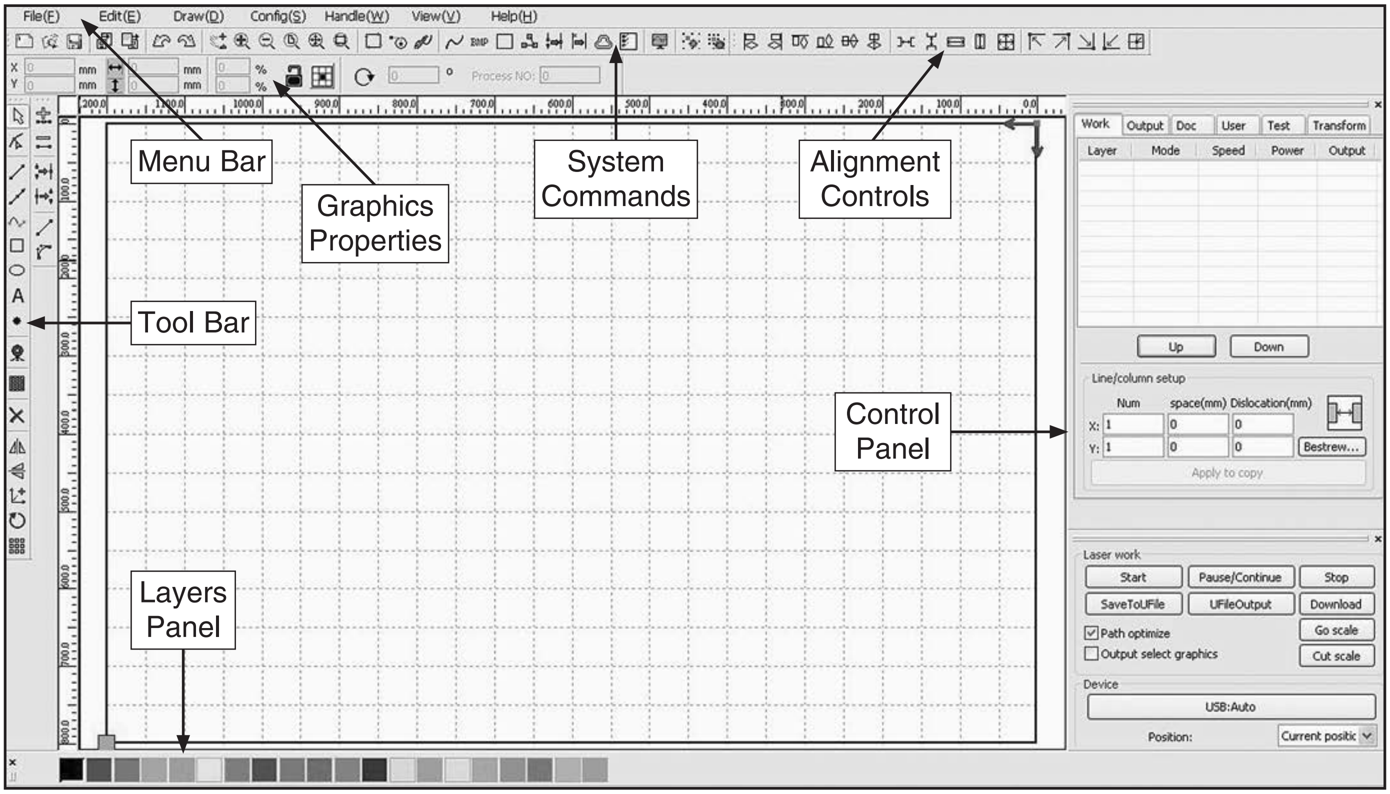


Figure 31. RDCam software main interface.

Menu Bar: Location of main software functions. Includes: File, Edit, Draw, Config, Handle, View, and Help.

Graphics Properties: Contains basic attributes of graphics operations, including location, size, scale, and reference points.

System Commands: Commonly used command buttons derived from Menu Bar.

Alignment Controls: Used for aligning objects and optimizing type settings.

Control Panel: Contains settings for toolpath layers, cut types, feeds, and cutting speeds.

Layers Panel: Assigns layer properties to an object based on color codes.

Tool Bar: Location of frequently-used tools for artwork design and editing.

Importing Artwork

For a complete guide to all of the capabilities of the RDCam software, refer to the **RDCam V8.0 User Manual** on the installation disc included with your machine.

Supported File Formats

- **Vector** ....dxf, ai, plt, dst, and dsb
- **Bitmap** .... bmp, jpg, gif, png, and mng

To import artwork into RDCam:

1. At the main interface, select "File" in the menu bar, and then select "Import" from the drop-down list.

2. Navigate to your desired file and select it. The file will now be highlighted (see **Figure 32**).

Note: *Select "Preview" to see a preview image of the selected file.*

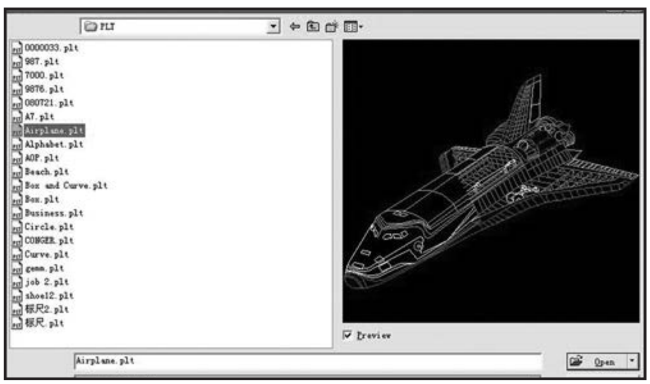


Figure 32. Selecting file to import.

3. Select "Open" to import the selected file.

Note: *Vector files will automatically import as separate layers. DST/DSB will be imported onto the currently selected layer.*

Saving Artwork as .RD File

The Model G0872 only accepts Ruby Document (.RD) files saved using the included RDCam software. Once the file is saved to a USB flash drive, it can be transferred to the machine as instructed in **Transferring Ruby Document (.RD) File** on **Page 31**.

Item Needed

USB Flash Drive 1

To save artwork as .RD file:

1. Select "Work" tab located in top section of control panel (see **Figure 33**).

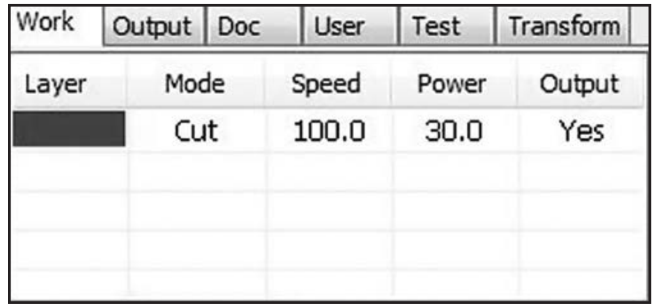


Figure 33. Control panel "Work" tab selected.

2. Double-click layer to open "Layer Parameter" dialog box (see **Figure 34**).

3. Set "Is Output" to "Yes" (see **Figure 34**).

Note: *Setting a layer's output to "No" will prevent that layer from being saved.*

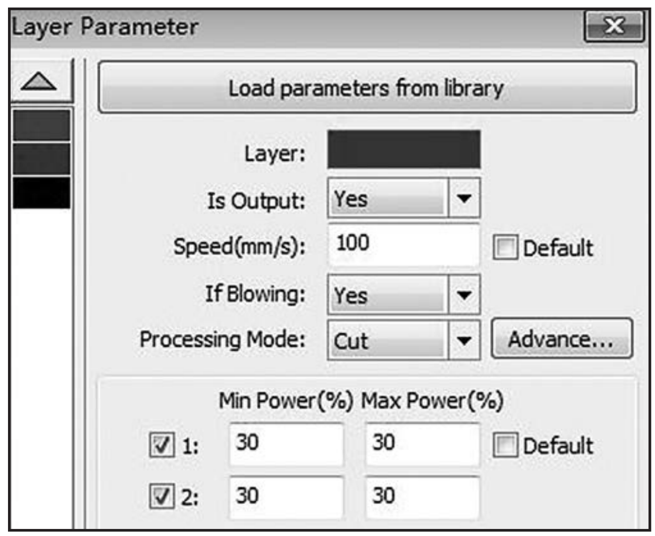


Figure 34. Layer Parameter dialog box.

4. Set desired cutting/engraving speed in mm/s (see **Figure 34**).

Note: *Slower speeds will create darker engravings and deeper cuts on the workpiece. This setting can be changed on the laser's control panel.*

5. Set "If Blowing" to "Yes."

6. Set "Processing Mode" to "Cut" or "Scan" as desired (see **Figure 35**).

   Note: *Chose "Cut" mode for outline burns through a workpiece, or "Scan" mode for fill layers used in engraving.*

7. Set "Min Power (%)" and "Max Power (%)" as desired (see **Figure 35**).

   Note: *Test workpiece with lower power settings (below 50%) and slowly increase percentage to determine best setting for workpiece material. This setting can be changed on the laser's control panel.*

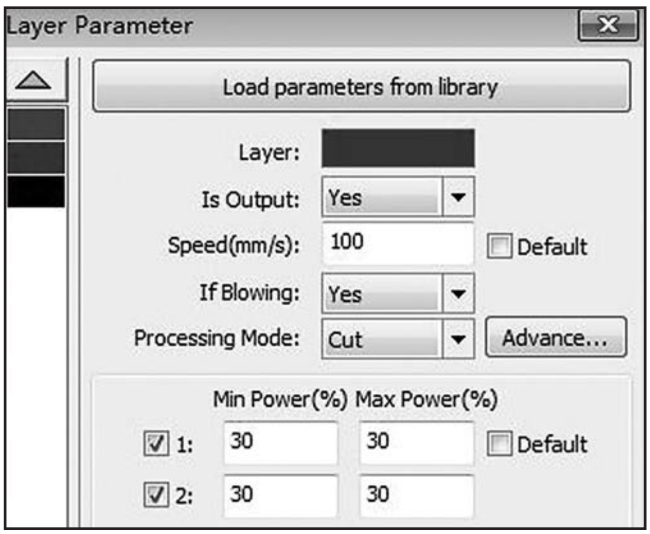


Figure 35. Layer Parameter dialog box.

8. Select "Ok" at bottom of dialog box to return to control panel.

9. Verify "Device" setting of output panel (located in bottom section of control panel) is set to "USB:Auto" (see **Figure 36**)

   — If "Device" is set to "USB:Auto," proceed to **Step 10**.

   — If "Device" is set to "Network," select "Network" button and change setting to "USB:Auto."

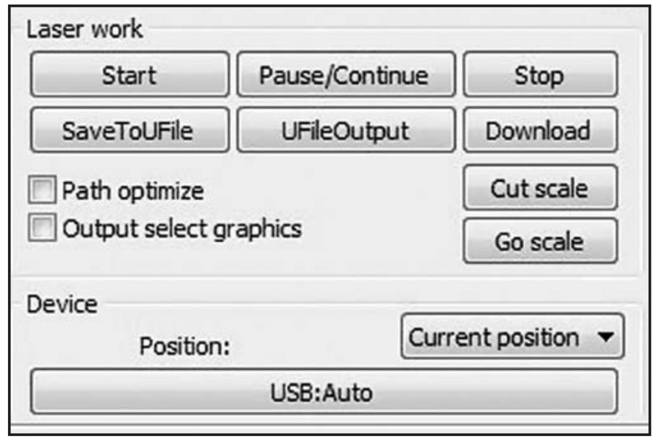


Figure 36. Output panel controls.

10. Insert USB flash drive into an open USB port on your personal computer.

11. Select "SaveToUFile" (see **Figure 36**), navigate to USB flash drive, and select "Save."

12. Eject USB flash drive from computer.

13. Proceed to **Transferring Ruby Document (.RD) File** on **Page 31**.

Water Chiller Overview

During operation, water is circulated through the laser tube cooling chambers. As the water flows through the laser tube, it absorbs heat along the way. The returned water dissipates heat through an air-cooled radiator before returning to the reservoir and being cycled through the laser tube again.

The water temperature must be kept below 122° F (50° C), or the water chiller will sound an audible alarm and suspend circulating water. If an overheat alarm occurs regularly, a larger water chiller system may be required, or a refrigeration-type water chiller may have to be installed in environments susceptible to extreme heat.

Air Pump Overview

Laser cutting operations require an air nozzle to focus a jet of air that blows gases, smoke, dust, and other materials away from the laser beam focal point. This jet of air also prevents combustion by continuously cooling the area and limiting fuel at the source of ignition.

During operation, air enters the rear of the pump through a barbed-fitting on the filter cover. The air passes through a washable-foam air filter, and exits out of a metal barbed fitting at the front of the pump. The air then travels through a hose to the AIR IN fitting on the auxiliary systems connection panel, before continuing on to the laser beam focal point.

Exhaust Fan Overview

The exhaust fan is a simple impeller-type fan that can draw a large volume of air away from an area.

During operation, the exhaust fan evacuates the laser machine and maintains a clean-air environment for unobstructed laser beam focus.

Smoke and particulates from laser cutting operations are drawn down through the lower cabinet and expelled through ducting to an area free of combustible material.

Inspecting Workpiece

Some materials are not safe for laser cutting, or may be outside the capabilities of this machine. **Before cutting/engraving, inspect all workpieces for the following:**

- ***Engraving Material:*** This machine is capable of engraving natural wood, glass, plastic, rubber, ceramic, leather, and cloth. This machine is NOT designed to engrave metal; engraving this material with this machine may lead to machine damage or personal injury.

- ***Cutting Material:*** This machine is capable of cutting natural wood, plastic, rubber, leather, and cloth. This machine is NOT designed to cut metal, glass, or ceramics; cutting these materials with this machine may lead to machine damage or personal injury.

- ***Foreign Objects:*** Nails, staples, dirt, rocks and other foreign objects are often embedded in wood. Always visually inspect your workpiece for these items. If they can't be removed, DO NOT use the workpiece.

Transferring Ruby Document (.RD) File

Before operations on the Model G0872 can begin, a Ruby Document (.RD) file must be created using the included RDCam design software, and then transferred from a flash drive to machine memory.

While navigating through machine operations, you will be using push buttons on the control panel shown in **Figure 37**.

Note: *For a complete list of CNC laser commands, a Command Tree diagram is located in **SECTION 10: APPENDIX** on **Page 73**.*

Figure 37. Control panel layout.

To transfer an .RD file from flash drive to machine memory:

1. Clear all setup tools away from machine.

2. Twist Emergency Stop button clockwise until it springs out to turn machine ***ON***.

   — After an audible beep, machine will power ***ON*** and move laser head from its current position to home (upper right corner of table), and then to origin.

   Note: *While performing startup, status screen will display information shown in* **Figure 38**.

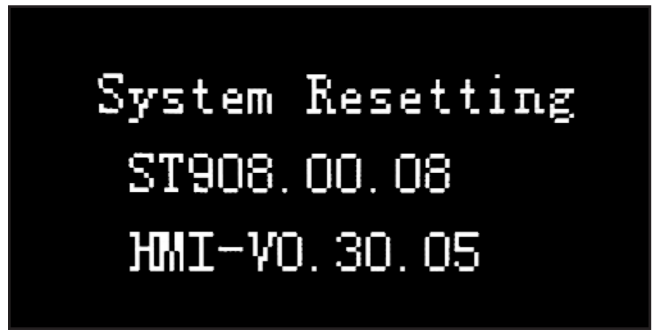


Figure 38. Status screen display during startup.

   — When machine startup is complete, status screen will display queued .RD file as a numeral (e.g. "File: 04"), and attributes assigned to current job (see **Figure 39**).

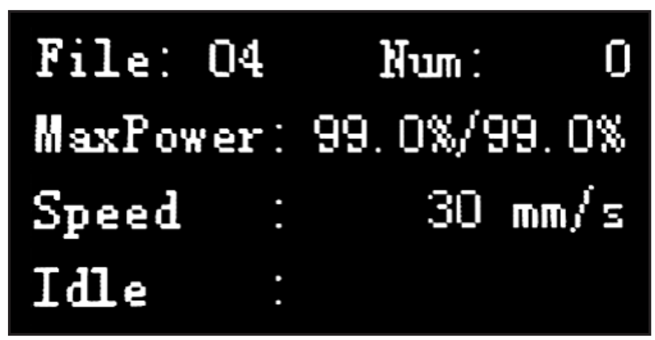


Figure 39. Example of displayed status screen.

3. Turn worklight ***ON***.

4. Insert USB flash drive into USB receptacle.

5. Press Menu button. Main menu screen will open (see **Figure 40**).

Figure 40. Main menu screen.

6. Select "File manage+" and press Enter. File management screen will open, as shown in **Figure 41**.

Figure 41. File management screen.

7. Use arrow nav buttons to select "Udisk Files" then press Enter (see **Figure 42**).

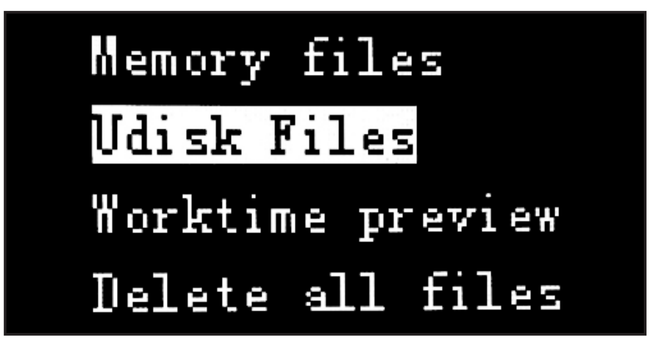


Figure 42. "Udisk Files" selected.

8. File list screen will open, displaying .RD files saved on flash drive (see **Figure 43**).

Figure 43. Example of file list screen.

9. Use navigation arrows to select desired file, then press Enter. Copy/delete screen will open (see **Figure 44**).

Figure 44. File copy/delete screen.

10. Select "To Memory" and press Enter. File will begin copying to machine memory.

Note: *When copying or deleting, task status is displayed on the screen, as shown in* **Figure 45**.

Figure 45. Copy progress displayed.

11. Once file is copied, screen will display "Completed" task status.

— "Name Over Lap" is displayed when a file with the same name is detected in machine memory (see **Figure 46**).

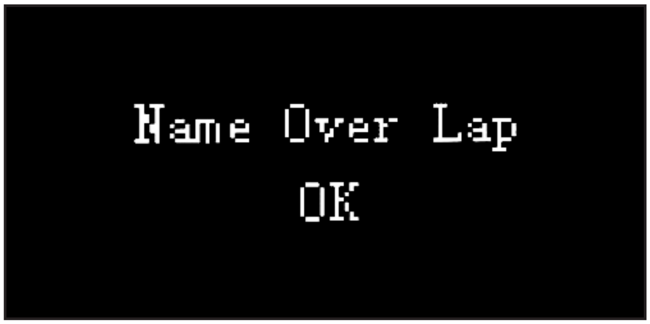


Figure 46. Completed task status.

12. Press Enter to return to file copy/delete screen (see **Figure 47**).

Figure 47. File copy/delete screen.

13. Press Esc four times to return to status screen (see **Figure 48**).

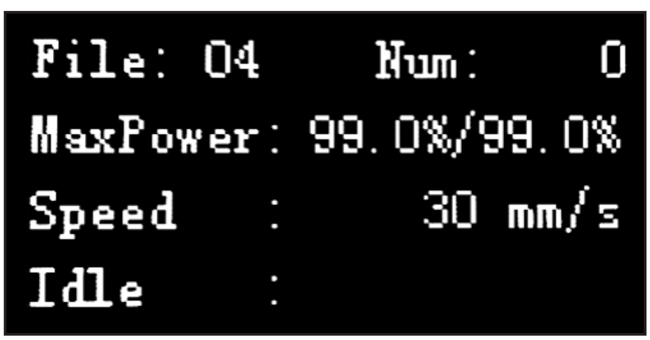


Figure 48. Example of displayed status screen.

File is successfully transferred and ready for laser operations.

Operating Laser

DANGER

Operator and bystanders MUST wear ANSI-approved safety glasses rated for use with a CLASS 4 laser when machine is operating. NEVER look directly into laser beam, or stare at laser contact point for more than a few seconds, or severe eye injury or blindness will occur.

WARNING

NEVER leave machine unattended; materials could catch fire during operation. Fires MUST be extinguished immediately to prevent personal injury and machine damage.

WARNING

To prevent exposure to toxic fumes and particulates during operation, CNC lasers MUST be equipped with an air filtration system that utilizes MERV 15+ or HEPA filters.

Once Ruby Document (.RD) file is transferred to machine memory, the image file must be queued before operating the laser.

Note: *For a complete list of CNC laser commands, a Command Tree diagram is located in* ***SECTION 10: APPENDIX*** *on* ***Page 73****.*

To queue image file and operate laser:

1. Clear all setup tools away from machine.

2. Twist Emergency Stop button clockwise until it springs out to turn machine ***ON***.

— After an audible beep, machine will power ***ON*** and move laser head to origin.

3. Open top loading door.

NOTICE

Before loading a workpiece, measure and verify that sufficient space exists between laser head tip and workpiece. Inadvertent contact with laser head assembly during workpiece loading can damage machine.

4. Select applicable table for current job:

— For light materials such as wood, plastic, cloth, leather, and thin veneers that need close support, use a honeycomb table.

— For heavy, rigid self-supporting materials, and also materials hard enough to damage honeycomb, use a solid table.

— For irregularly-shaped workpieces incapable of being self-supported, build a custom table or design a jig to prevent workpiece movement during operation.

5. Measure thickness of workpiece and verify sufficient space exists between workpiece and laser head tip.

Note: *Excess space will be adjusted later when setting focal length.*

6. Lower table by attaching Z-Axis hand crank to adapter and rotating counterclockwise.

7. Set focal length as instructed in **Setting Focal Length** on **Page 36**.

8. Move laser head assembly to origin, and push Origin button on control panel.

9. Press Menu button. Main menu screen will open (see **Figure 49**).

Figure 49. Main menu screen.

10. Select "File manage+" and press Enter. File management screen will open, as shown in **Figure 50**.

Figure 50. File management screen.

11. Select "Memory files", then press Enter. File list screen displays files saved in machine memory (see **Figure 51**).

Figure 51. Example of file list screen.

12. Select file, then press Enter. Job setup screen will open (see **Figure 52**).

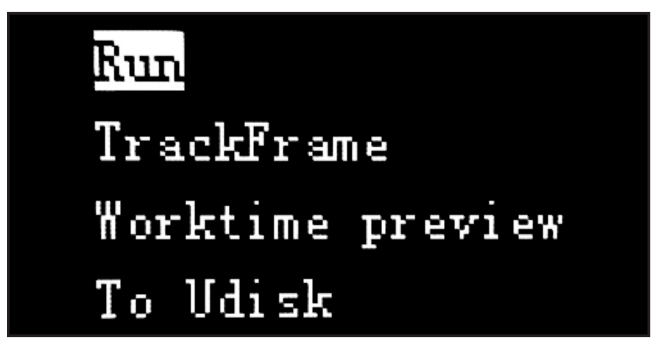


Figure 52. Job setup screen.

13. Perform "Worktime preview" as instructed in **Performing Worktime Preview** on **Page 37**.

14. Perform "TrackFrame" as instructed in **Performing TrackFrame** on **Page 37**.

15. Press Menu button. Main menu screen will open (see **Figure 53**).

Figure 53. Main menu screen.

16. Highlight "Speed set+" and press Enter.

17. Set desired laser movement speed in mm/s using arrow nav buttons, then press Enter to return to main menu.

    Note: *Slower speeds will create darker engravings and deeper cuts on the workpiece.*

18. Highlight "Max Power+" and press Enter, then highlight "MaxPower 1" and press Enter.

    Note: *When adjusting power settings for the Model G0872, use "MaxPower 1" and "MinPower 1". The secondary "MaxPower 2" and "MinPower 2" settings are unused.*

19. Set desired maximum power using arrow nav buttons and press Enter.

    Note: *Test workpiece with lower power settings (below 50%) and slowly increase percentage to determine best setting for workpiece material.*

20. Highlight "Min Power+" and press Enter, then highlight "MinPower 1" and press Enter.

21. Set desired minimum power using arrow nav buttons and press Enter.

22. Press Esc until status screen displays currently queued image file in upper left corner (e.g. "File: 04"), as shown in **Figure 54**.

    Note: *The status screen also displays additional cutting parameters such as* ***MaxPower****, cutting* ***Speed****, number (****Num****) of units cut, and machine status.*

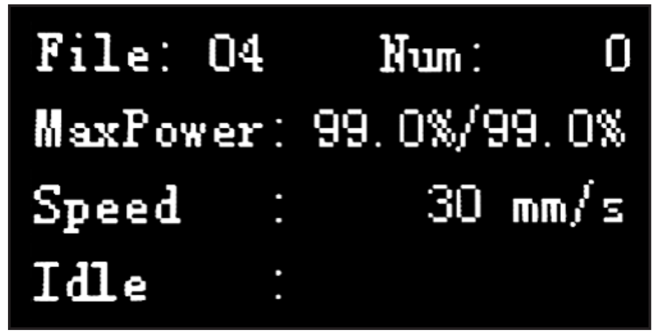


Figure 54. Status screen job attributes.

23. Turn ***ON*** all auxiliary systems and verify operation.

24. Put on Class 4 laser safety glasses.

25. Close top loading door and press Start/Pause button to begin laser operations.

    Note: *If a problem arises, or the job must be stopped for a period of time, press the Start/Pause button to pause or resume operations. In case of a power interruption, or Emergency Stop button is pressed, the machine will resume operations exactly where the event occurred once power is restored.*

26. During operation, verify ammeter indicator does not exceed the mA rating listed above ammeter (see **Figure 55**).

    IMPORTANT: An ammeter indication exceeding the mA rating may be caused by impending laser tube failure, or electrical faults.

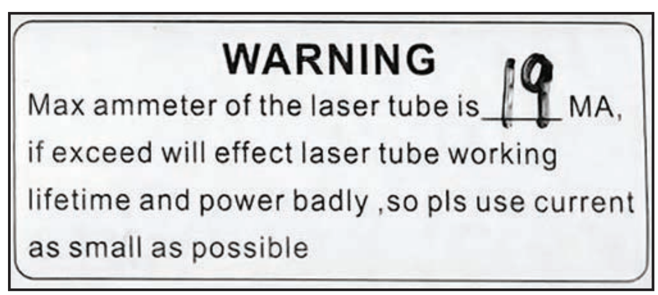


Figure 55. Example of ammeter mA rating.

27. When laser operations are completed, turn machine ***OFF***.

28. Turn ***OFF*** all auxiliary systems.

29. Remove workpiece and scrap parts, put all tools away, and clean area for next operation.

Setting Focal Length

⚠DANGER

Operator and bystanders MUST wear ANSI-approved safety glasses rated for use with a CLASS 4 laser when machine is operating. NEVER look directly into laser beam, or stare at laser contact point for more than a few seconds, or severe eye injury or blindness will occur.

Focal length is determined by the distance between the tip of the laser head and the workpiece (see **Figure 56**). It must be set before every cutting/engraving operation for optimum laser performance.

Note: *Excess distance will diffuse laser beam focus and cause poor cutting results. Insufficient distance will create dense clouds of smoke and particulates in the laser beam path, which will also diffuse focus.*

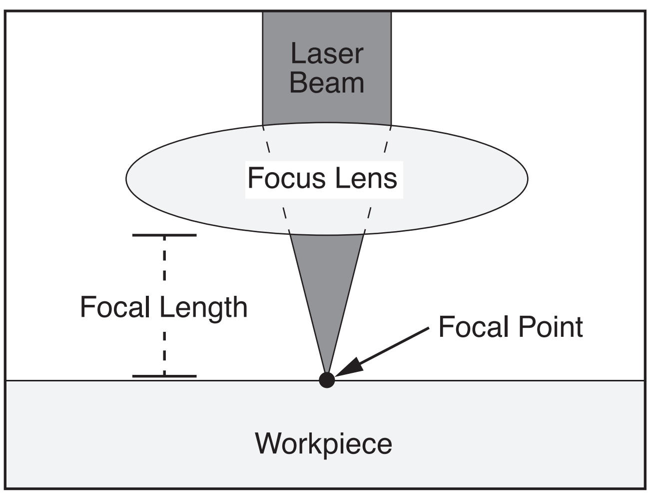


Figure 56. Focal point of laser on a workpiece.

Items Needed	**Qty**
Class 4 Laser Safety Glasses (per person)	1
Focus Gauge	1
Workpiece	As Needed

To set focal length:

1. Open top loading door and move laser head assembly to an easily accessible area.

2. Load a flat, even workpiece onto table under laser head assembly.

   Note: *For workpieces with varying height, it may be necessary to perform separate cutting/engraving operations for each tier to maintain consistent focal length.*

3. Use Z-Axis crank handle to raise or lower table until distance between workpiece surface and underside of laser head tip is approximately 9mm.

4. Place bottom of focus gauge on workpiece, and use Z-Axis crank handle to raise or lower table until tab on focus gauge rests on top of step on focus barrel, as shown in **Figure 57**.

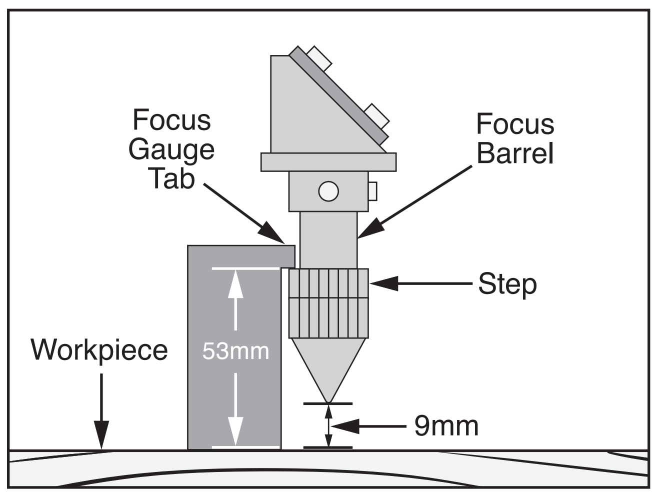


Figure 57. Focus gauge resting on workpiece and upper step of laser head focus barrel.

5. Focal length is now correctly set.

6. Remove focus gauge and close top loading door.

Performing Worktime Preview

The "Worktime preview" function instructs the computer to review all toolpath settings of currently queued files and estimate total time for completing cutting/engraving operations.

Note: *This optional function is useful for calculating cost-per-unit or for estimating remaining life of machine consumables.*

To perform "Worktime preview" function:

1. Press Menu button, then select "File manage+" and press Enter.

2. Highlight "Worktime preview" in job setup screen (see **Figure 58**).

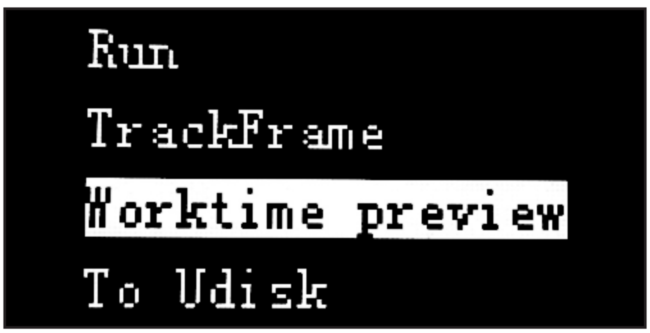


Figure 58. Job setup screen.

3. Press Enter button. Computer will begin making calculations (see **Figure 59**).

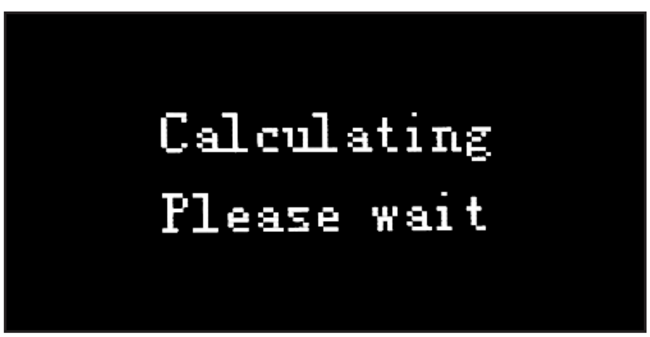


Figure 59. Calculation status screen.

4. When computer finishes calculating total cutting/engraving time, job duration is displayed on screen, as shown in **Figure 60**.

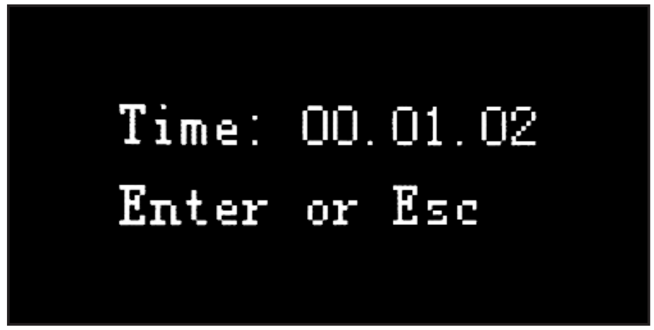


Figure 60. Job duration displayed on screen.

5. Press Enter or Esc button to return to job setup screen.

Performing TrackFrame Function

The "TrackFrame" function verifies the working envelope of the currently queued file to prevent exceeding available table area.

Note: *This optional function is useful for setting origin, and for reducing delays from travel errors during cutting/engraving operations.*

To perform "TrackFrame" function:

1. Press Menu button, then select "File manage+" and press Enter.

2. Highlight "TrackFrame" in job setup screen (see **Figure 61**).

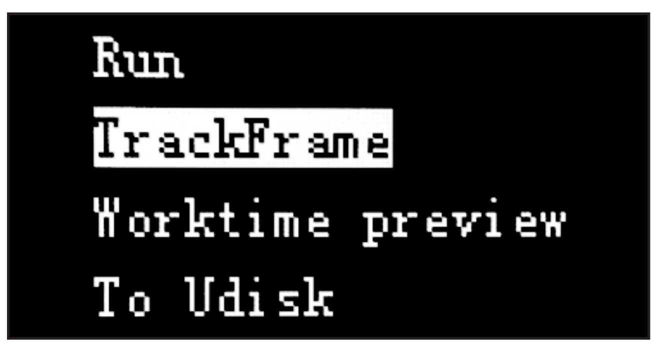


Figure 61. TrackFrame selected.

3. Review .RD file using RDCam, and position workpiece so queued image fits within working envelope.

4. Move laser head assembly to location of origin defined in RDCam, in reference to working envelope.

   Note: *For example, if operator set origin to center of image in RDCam software, position laser head assembly over center of table.*

5. Press Origin button. Machine will beep to indicate origin has been set.

6. Press Enter and observe perimeter of toolpaths being traced by laser head movement. Verify display shows "Tracking Frame" during operation (see **Figure 62**).

   — If display returns to job setup screen, "TrackFrame" function has completed successfully.

   — If "XSlop Over," "YSlop Over," or "Frame slop" errors are displayed on screen, repeat **Steps 1–6**.

   Note: *Once "TrackFrame" operation is completed, you may physically reposition workpiece in reference to position of laser head. This helps ensure laser head has available room for maneuvering over workpiece.*

Figure 62. "Tracking Frame" displayed.

IMPORTANT: If "XSlop Over," "YSlop Over," or "Frame slop" errors are displayed on screen, the machine has determined that the currently queued image is *outside* the working envelope, based on current origin (see **Figures 63–64**).

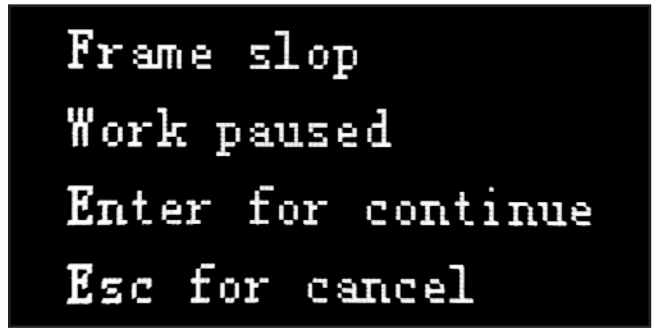


Figure 63. "Frame slop" error displayed.

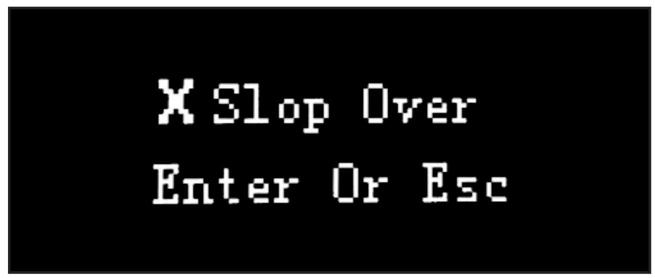


Figure 64. "XSlop Over" error displayed.

SECTION 5: ACCESSORIES

⚠WARNING

Installing unapproved accessories may cause machine to malfunction, resulting in serious personal injury or machine damage. To reduce this risk, only install accessories recommended for this machine by Grizzly.

NOTICE

Refer to our website or latest catalog for additional recommended accessories.

T30342—60W Laser Tube for G0872
Replacement 60W laser tube for use with the Model G0872 CNC Laser Cutter/Engraver.

Figure 65. Model T30342 60W Laser Tube.

T30334—17" x 23" Honeycomb Table
Replacement honeycomb table for use with the Model G0872 CNC Laser Cutter/Engraver.

Figure 66. Model T30334 Honeycomb Table.

T30337—Water Chiller System for G0872
Replacement water chiller system for use with the Model G0872 CNC Laser Cutter/Engraver.

Figure 67. Model T30337 Water Chiller System.

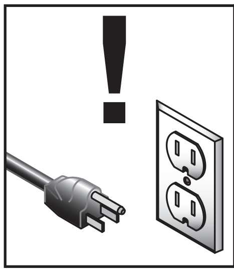
T30336—Extraction Fan for CNC Lasers
Replacement exhaust fan for use with the Model G0872 CNC Laser Cutter/Engraver.

Figure 68. Model T30336 Extraction Fan.

T30335—Air Pump for CNC Lasers
Replacement air pump for use with the Model G0872 CNC Laser Cutter/Engraver.

Figure 69. Model T30335 Air Pumo.

SECTION 6: MAINTENANCE



⚠WARNING

To reduce risk of shock or accidental startup, always disconnect machine from power before adjustments, maintenance, or service.

Schedule

For optimum performance from this machine, this maintenance schedule must be strictly followed.

Ongoing
To minimize your risk of injury and maintain proper machine operation, shut down the machine immediately if you ever observe any of the items below, and fix the problem before continuing operations:

- Loose mounting bolts.
- Damaged laser optics.
- Worn or damaged wires.
- Any other unsafe condition.

Daily Maintenance
- Wipe down interior cabinet area before and after use.
- Clean and vacuum dust buildup on table, inside cabinet, and on shafts and rails.
- Verify laser tube connections are secure and no air bubbles are present in tube.
- Check auxiliary systems for proper function and fill water reservoir, as required.

Weekly Maintenance
- Clean and vacuum dust buildup on auxiliary systems and air pump filter.
- Wipe shaft and rail metal surfaces with light sewing machine oil.

Monthly Check
- Verify fasteners on moving parts are secure.
- Inspect laser optics and clean as required.
- Inspect condition of electrical system.
- Verify limit switches are operational.
- Check synchronous belts for wear.

Cleaning & Protecting

The Model G0872 only requires a general cleaning before and after each use. After one month of use, perform a comprehensive inspection and verify all fasteners on all moving parts have not come loose.

Over the life of the machine, longevity is reduced by moisture, abrasive material, corrosion, and dust and gases generated by laser operations.

NOTICE

DO NOT clean belts with acids, alkalis, oils, or solvents. Contact with chemicals and oils will accelerate belt deterioration.

NOTICE

Avoid harsh solvents like acetone or brake parts cleaner that may damage painted surfaces. Always test on a small, inconspicuous location first.

Lubrication

Sewing machine oil (ISO 10 or less) is recommended by the factory for its light viscosity and odorless application.

General machine oil or mineral oil is NOT recommended for use on this machine due to the high volume of particulates produced by laser cutting and engraving.

The following sections summarize the cleaning and lubrication required for each area.

Items Needed	Qty
Soft Paintbrush	1
Stiff-Bristled Brush	1
Window Cleaner	As Needed
Mineral Spirits	As Needed
Sewing Machine Oil	As Needed
Clean Shop Rags	As Needed

Cabinet Cleaning Best Practices

Vacuum accumulated dust, ash, and scrap material. DO NOT use compressed air as it contains atomized oil and water, and will obscure machine optics over time.

Note: *A soft paintbrush paired with a vacuum nozzle works well for cabinet cleaning.*

The cabinet interior should be wiped down before and after every use with a quality window cleaner. DO NOT use solvents or soap and water. Solvent will damage paint, Plexiglas, and labels. Water can short electronics and corrode parts.

⚠WARNING

If materials that produce toxic dust or ash are cut, verify that a MERV 15+ or HEPA filter is installed in your vacuum. Hazardous particles can quickly spread through enclosed areas when expelled from a vacuum.

⚠WARNING

Under certain conditions, residual ash particles and wood chips may continue to smolder long after igniting. To help prevent vacuum container from smoldering and possibly catching fire, incorporate these safety steps in your cleanup schedule:

- **After machine use, wait for a period of time to allow any smoldering chips to naturally expire.**
- **Immediately empty and dispose of the collected waste in a steel container.**
- **Use a vacuum with a steel drum, and avoid using vacuums with paper bags or existing combustible material.**

Shaft and Rail Metal Surfaces

By design, the shafts, rails, and other metal parts on this machine have hardened surfaces that are highly resistant to corrosion and wear. However, periodically wipe metal parts with light sewing machine oil to extend their life.

Note: *Lubrication can cause sludge build-up that will bind moving parts, and corrosion can still occur if catalysts are trapped beneath lubricant. Always clean surfaces before applying any form of lubrication.*

Moving Parts

Use a flashlight to inspect belts, pulleys, guide rails, drive couplings, tracks, and slides before and after every use. Look for evidence of loose or missing parts, ensure mechanical connections and fasteners are tightened, and verify that the X, Y, and Z axes operational paths are clean and unobstructed.

Tip: *If you must vacuum immediately after laser operations, and if using a wet/dry shop vacuum, the collection drum can be filled with 1" of water at the bottom to extinguish any smoldering particles.*

Tip: *Using a flashlight, even in well-lit areas, greatly improves the quality of inspection. In busy and visually-distracting areas, the light beam serves as a focal point and highlights the area being inspected.*

X & Y Crossbeam Linear Shafts

IMPORTANT: When performing lubrication procedures, make sure to protect laser optics and belts from any cleaning fluid or lubricating oil.

To clean and lubricate X & Y crossbeam linear shafts:

1. DISCONNECT MACHINE FROM POWER!
2. Place sheet of cardboard or cloth under areas to be cleaned and lubricated.

3. Slowly push laser head assembly to left-most X-Axis position.

   Note: *If you move the laser head, or any component driven by a stepper motor too quickly, it will generate electricity and promptly stop movement. Once the charge dissipates, try moving the component again at a slower pace than before.*

4. Using a clean, cotton cloth lightly covered with sewing machine oil, wipe crossbeam linear shaft until clean (see **Figure 70**).

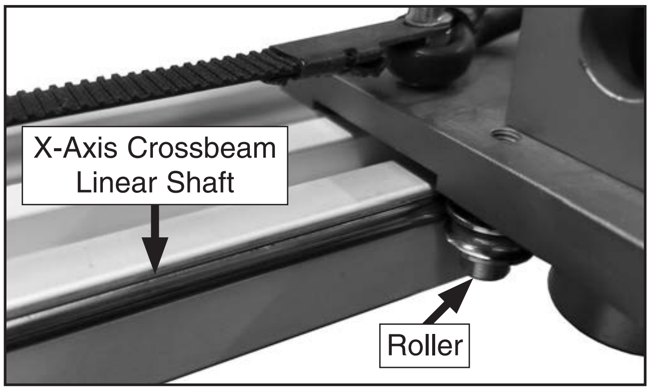


Figure 70. X-Axis crossbeam linear shaft and roller.

5. Slowly push laser head assembly to right-most X-Axis position.

6. Using a clean, cotton cloth lightly covered with sewing machine oil, wipe crossbeam linear shaft until clean.

7. Dip rag in oil and apply a thin coating across entire surface of crossbeam linear shaft.

8. Repeat **Steps 3–7** on left and right Y-Axis crossbeam linear shafts (see **Figure 71**).

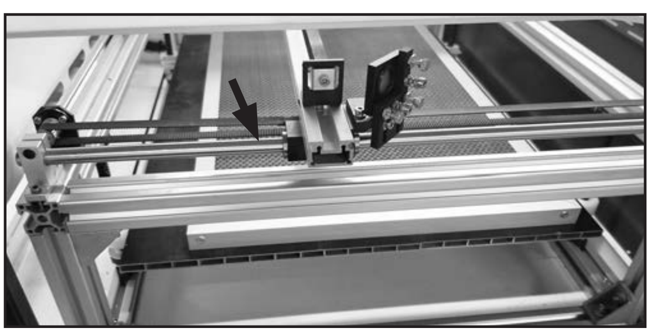


Figure 71. Left Y-Axis crossbeam linear shaft.

9. Remove any cardboard or cloth from area when finished.

   Tip: *Should the laser head assembly develop play on the X-Axis linear shaft, the roller cap screws may be loosened and the eccentrics rotated until play is removed (see* ***Figure 72****).*

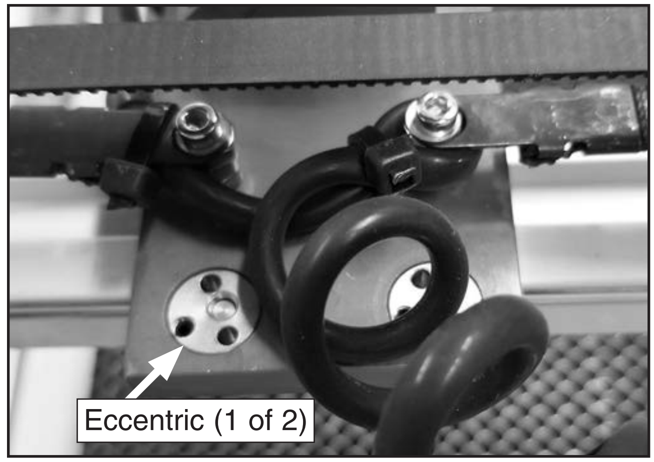


Figure 72. X-Axis roller eccentrics.

Z-Axis Table Lift

The Model G0872 table is raised manually with a hand crank. At each corner of the table, a leadscrew nut threaded with a leadscrew raises and lowers the table depending on which direction the hand crank is rotated.

All four leadscrews are timed with one another by a synchronous belt and pulley (see **Figure 73**).

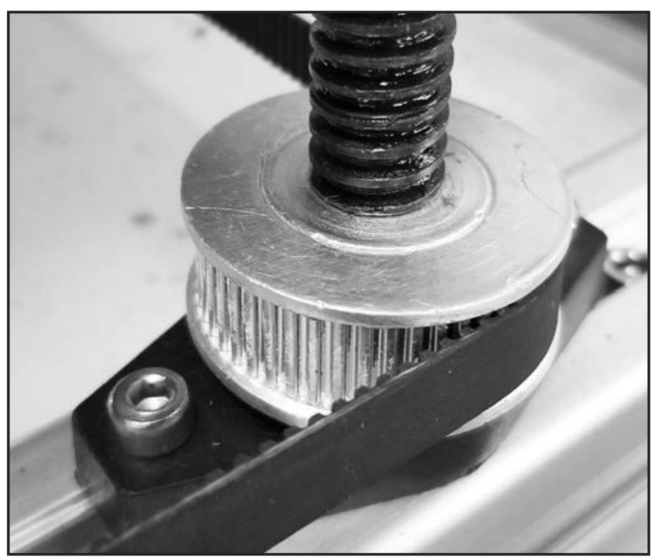


Figure 73. Z-Axis leadscrew belt and pulley.

Z-Axis Table Lift Components

To clean and lubricate Z-Axis table lift components:

1. DISCONNECT MACHINE FROM POWER!

2. Access lower cabinet by opening any lower panel or door.

3. Use a stiff brush to loosen existing build-up, and vacuum all contaminants from leadscrews, nuts, pulleys, and belts.

4. Use cardboard or cloth rags to protect leadscrew belt (see **Figure 74**) from cleaning fluids and lubricating oil.

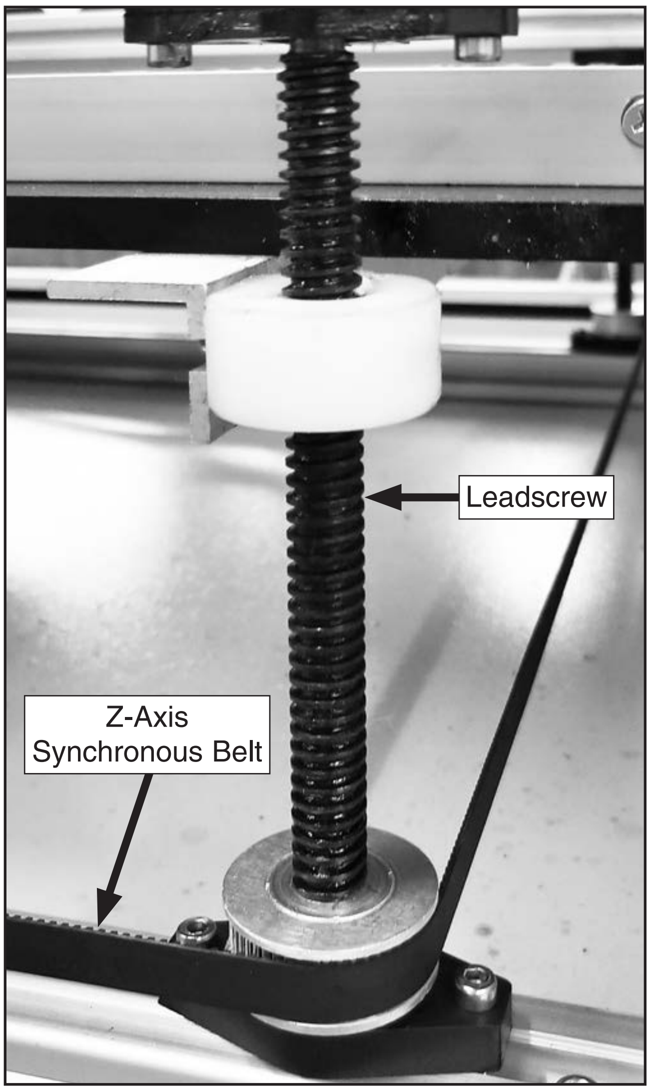


Figure 74. Z-Axis leadscrew location (1 of 4).

— If the leadscrews have heavy contamination, use a stiff-bristled brush and a rag with mineral spirits to clean threads.

5. Allow leadscrews to dry after cleaning, and apply a light coating of sewing machine oil.

6. Move table up and down the full range of movement to distribute oil over leadscrews.

Belts & Pulleys

The rotational direction of pulleys is controlled by the X and Y stepper motors, which are given directional coordinates by the CNC software.

The pulleys are toothed and engage with cogged synchronous belts (see **Figure 75**) to prevent slipping when the laser head moves along the linear guide rails. The bearings for these pulleys are sealed and maintenance-free.

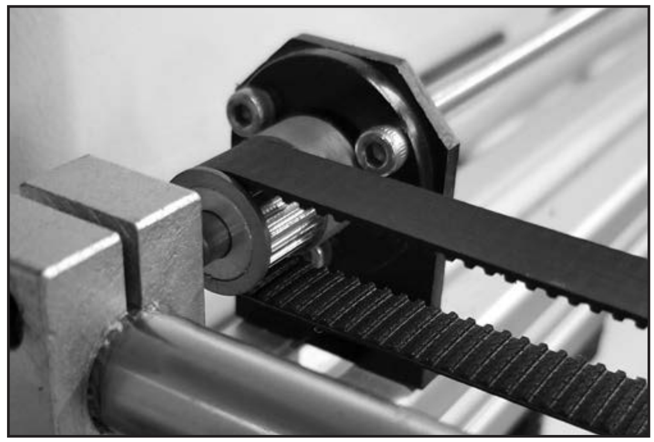


Figure 75. Synchronous belt and pulley.

NOTICE

DO NOT clean belts with acids, alkalis, oils, or solvents. Contact with chemicals and oils will accelerate belt deterioration.

To inspect and clean pulley belts and cogs:

1. DISCONNECT MACHINE FROM POWER!

2. Slowly move laser head along X- and Y-axes while inspecting condition of belts and pulley cogs with a flashlight.

3. Check for belt deflection as instructed in **Adjusting/Replacing Synchronous Belts** on **Page 51**.

4. Verify that left and right belts (where used) have same deflection, or binding may occur.

5. Remove any contaminants in crevices of cogs or teeth by using a firm-bristled toothbrush while supporting underside of belt with your finger to prevent stretching.

   Note: *Material build-up on cogs or teeth can cause a bump or notch in a cutting path.*

6. If any belts have cracks or damaged teeth, replace as instructed in **Adjusting/Replacing Synchronous Belts** on **Page 51**.

Maintaining Laser Components

The laser tube on the Model G0872 has no maintenance requirements other than protecting it from freezing when filled with water, and periodic inspection to verify water connections are sealed, air bubbles are not present in the tube, and electrical contacts are clean and free from corrosion. All laser tubes are tested before shipping.

Laser tube, lenses, and mirrors are consumable parts, and are covered under warranty only for defects that occur during shipping. Periodic replacement is required and often done as a complete set.

Life expectancy of the laser tube is affected by laser burn-time, laser strength setting, heating and cooling cycles, and the natural dissipation of gas from the laser tube.

IMPORTANT: Grizzly Industrial *does not* recommend having a second laser tube on hand for longer than **three months**. Laser tubes are perishable and will lose their efficiency over time by just sitting on the shelf.

Tip: *To cover the operating cost of a replacement laser tube, divide the price of a replacement laser tube by the amount of units you expect to produce. This calculation does not take into account length of laser run time, or power settings required for different types of widgets, but it will provide a baseline value to add to each unit to cover laser tube operating costs.*

Laser Optics Introduction

CNC laser cutters and engravers are a new generation of highly sensitive machines available for the general consumer market, and require special attention to properly maintain.

Note: *When replacing the laser tube, it is highly recommended that lenses and mirrors are also replaced.*

The beam emitted from the laser tube is reflected three times before reaching the focal point on the workpiece. This alternating direction of laser travel is achieved by mirrors. Optimum performance of the machine is largely affected by the ability of these lenses and mirrors to focus and reflect the beam without diffusion.

Over time, smoke, fumes, and dust will coat the reflective surfaces and diffuse the beam, resulting in low-quality cuts.

If the lenses and mirrors are not kept clean, any ash or dust on the surface will absorb laser energy. This energy converts to heat, causing a "thermal lens effect," where the optics will become permanently pitted after normal use. As a result, laser machine optics are considered consumable parts.

Once the optics have begun degrading, the user will have to increase laser power over time to achieve the same cutting results. The longer the laser tube is operated at higher power settings, the sooner the tube will require replacing.

Treat laser optics with care and protect delicate surfaces from improper cleaning methods, and ambient contaminants. Even the smallest dust particle can cause microscopic scratching if the surface is wiped by hand. Microscopic scratches will increase the damage caused by "thermal lens effect."

Follow these best practice guidelines when cleaning laser optics:

- DO NOT use compressed air to clean optics. Compressed air contains microscopic oil and water particles.
- DO NOT use a vacuum nozzle with an attached brush. Even soft brushes can scratch laser lenses and mirrors.
- DO NOT blow on the optics. Moisture droplets can stain the surface.
- DO NOT touch the film layer of the lens when removing. Hold lenses and mirrors by the edges when transporting.
- Keep all optics in a clean, dry container protected with soft lens paper until ready.
- Prepare all cleaning surfaces with several layers of new lens cotton or paper. DO NOT use shop rags, newspapers, or paper towels, which might be contaminated with microscopic abrasives.
- Set up the cleaning area indoors, away from wind and airborne particulates.
- DO NOT eat or drink while cleaning optics.

Cleaning Laser Optics

The most important step when working around optics is to thoroughly wash and dry hands. In the event you have to handle lenses or mirrors, wearing sterile, disposable gloves is recommended.

Items Needed	Qty
Small Bellows	1
Optics-Grade Cotton Swabs	As Needed
Lens Paper	As Needed
Denatured/Isopropyl Alcohol	As Needed
Clean Shop Rags	As Needed

Method 1 (Optics Installed)

When lenses and mirrors are installed, they can be cleaned using a cotton swab (see **Figure 76**).

> **IMPORTANT:** While cleaning optics, make sure to protect the pulley belts from all cleaning fluids. Belts will deteriorate from contact with chemicals and oils.

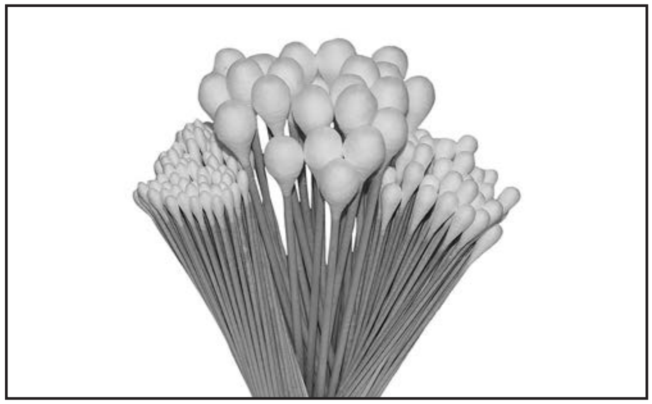


Figure 76. Typical optics-grade cotton swabs.

To clean installed laser optics:

1. DISCONNECT MACHINE FROM POWER!
2. Use a small bellows to blow any particulates off lens/mirror surface.

   IMPORTANT: DO NOT physically wipe lenses or mirrors.
3. Soak ends of several cotton swabs in denatured or isopropyl alcohol. Saturate swabs with enough solution to be moist, but when pressed against a surface, solution *does not* squeeze out and drip.
4. With a gentle rolling motion, roll swab across surface of lens/mirror to absorb any particulates stuck to surface (see **Figure 77**).

   Tip: *Dislodge and lift particulates off of surface by rolling cotton swab, which exposes clean cotton as you progress. Rubbing or dragging swab across surface without rolling it can create scratches and damage lens and mirrors.*

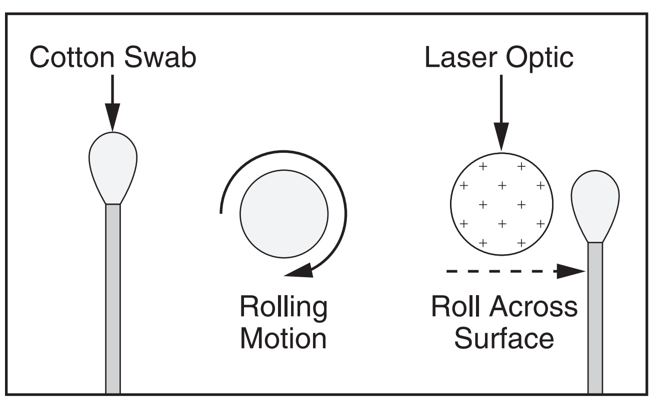


Figure 77. Rolling swab across surface.

5. Moisten dry swab with 1–3 drops of same solution, roll swab across surface of lens/mirror, and allow surface to air dry.

6. Inspect lens/mirror surface:

— If any particulates or surface stains remain, repeat **Steps 2–5**.

— If any particulates or surface stains are still present after second cleaning, they are most likely permanently burned into surface. Replace optics as instructed in **Laser Optics Removal/Installation** on **Page 61**.

Method 2 (Optics Removed)
When lenses and mirrors are removed, they can be cleaned using the "Drop and Drag" method used for cleaning camera and microscope lenses.

To clean removed laser optics:

1. DISCONNECT MACHINE FROM POWER!

2. Use a small bellows to blow any particulates off lens/mirror surface.

   IMPORTANT: DO NOT physically wipe lenses or mirrors.

3. Place small drop of denatured/isopropyl alcohol on center of lens cleaning paper.

   Note: *Flooding paper with solvent only leaves residual streaks. Only place enough solvent so that paper conforms to surface, and solvent does not absorb to edge of paper.*

4. Grab edge of paper and slowly pull across lens/mirror surface (see **Figure 78**). Allow surface to air dry.

   Note: *Dry edge of paper removes any residual solvent and streaking. This method does not scratch optics since no downward force is being applied to paper.*

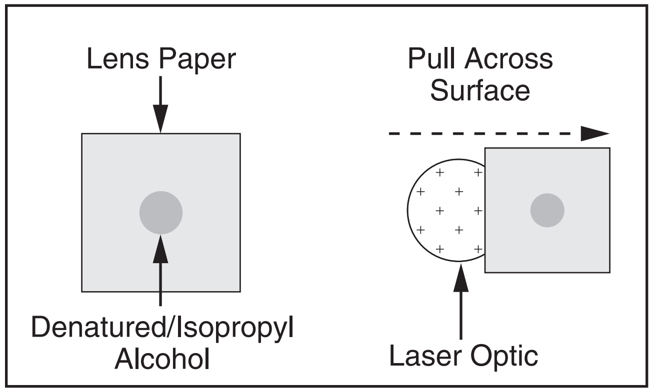


Figure 78. Pulling lens paper across surface.

5. Inspect lens/mirror surface:

— If any particulates or surface stains remain, repeat **Steps 2–4** using clean sheet of lens cleaning paper for every pass.

— If any particulates or surface stains are still present after second cleaning, they are most likely permanently burned into surface. Replace optics as instructed in **Laser Optics Removal/Installation** on **Page 61**.

Water Chiller System

Water quality and effective cooling directly contribute to the operational life of the laser tube. The cooling system requires distilled water to prevent scaling and contaminant build-up. Due to oxygenation and warm temperatures inherent in the system, water needs to be checked regularly for evidence of algae or unpleasant odors. Mildew growth can reduce cooling efficiency and decrease tube life.

Operational load determines actual maintenance intervals. General maintenance of the water chiller system is detailed in the following section.

To inspect and maintain water chiller system:

1. DISCONNECT WATER CHILLER SYSTEM FROM POWER!

2. Inspect radiator fan intake screen for blockage, and clean as required.

3. Inspect hoses and connections for leaks or damage, and repair or replace as required.

4. Inspect water for discoloration and evidence of algae. If contaminated, drain water, clean reservoir, and refill with distilled water.

5. Reconnect water chiller system to power and turn ***ON***. Allow water to cycle for 5 minutes and verify air bubbles have released from laser tube.

6. Remove reservoir cap and verify water level is 3–6" from reservoir opening.

   Note: *For additional information on alarm codes and troubleshooting guidelines, see user's manual included with water chiller.*

Cleaning Air Pump Filter

The air pump requires no internal maintenance, but uses a foam filter to maintain clean air delivery to the laser focal point.

Operating time and cleanliness of the air drawn into the pump determines actual maintenance intervals.

To clean air pump filter:

1. DISCONNECT AIR PUMP FROM POWER!

2. At intake-end of air pump, loosen Phillips head screw and remove cover plate.

3. Remove foam filter (see **Figure 79**), and soak in a solution of warm water and dish soap.

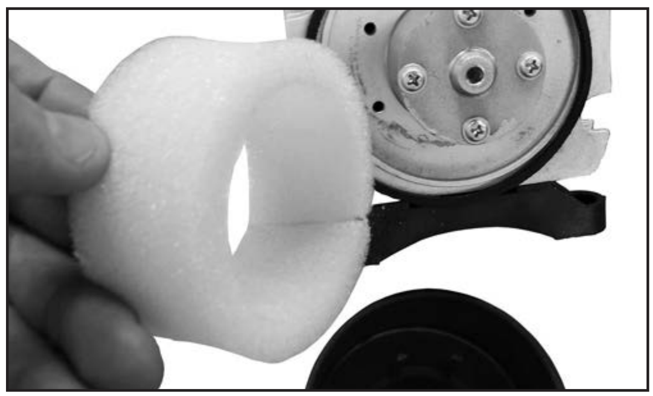


Figure 79. Typical air pump foam filter.

4. Wrap clean shop rag around foam filter and dry with a few firm squeezes.

   Note: *DO NOT use compressed air to dry the filter, or damage may occur.*

5. Inspect foam filter for holes, rips, or tears. Replace as required.

6. Inspect tubing and filter cover for cracks and leakage. Replace as required.

7. Re-install filter and secure cover with screw removed in **Step 2**.

Exhaust Fan

The exhaust fan requires no internal maintenance, and only regular visual inspections and light cleaning to remain operational.

Operating time and cleanliness of the air drawn through the fan determines actual maintenance intervals.

To inspect and clean exhaust system:

1. DISCONNECT LASER MACHINE AND EXHAUST FAN SYSTEM FROM POWER!

2. Inspect ducting for evidence of leaks. Patch or replace ducts as required.

   Note: *If screens have been attached to the inlet/outlet ducts to prevent pest infestation, inspect and clean them as required.*

3. Verify duct clamps and power cord are secure and undamaged.

4. Verify exhaust fan is securely mounted in an accessible location where ducting can be removed and cleaned during maintenance.

5. Disconnect ducting and separate from exhaust fan far enough to gain access to input and output ports.

6. Vacuum chips, dust, and ash from exhaust fan intake screen.

7. Vacuum deposits at bends in ducting.

8. Re-attach ducting to exhaust fan and secure.

Storing Machine

For long-term machine storage, or when not in operation during winter months, it is MANDATORY that ALL water is drained from the laser tube. Freezing temperatures can be encountered even in heated buildings or storage facilities from power outages. Water left in the laser tube may freeze and break the internal glass cooling coils. Damage to the laser tube after shipping is NOT covered under warranty.

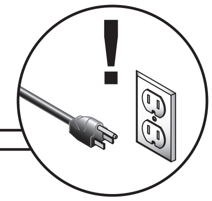
Grizzly Industrial recommends that the Model G0872 be stored in a sealed, wooden crate for long-term storage over a year. Place generous quantities of desiccant bags in the laser cabinet, electrical boxes, and in the crate before sealing.

Perform ALL maintenance procedures for cleaning and lubrication as outlined in **SECTION 6: MAINTENANCE** on **Page 40** before placing machine into storage.

SECTION 7: SERVICE

Review the troubleshooting procedures in this section if a problem develops with your machine. If you need replacement parts or additional help with a procedure, call our Technical Support. **Note:** *Please gather the serial number and manufacture date of your machine before calling.*

Troubleshooting



Electrical

Symptom	Possible Cause	Possible Solution
Machine does not start, or breaker immediately trips after startup of machine or auxiliary systems.	1. Emergency stop button depressed/at fault. 2. Incorrect power supply voltage or circuit size. 3. Power supply circuit breaker tripped or fuse blown. 4. Air pump, water chiller, or exhaust fan have a short. 5. Wiring broken, disconnected, or corroded. 6. Machine chassis ground at fault. 7. Laser tube at fault. 8. Power supply or controller at fault. 9. Computer board at fault.	1. Rotate emergency stop button head to reset. Replace if at fault. 2. Ensure correct power supply voltage and circuit size. 3. Ensure circuit is sized correctly and free of shorts. Reset circuit breaker or replace fuse. 4. Inspect/replace if at fault. 5. Fix broken wires or disconnected/corroded connections. 6. Machine must be connected to a dedicated ground rod at/or near machine (**Page 15**). 7. Inspect for evidence of arcing at laser tube connections. Verify wire insulation is preventing arcing and discharge to machine frame. 8. Inspect/replace if at fault (**Page 63**). 9. Inspect/replace if at fault (**Page 63**).
Machine has vibration or noisy operation.	1. Stepper motor or component loose. 2. Belt(s) worn, loose, pulleys misaligned or belt slapping cover. 3. Incorrectly mounted to workbench. 4. Workpiece loose. 5. Stepper motor bearings at fault.	1. Replace damaged or missing bolts/nuts or tighten if loose. 2. Inspect/replace belts with a new matched set. Realign pulleys if necessary. 3. Adjust feet, shim, or tighten mounting hardware. 4. Use correct holding fixture and reclamp workpiece. 5. Test by rotating shaft; rotational grinding/loose shaft requires bearing replacement.

Operations

Symptom	Possible Cause	Possible Solution
Flash drive is not read by laser machine, or file unable to load from flash drive.	1. Incorrect file structure or wrong file format. 2. Artwork file not saved using included RDCam software.	1. Use a formatted flash drive with artwork files saved in RDCam Ruby Document (.RD) format. 2. Save artwork file to flash drive using included RDCam software (**Page 28**).
Machine laser has poor cutting or engraving results.	1. Laser speed, path, or other CNC error exists. 2. Incorrect laser speed or laser power setting. 3. Focus is not set to correct height. 4. Laser is skewed or off-center. 5. Laser path is obstructed by smoke or material. 6. Laser unable to cut or scan workpiece. 7. Laser kerf too wide, and material is poorly cut. 8. Laser head has vibration or lash (workpiece path shows distortion/overlap). 9. Workpiece buckling or moving. 10. Laser output creating sawtooth pattern on cuts or engravings. 11. Laser tube at fault. 12. Transformer, power supply, or controller at fault.	1. Review RDCam path settings and verify that software is free of errors. 2. Review RDCam power settings and verify that speed/power is set as desired (**Page 28**). Adjust laser speed/power at machine (**Page 35**). 3. Inspect/adjust focal length (**Page 36**). Inspect/adjust table parallelism (**Page 52**). 4. Verify mirrors are secure, and reflective side is facing outward (**Page 61**). Align laser beam (**Page 57**). Clean laser optics (**Page 45**). 5. Verify air supply hose is unobstructed and connected to air nozzle. Inspect/replace air pump (**Page 22**). Verify exhaust ducting is unobstructed and functional (**Page 22**). 6. Workpiece material beyond machine capability. 7. Verify mirrors are secure, and reflective side is facing outward (**Page 61**). Align laser beam (**Page 57**). Inspect/adjust focal length (**Page 36**). Inspect/adjust table parallelism (**Page 52**). 8. Adjust/replace belts (**Page 51**). Inspect/adjust linkage, tracks, and guides for loose fasteners or binding. Reset origin (**Page 34**). 9. Use honeycomb table for thin workpiece support. Use clamps to secure large or irregular workpieces. 10. Increase laser power (**Page 35**). Workpiece may contain impurities that ignite, ejecting particles of molten material. 11. Replace laser tube (**Page 52**). Replace laser tube power supply. 12. Inspect/replace transformer, power supply, or controller, as required.
Laser tube inoperative or laser powers down while machine is operating.	1. Water chiller system not cooling laser tube; thermal kill switch activates or alarm sounds. 2. Air pump, water pump, or exhaust fan have a short. 3. One or more laser tube power supply components have failed. 4. Laser tube electrical connections at fault. 5. Electrical system at fault. 6. Laser tube at fault.	1. Inspect/replace water chiller system (**Page 21**). Reduce ambient temperature of machine operating environment. Add ice to reservoir, as required. Add additional water chilling equipment, as required. 2. Inspect/replace if at fault. 3. Replace laser tube power supply and machine electrical components, as required (**Page 63**). 4. Verify laser tube electrical connections are correct and secure. 5. Test/replace electrical system components, as required (**Page 63**). 6. Replace laser tube (**Page 52**).

Adjusting/Replacing Synchronous Belts

The Model G0872 uses toothed synchronous belts that engage with a cogged drive pulley.

After long-term use, one or more synchronous belts may have to be replaced. To maximize belt life, and maintain adequate cut quality, follow these recommended guidelines:

- Unlike V-belts, the synchronous belts *must not* be allowed to slip under any circumstances, or the workpiece will be ruined due to laser coordinates straying from the programmed path. Synchronous belt tension only needs to keep the belt teeth engaged with the pulley cogs under a light load.
- Excessive belt tension will wear out delicate pulley bearings and stretch the belt.
- Insufficient belt tension will cause noisy directional changes due to belt slip and reduce resolution quality of the cut.

The following sections show proper belt tension deflection values for each axis. See **Figure 80** for a typical method of testing belt deflection.

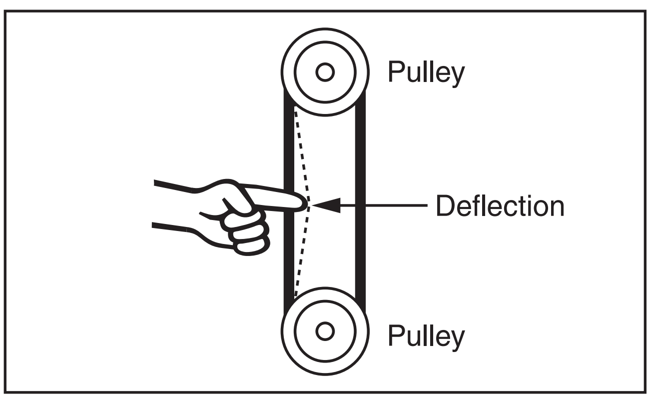


Figure 80. Testing belt deflection.

Tools Needed	Qty
Hex Wrenches 2.5, 3mm	1 Ea.
Phillips Screwdriver #2	1
Flat Head Screwdriver #2	1
Adjustable Wrench	1
Clean Shop Rags	As Needed

X-Axis Belt Adjustment

Test for 6mm of deflection on top side of belt at its center.

To *increase* belt tension, tighten adjustment screw at idler pulley bracket (see **Figure 81**).

To *decrease* belt tension, loosen adjustment screw at idler pulley bracket (see **Figure 81**).

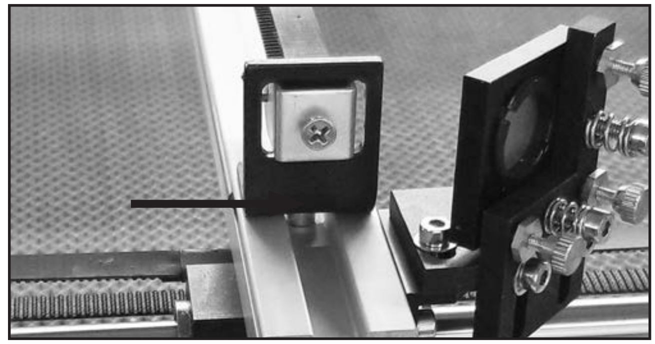


Figure 81. X-Axis belt tension adjustment screw location.

Y-Axis Belt Adjustment

Center laser head assembly over table, and test for 8mm deflection on bottom side of belt at its center between laser head and pulley.

To *increase* belt tension, tighten adjustment screw at idler pulley bracket (see **Figure 82**).

To *decrease* belt tension, loosen adjustment screw at idler pulley bracket (see **Figure 82**).

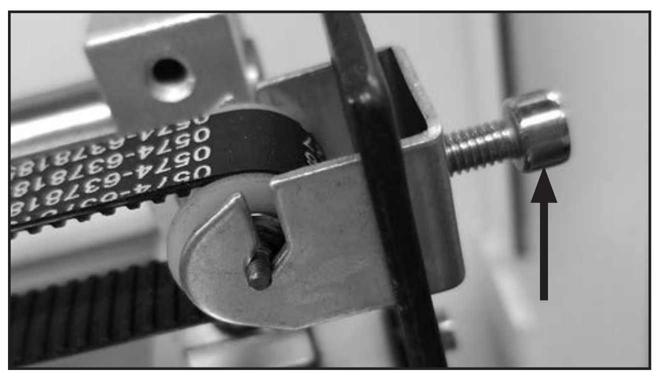


Figure 82. Y-Axis belt tension adjustment screw location.

Z-Axis Belt Adjustment

This belt is considered a low-usage component and is NOT adjustable.

Note: *If the belt ever becomes loose, the table will be out-of-square with laser head rack. Using a dial indicator at each corner, alternately rotate each leadscrew to square table with laser rack. Once all lead screws are synced and table is square, re-install synchronous belt on pulleys.*

Replacing X-Axis and Y-Axis Belts

1. DISCONNECT MACHINE FROM POWER!

2. Loosen belt tension adjustment screw (see **Figures 81–82 on Page 51**).

3. Remove fastener anchoring end of belt (see **Figure 83**), then remove belt.

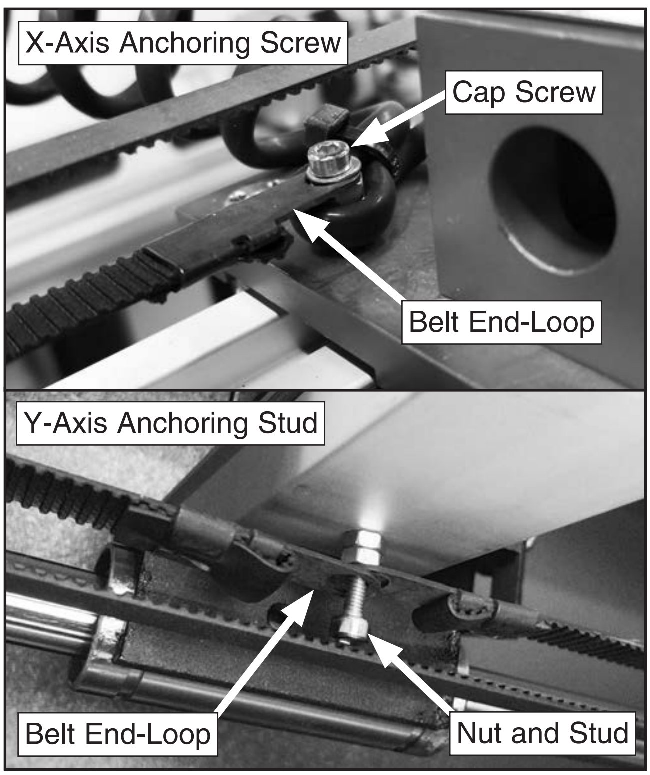


Figure 83. Belt anchoring fasteners locations.

4. Install new belt and align teeth with pulley cogs. Re-install fasteners previously removed.

5. Tighten fasteners and check belt tension. Adjust as required.

Removing/Replacing Laser Tube

The CO_2 gas-filled laser tube is supported by two soft-mount saddles and straps. The tube has four connection points: two for water inlet/outlet hoses, and two for electrical connections.

The following procedure includes steps required for the purpose of long-term storage or machine shipping. If the laser tube has reached the end of its service-life and will be discarded, the tube should be placed in a container and tagged for appropriate disposal.

Items Needed	Qty
Hex Wrenches 2.5, 3mm	1 Ea.
Phillips Screwdriver #2	1
Flat Head Screwdriver #2	1
Utility Knife	1
Water Bucket	1
Container (for Removed Laser Tube)	1
Cloth Rags or Towels	As Needed

Removing Laser Tube

1. DISCONNECT MACHINE FROM POWER!

2. Open both laser tube access doors.

3. Loosen and remove (3) cap screws from both laser tube covers on each side of main tube cabinet (see **Figure 84**).

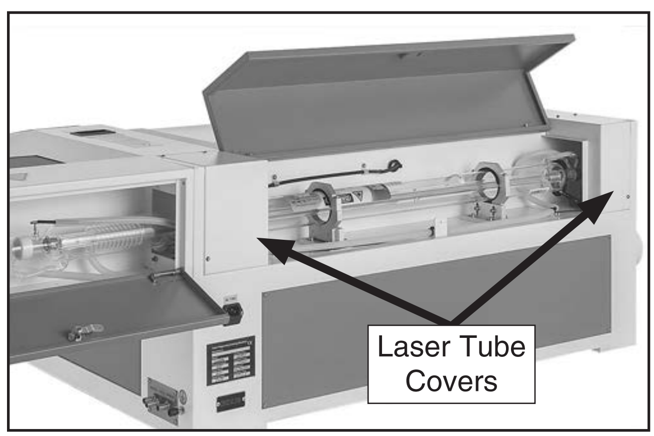


Figure 84. Location of laser tube covers.

4. Disconnect water hose at INLET port of water chiller, and place end into a bucket to collect water runoff.

5. Disconnect water hose at OUTLET port of water chiller, and manually blow through hose several times to purge laser tube and hoses of water.

   IMPORTANT: Remove as much residual water from laser tube as possible. If freezing conditions occur, laser tube glass will crack.

6. Install caps or tape on all open water ports and hoses to keep system sealed and free from contaminants and obstructions.

7. Disconnect anode and cathode wires from laser tube (see **Figure 85**).

8. At laser tube INLET and OUTLET water spouts (see **Figure 85**), cut off water hose at end of spout, and with a utility knife, carefully slit remaining hose until it can be peeled off.

   IMPORTANT: DO NOT attempt to pull hose directly off of spout. Doing so could break glass water spout.

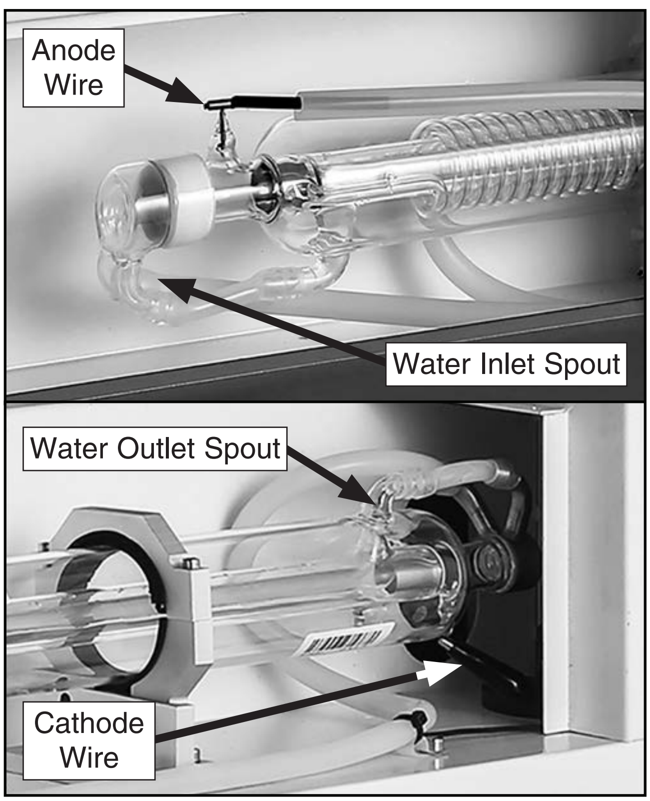


Figure 85. Laser tube end components.

9. Install caps or tape on both open water hoses to keep system sealed and free from contaminants and obstructions.

10. Remove saddle-strap cap screws in an alternating pattern, and then remove saddle straps, as shown in **Figure 86**.

    IMPORTANT: When removing saddle straps for laser tube removal, DO NOT loosen or reposition saddles. The saddles are factory-aligned and require many additional steps to re-align if initial alignment is altered.

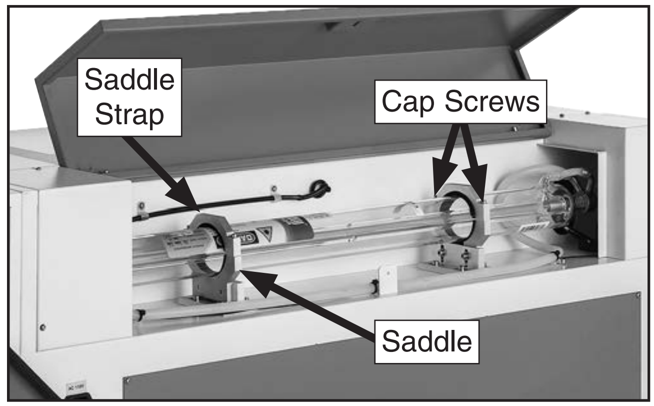


Figure 86. Laser tube saddle components.

11. Install caps or tape on open water spouts to keep laser tube sealed and free from contaminants and obstructions.

12. Prepare container to hold laser tube after removal from machine.

NOTICE

To avoid damaging laser tube, remove tube from saddles ONLY when ready to place in a container. Handle tube carefully in a pre-planned manner to prevent tube contacting machine cabinet.

13. Carefully lift and remove laser tube out of saddles and place into container.

14. Install caps or tape on (2) open water spouts to keep tube sealed and free from contaminants and obstructions, then seal container and tag as required.

Installing Laser Tube

1. Inspect laser tube saddles (see **Figure 87**) and rubber pads for any debris or foreign material, and clean as required.

   IMPORTANT: When inspecting and setting up machine for laser tube installation, DO NOT loosen or reposition laser tube saddles. The saddles are factory-aligned and require many additional steps to re-align if initial alignment is altered.

2. Carefully open laser tube shipping container.

   Note: *DO NOT destroy or discard laser tube shipping container. Retain the original box for ordering replacement tubes, or in the event of damage caused during shipping.*

3. Remove dust caps or tape (if installed) from laser tube.

   IMPORTANT: DO NOT force caps off water spout if they cannot be removed by hand. Carefully cut them off using a utility knife.

> **NOTICE**
> **To avoid damaging laser tube, remove tube from saddles ONLY when ready to place in a container. Handle tube carefully in a pre-planned manner to prevent tube contacting machine cabinet.**

4. Carefully remove laser tube from shipping container, and orient anode-end on anode side of tube compartment (see **Figure 87**).

5. Gently place laser tube in saddles.

6. Position cathode-end (see **Figure 87**) of laser tube so there is a gap of 1–1½" between cathode-end and center of first mirror.

7. Verify inlet and outlet hoses will not interfere with closed access doors or installed laser tube covers, and ensure hoses are free of kinks that could potentially restrict water flow to laser tube.

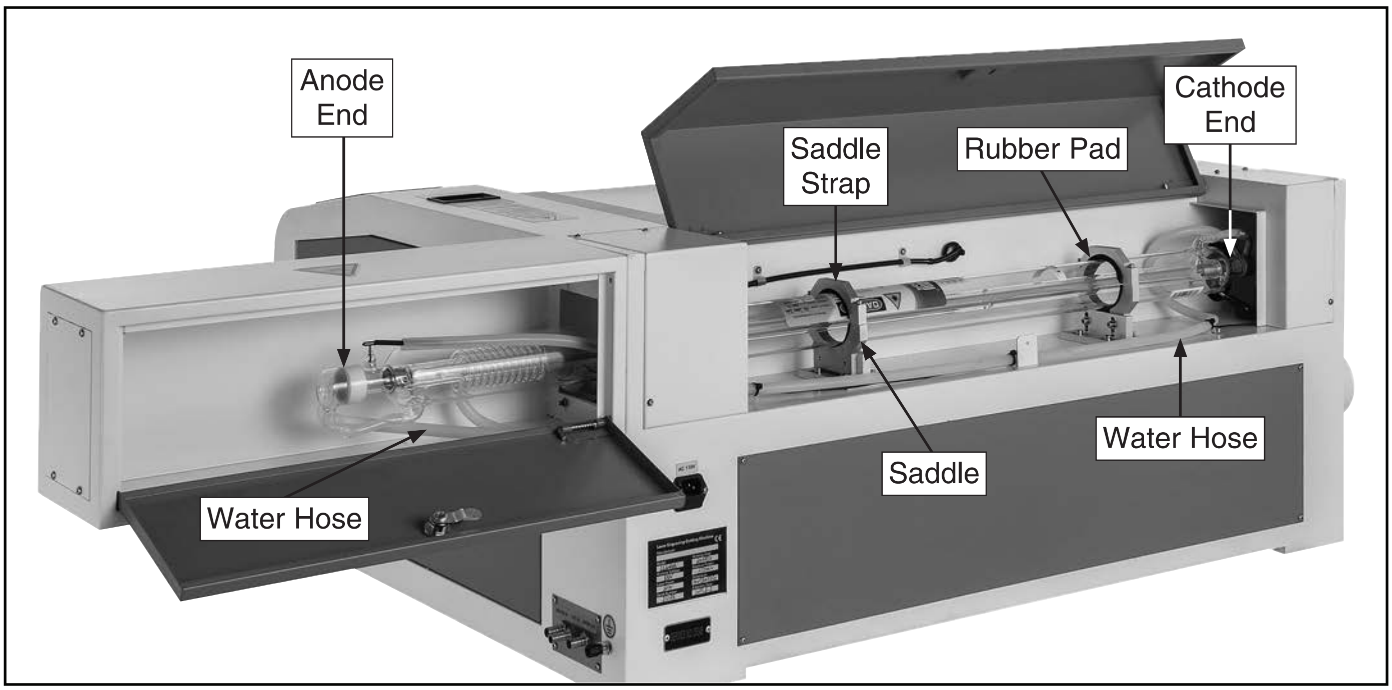


Figure 87. Laser tube compartment identification.

> **NOTICE**
> **DO NOT use hose or spring clamps to secure water hoses to laser tube spouts. Compression from clamping can cause glass spout to crack during operation.**

8. Carefully install inlet and outlet water hoses over appropriate glass spouts marked INLET and OUTLET until barbed-end is fully covered by hose.

   Note: *Water-based lubricant may be applied to exterior of spout to aid installation. DO NOT apply any lubricant to interior of spout where it will enter water chiller system.*

   IMPORTANT: If a water hose must be removed for any reason, DO NOT attempt to pull hose directly off of spout. Doing so could break glass water spout. Cut off water hose at end of spout, and carefully slit remaining hose segment with a utility knife until it can be peeled off.

9. Rotate and center laser tube in saddles so inlet spout is pointed down, and outlet spout is pointed up, as shown in **Figure 88**.

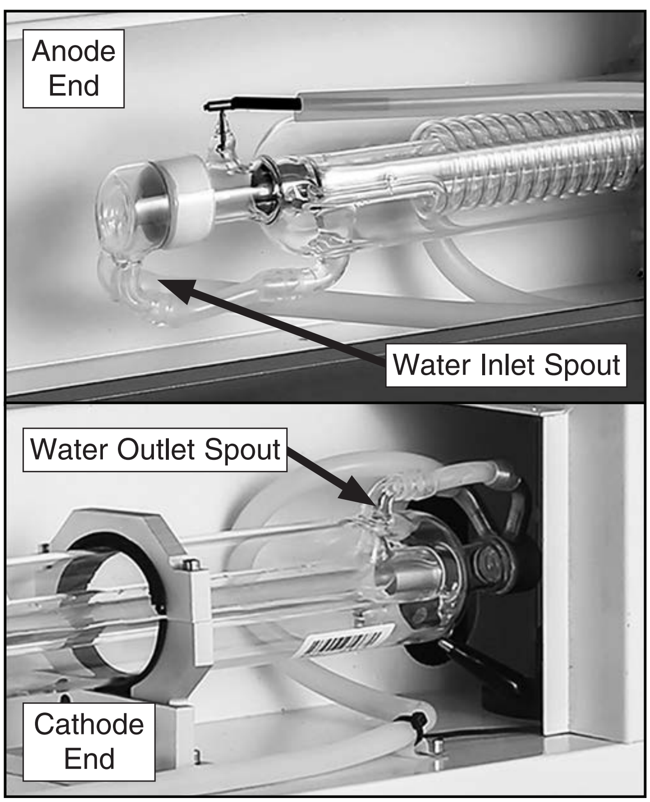


Figure 88. Laser tube water inlet/outlet spouts.

10. Verify water chiller system inlet and outlet hoses are connected to auxiliary systems connection panel according to **Installing Water Chiller** on **Page 21.**

    IMPORTANT: If you are replacing laser tube, replace the distilled water in water chiller system as well. This will prevent accumulation of contaminants in laser tube.

11. Turn water chiller system ***ON*** and allow system to run. DO NOT turn ***ON*** laser at this time.

12. Verify laser tube cooling chambers fill with water and no leaks are visible throughout water chiller system.

13. Wait one minute for water to flow through system, then slowly rotate laser tube while slightly raising tube at cathode-end to release air bubbles.

    IMPORTANT: DO NOT tap on laser tube to release air bubbles. Trapped air will dissipate from cooling chambers once tube begins to warm up during first use.

14. Once all air pockets and large bubbles are released (see **Figure 89**), laser tube is primed with cooling water.

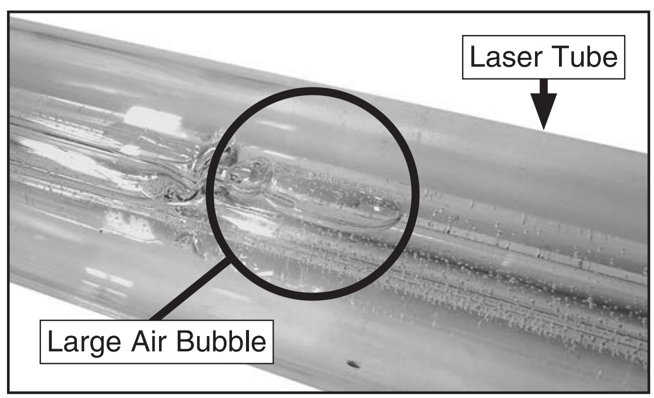


Figure 89. Example of large air bubbles when priming laser tube.

15. Remove reservoir cap and verify water level is 3–6" from reservoir opening.

16. Turn water chiller system ***OFF***.

NOTICE

DO NOT turn *ON* laser with water chiller system *OFF*. Laser tube will quickly overheat during laser operations without a functioning cooling system.

17. Install anode wire on positive (+) anode terminal, and tighten set screw on end of wire, as shown in **Figure 90**.

18. Install cathode wire on negative (-) cathode terminal by using attached alligator clip on wire, as shown in **Figure 90**.

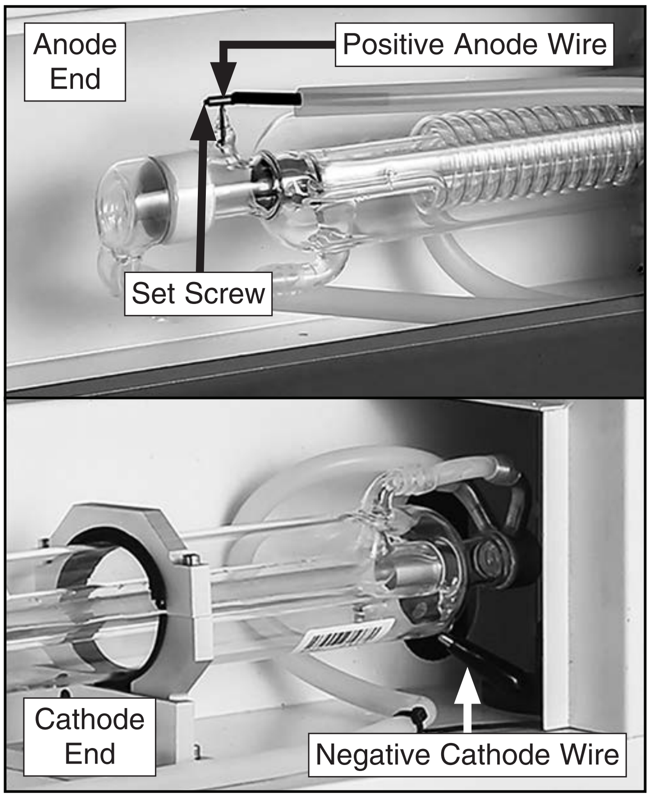


Figure 90. Location of laser tube anode/cathode wires.

19. Verify gap of 1–1½" is still present between cathode-end of laser tube and center of first mirror.

20. Verify laser tube is centered evenly in rubber saddle pads, and no contaminants are visible on pad surfaces.

21. Clean rubber pads on saddle straps and place on laser tube above saddles, as shown in **Figure 91**.

Note: *If there is a gap between saddle strap and saddle, it may be necessary to remove a layer of rubber padding to ensure a proper fit.*

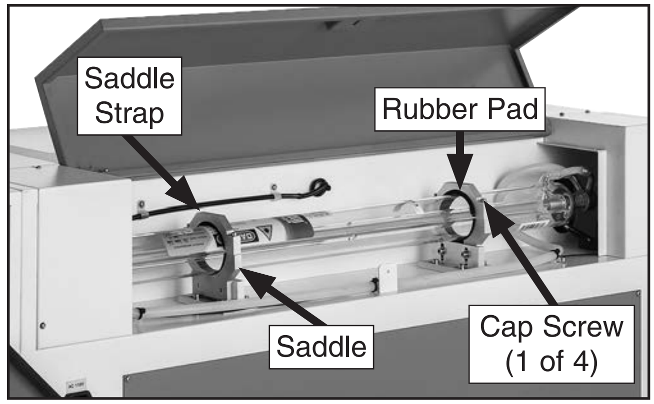


Figure 91. Laser tube saddles installed.

NOTICE

Laser tube saddles should gently hold the laser tube in position. Tightening saddle straps with excessive force WILL break the laser tube.

22. Carefully secure saddle straps to saddles by gently tightening cap screws in an alternating pattern until laser tube can no longer be rotated by hand.

23. Align laser tube beam path by performing **Aligning Laser Beam Path** on **Page 57**.

24. When laser alignment is complete, re-install (2) laser tube covers, and (6) cap screws removed in **Step 3** on **Page 54**.

Aligning Laser Beam Path

⚠DANGER

Operator and bystanders MUST wear ANSI-approved safety glasses rated for use with a CLASS 4 laser when machine is operating. NEVER look directly into laser beam, or stare at laser contact point for more than a few seconds, or severe eye injury or blindness will occur.

The laser beam path is directed by three mirrors before passing through a focus lens to the focal point on the workpiece, as shown in **Figure 92**. The included low-power reference laser connected to Mirror #1 can be used as a reference point when adjusting alignment.

Proper alignment of the laser beam path is critical to power efficiency, cutting/engraving quality, and laser tube service-life.

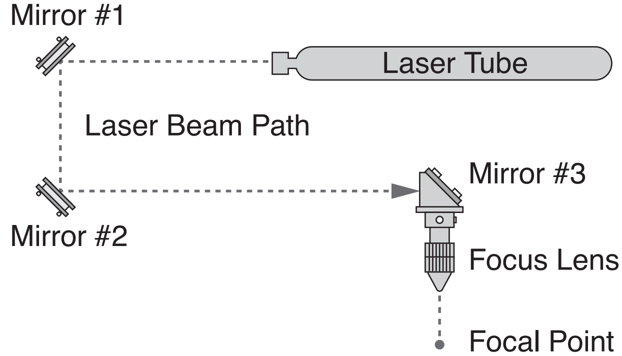


Figure 92. Laser beam path through cabinet.

Items Needed	**Qty**
Class 4 Laser Safety Glasses (per person)	1
Laser Beam Alignment Gauge	1
Mirror Alignment Gauge	1
Hex Wrenches 2.5, 3mm	1 Ea.
Manila Folder	As Needed
Sheet of Paper	As Needed
Masking Tape	As Needed

To align laser beam path:

1. Prepare machine for operation according to **SECTION 3: SETUP** on **Page 16**.

2. Open top loading door and main laser tube access door.

3. Loosen and remove (3) cap screws from both laser tube covers on each side of main tube cabinet to access Mirror #1.

4. Turn machine and water chiller system ***ON***. Allow machine to complete boot cycle before proceeding.

5. Press Menu button. Main menu screen will open (see **Figure 93**).

Figure 93. Main menu screen.

6. Use arrow navigation buttons to highlight "Max Power+" and press Enter.

7. Highlight "MaxPower 1" and press Enter.

   Note: *When adjusting power settings for the Model G0872, use "MaxPower 1" and "MinPower 1". The secondary "MaxPower 2" and "MinPower 2" settings are for a future model and do not function at this time.*

8. Set maximum power to 99% and press Enter.

9. Highlight "Min Power+" and press Enter.

10. Highlight "MaxPower 1" and press Enter.

11. Set minimum power to 50% and press Enter.

    Note: *Machine pulse function will not provide a clear test mark if set below 50% power.*

12. Press Esc to exit main menu and return to status screen (see **Figure 94**).

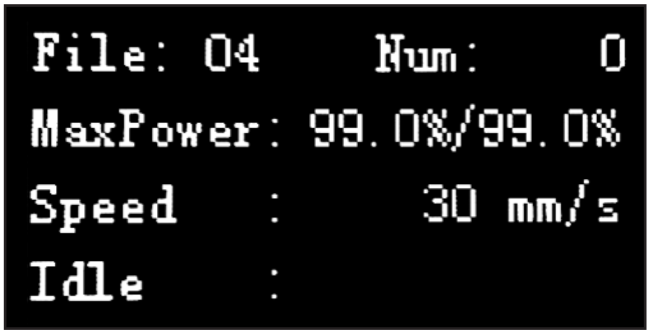


Figure 94. Example of status screen.

13. Move laser head assembly to upper left corner of table (see **Figure 95**).

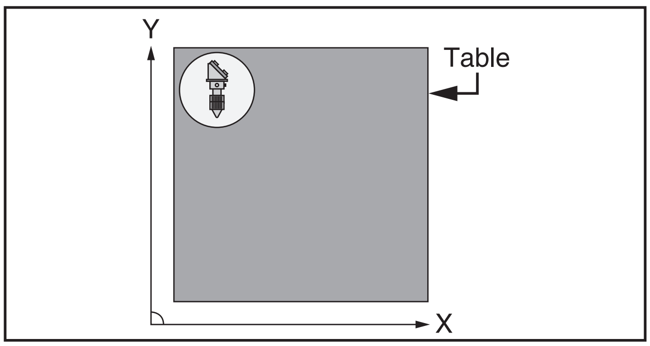


Figure 95. Laser head assembly in upper left corner of table.

14. Insert a 1" piece of paper or manila folder into small slot on laser beam alignment gauge behind wire crosshairs, and install gauge in beam inlet on laser head assembly (see **Figure 96**).

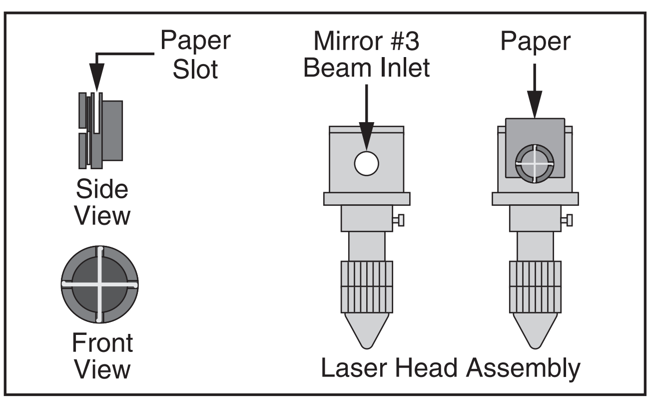


Figure 96. Laser beam alignment gauge installed in beam inlet.

15. Close top loading door and check alignment by pressing Pulse button on control panel. Compare results with **Figure 97** below.

IMPORTANT: Pulse function will only operate with top loading door closed.

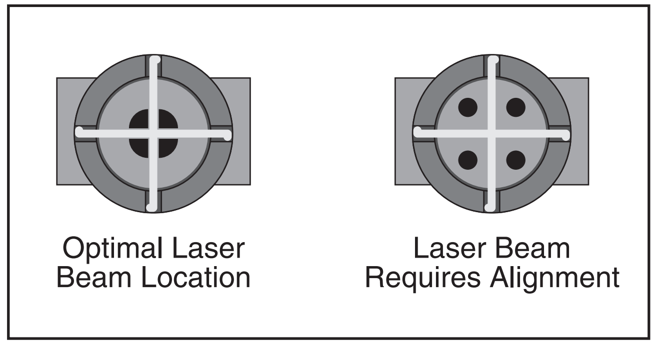


Figure 97. Laser beam alignment marks.

— If laser beam falls within center of crosshairs, alignment is optimal. Proceed to **Step 19**.

— If laser beam falls *outside* center of crosshairs, adjustment is required. Proceed to **Step 16**.

16. Loosen lock nuts on (3) brass thumbscrews located behind Mirror #1 (see **Figure 98**).

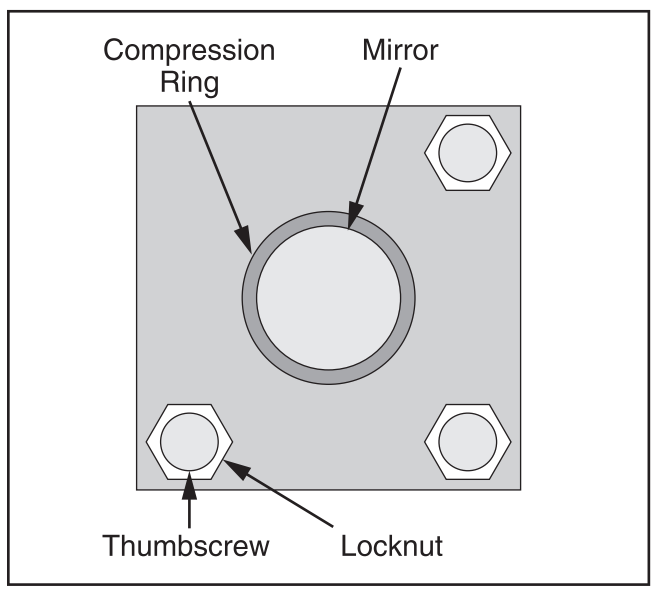


Figure 98. Typical mirror assembly.

17. Adjust direction of beam path by tightening thumbscrews depending on desired direction of beam travel, as shown in **Figure 99**.

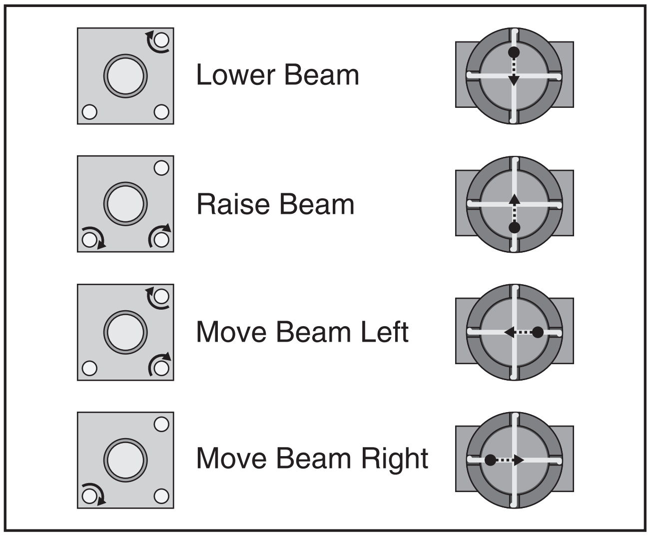


Figure 99. Mirror thumbscrew adjustment.

18. Verify mirror adjustment by pressing Pulse button on control panel. Compare results with those shown in **Figure 97** on **Page 58**.

— If laser beam falls within center of cross-hairs, alignment is optimal. Proceed to **Step 19**.

— If laser beam falls *outside* center of cross-hairs, adjustment is required. Repeat **Steps 16–18**.

19. Move laser head assembly to lower left corner of table (see **Figure 100**).

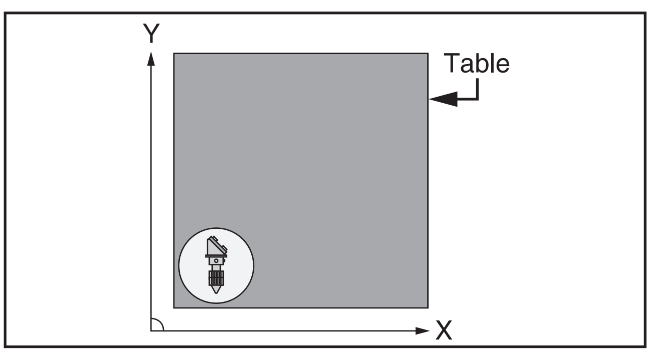


Figure 100. Laser head assembly in lower left corner of table.

20. Check alignment by pressing Pulse button on control panel. Compare results with those shown in **Figure 97** on **Page 58**.

— If laser beam falls within center of cross-hairs, alignment is optimal. Proceed to **Step 26**.

— If laser beam falls *outside* center of cross-hairs, adjustment is required. Proceed to **Step 21**.

21. Place mirror alignment gauge in front of Mirror #2 and hold in place with masking tape, as shown in **Figure 101**.

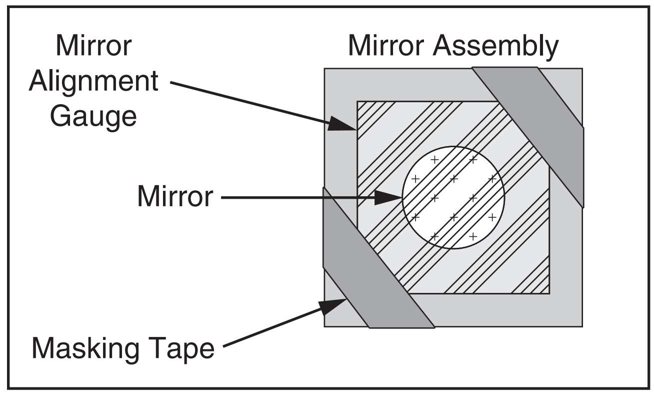


Figure 101. Mirror alignment gauge installed.

22. Verify Mirror #1 and Mirror #2 alignment by pressing Pulse button on control panel.

— If laser beam is within center of mirror, alignment is optimal. Remove mirror alignment gauge from Mirror #2 and proceed to **Step 26**.

— If laser beam is *outside* center of mirror, adjustment is required. Proceed to **Step 23**.

23. Loosen lock nuts on (3) brass thumbscrews located behind Mirror #1 assembly (see **Figure 98** on **Page 58**).

24. Adjust direction of beam path by tightening thumbscrews depending on desired direction of beam travel, as shown in **Figure 99**.

25. Repeat **Step 22.**

26. Move laser head assembly to lower right corner of table (see **Figure 102**).

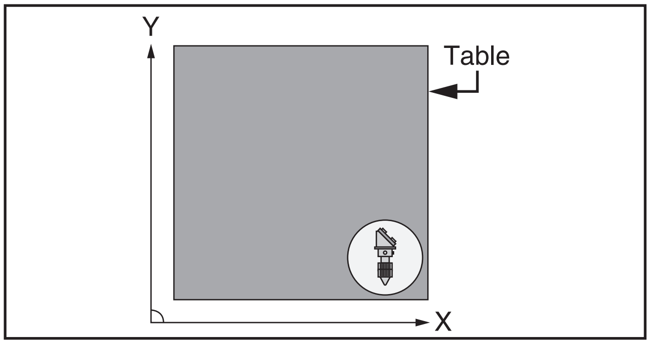


Figure 102. Laser head assembly in lower right corner of table.

27. Verify Mirror #2 and Mirror #3 alignment by pressing Pulse button on control panel.

— If laser beam falls within center of crosshairs, alignment is optimal. Proceed to **Step 31**.

— If laser beam falls *outside* center of crosshairs, adjustment is required. Proceed to **Step 28**.

28. Loosen lock nuts on (3) brass thumbscrews located behind Mirror #2 assembly (see **Figure 103**).

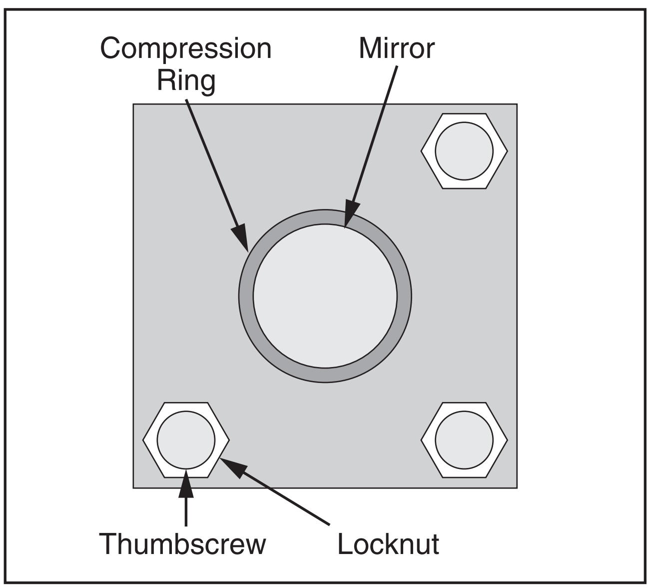


Figure 103. Typical mirror assembly.

29. Adjust direction of beam path by tightening thumbscrews depending on desired direction of beam travel (see **Figure 104**).

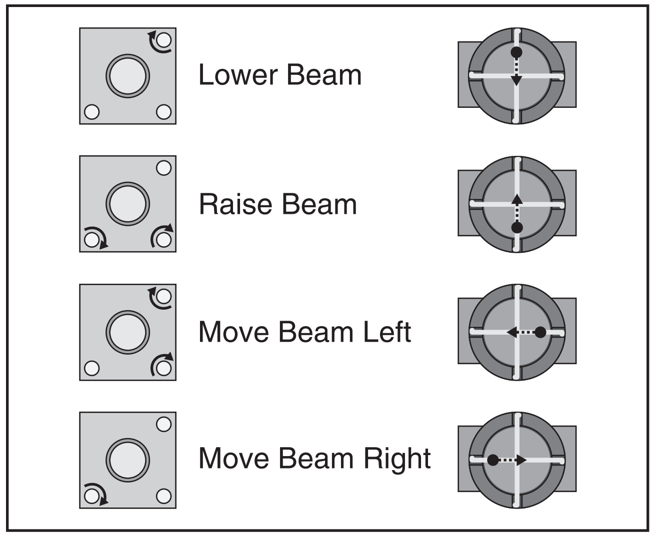


Figure 104. Mirror thumbscrew adjustment.

30. Repeat **Step 27.**

31. Tighten all loose lock nuts on mirror assemblies. DO NOT tighten thumbscrews!

32. Re-install (3) cap screws and (1) laser tube cover removed in **Step 3**, and close all doors opened for alignment.

33. Turn ***OFF*** machine by pressing Emergency Stop Button.

34. Turn ***OFF*** water chiller system.

35. Put all tools and gauges away, and clean area for next operation.

Removing/Replacing Laser Optics

Before removing and replacing optical components on the Model G0872, thoroughly wash and dry hands before putting on sterile disposable gloves. This will help ensure laser optics remain clean during removal and replacing.

Follow all best practices guidelines described in **Laser Optics Maintenance** on **Page 45** for handling sensitive optical components.

Items Needed	**Qty**
Optics Disassembly Tool	1
Hex Wrench 2mm	As Needed
Duct Tape	As Needed
Sterile Disposable Gloves	As Needed
Lens Cleaning Paper	As Needed

Remove and Replace Mirrors

1. DISCONNECT MACHINE FROM POWER!

2. Remove any covers and open any doors to gain access to mirror assembly.

   IMPORTANT: To avoid re-aligning laser beam, DO NOT remove mirror assembly from its location.

3. Locate compression ring on rear of mirror (see **Figure 105**) and remove with optics disassembly tool.

   Tip: *If optics disassembly tool is too large to fit behind mirror assembly, a 2mm hex wrench can be used as a substitute.*

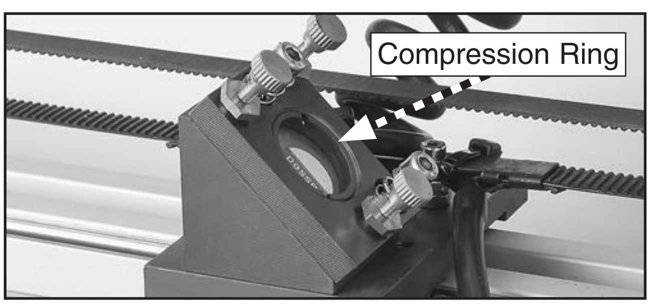


Figure 105. Typical compression ring location.

4. Remove mirror by applying duct tape to non-reflective rear surface and lifting.

5. Install replacement mirror in mirror assembly and re-install compression ring.

6. Close covers and doors opened in **Step 2**.

Remove and Replace Focus Lens

1. DISCONNECT MACHINE FROM POWER!

2. Open top loading door and move laser head assembly to an easily accessible area.

3. Disconnect air supply hose, loosen air nozzle (see **Figure 106**), and set aside.

4. Loosen thumbscrews and carefully remove focus canister (see **Figure 106**).

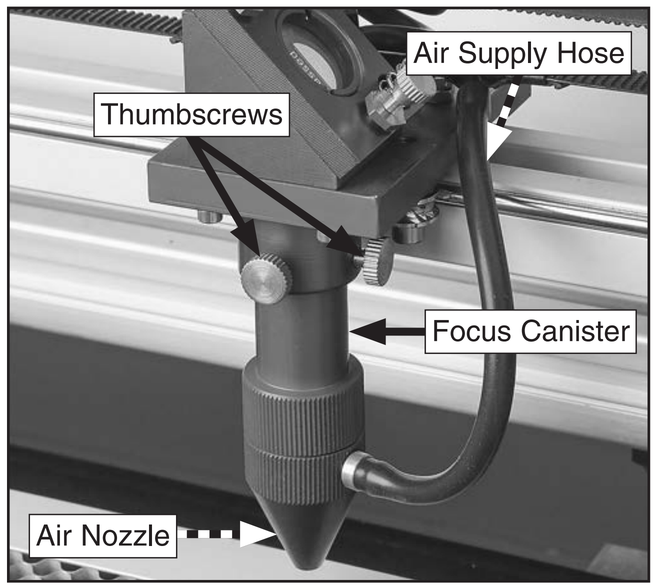


Figure 106. Laser head assembly components.

5. Locate compression ring holding focus lens and remove with optics disassembly tool.

6. Carefully separate focus lens from cushion, and set cushion on a sheet of cleaning paper.

7. Install replacement focus lens with curved side facing up, and install lens cushion and compression ring.

8. Install focus canister and air nozzle, then connect air supply hose.

9. Close top loading door.

SECTION 8: WIRING

These pages are current at the time of printing. However, in the spirit of improvement, we may make changes to the electrical systems of future machines. Compare the manufacture date of your machine to the one stated in this manual, and study this section carefully.

If there are differences between your machine and what is shown in this section, call Technical Support at (570) 546-9663 for assistance BEFORE making any changes to the wiring on your machine. An updated wiring diagram may be available. **Note:** *Please gather the serial number and manufacture date of your machine before calling. This information can be found on the main machine label.*

⚠WARNING

Wiring Safety Instructions

SHOCK HAZARD. Working on wiring that is connected to a power source is extremely dangerous. Touching electrified parts will result in personal injury including but not limited to severe burns, electrocution, or death. Disconnect the power from the machine before servicing electrical components!

MODIFICATIONS. Modifying the wiring beyond what is shown in the diagram may lead to unpredictable results, including serious injury or fire. This includes the installation of unapproved aftermarket parts.

WIRE CONNECTIONS. All connections must be tight to prevent wires from loosening during machine operation. Double-check all wires disconnected or connected during any wiring task to ensure tight connections.

CIRCUIT REQUIREMENTS. You MUST follow the requirements at the beginning of this manual when connecting your machine to a power source.

WIRE/COMPONENT DAMAGE. Damaged wires or components increase the risk of serious personal injury, fire, or machine damage. If you notice that any wires or components are damaged while performing a wiring task, replace those wires or components.

MOTOR WIRING. The motor wiring shown in these diagrams is current at the time of printing but may not match your machine. If you find this to be the case, use the wiring diagram inside the motor junction box.

CAPACITORS/INVERTERS. Some capacitors and power inverters store an electrical charge for up to 10 minutes after being disconnected from the power source. To reduce the risk of being shocked, wait at least this long before working on capacitors.

EXPERIENCING DIFFICULTIES. If you are experiencing difficulties understanding the information included in this section, contact our Technical Support at (570) 546-9663.

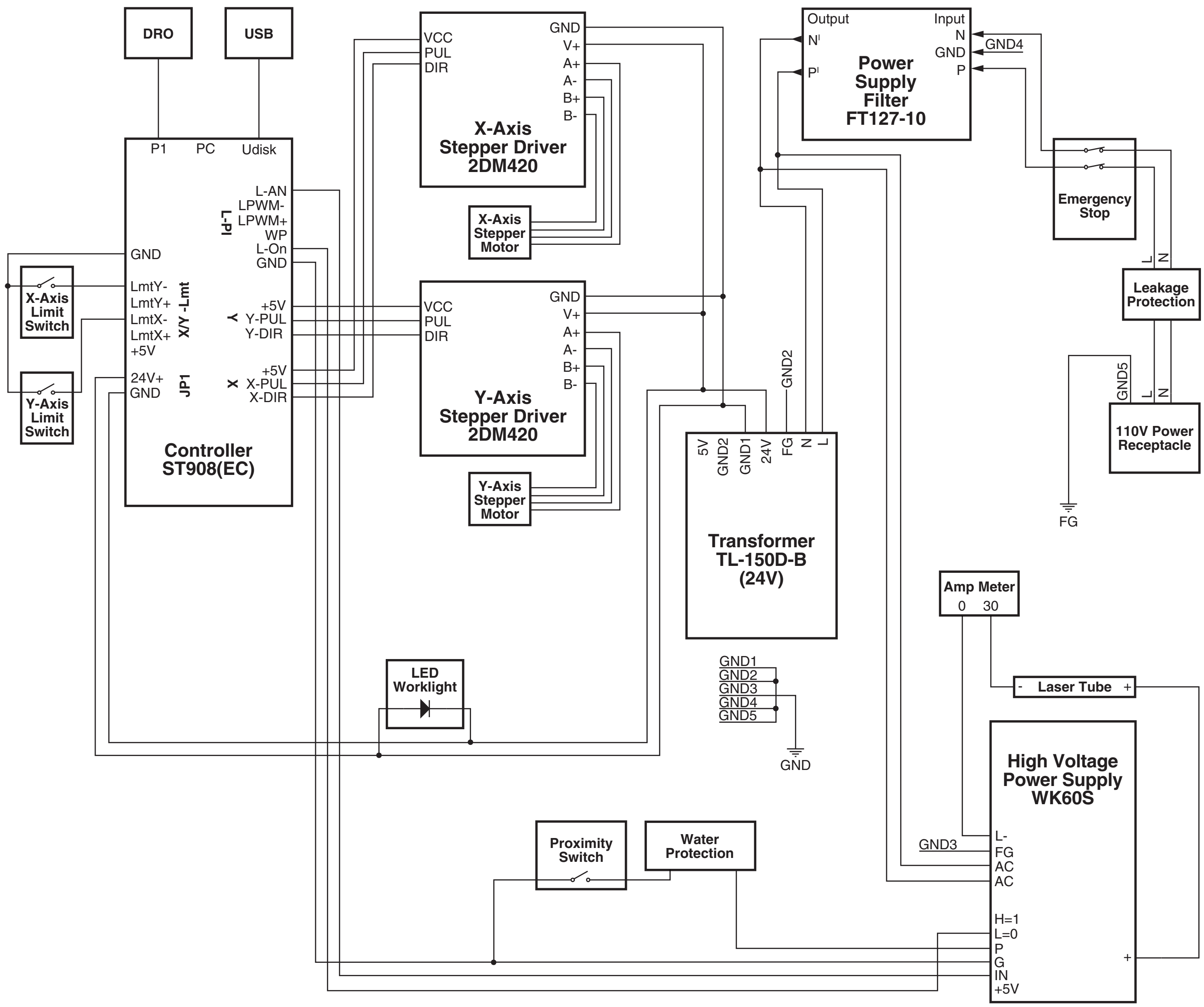
NOTICE

The photos and diagrams included in this section are best viewed in color. You can view these pages in color at www.grizzly.com.

COLOR KEY

BLACK	Bk	BLUE	Bl	YELLOW	Yl	LIGHT BLUE	Lb
WHITE	Wt	BROWN	Br	YELLOW GREEN	Yg	BLUE WHITE	Bw
GREEN	Gn	GRAY	Gy	PURPLE	Pu	TUR-QUOISE	Tu
RED	Rd	ORANGE	Or	PINK	Pk		

Wiring Schematic



STOP **READ ELECTRICAL SAFETY ON PAGE 62!**

Electrical Components

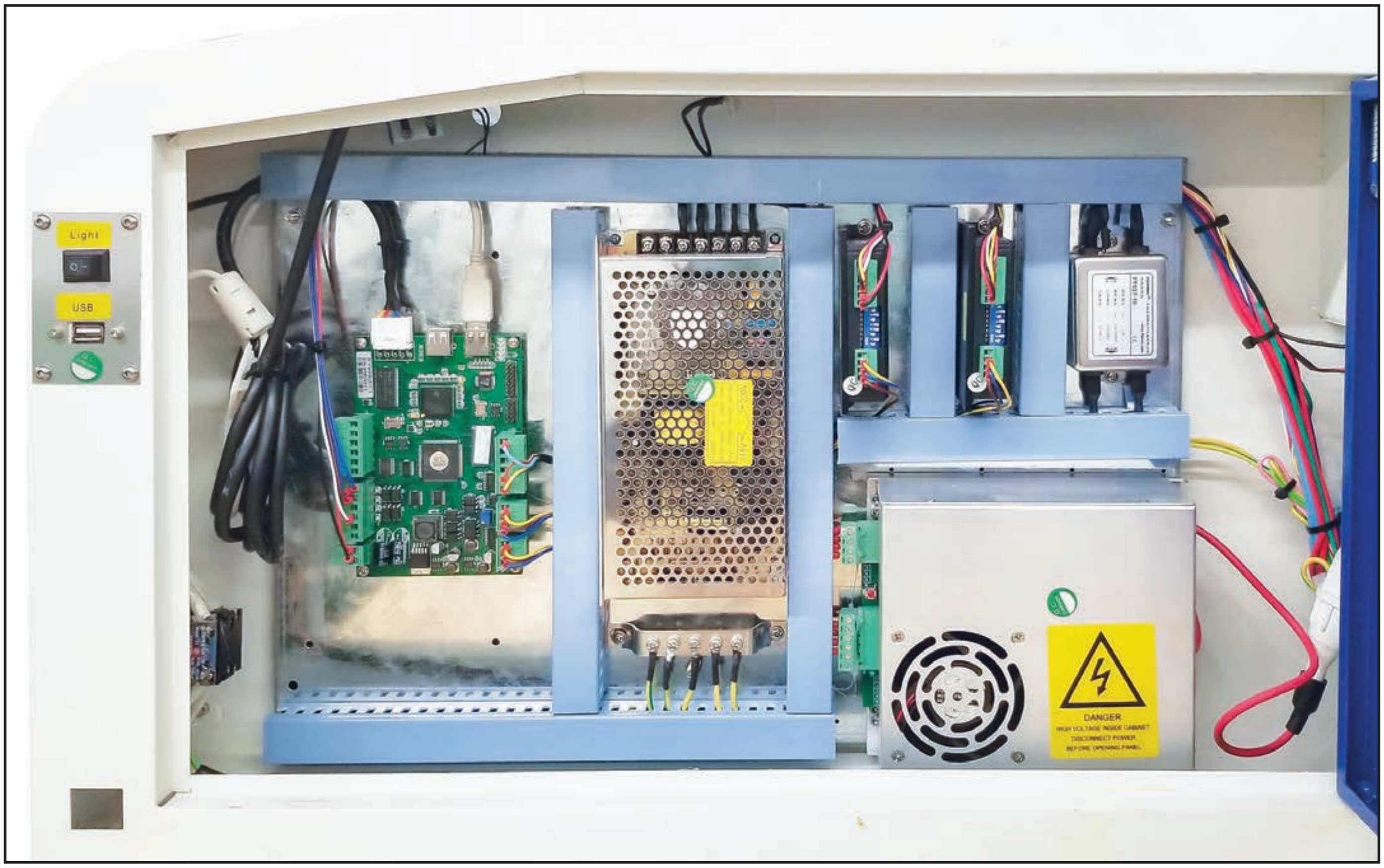


Figure 107. Electrical box components.

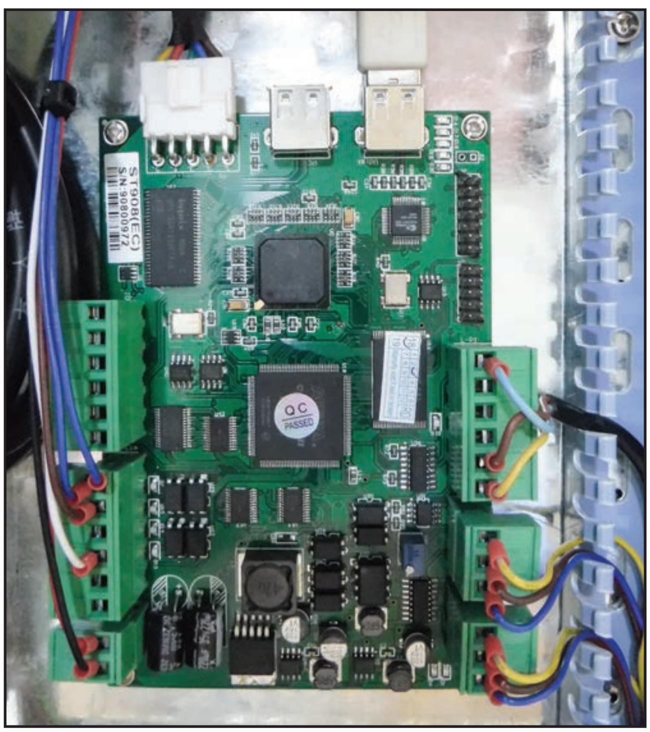


Figure 108. Controller.

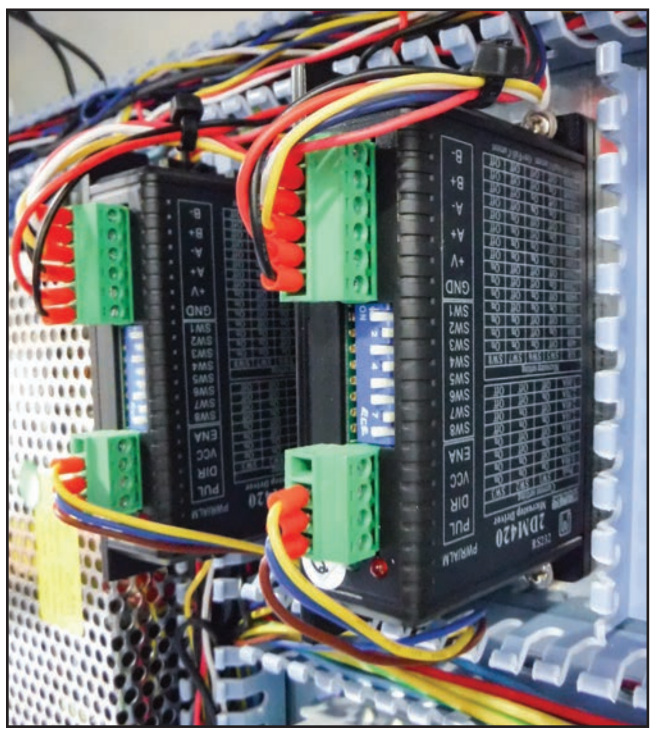


Figure 109. X-/Y-axis stepper drivers.

READ ELECTRICAL SAFETY ON PAGE 62!

Figure 110. High voltage power supply.

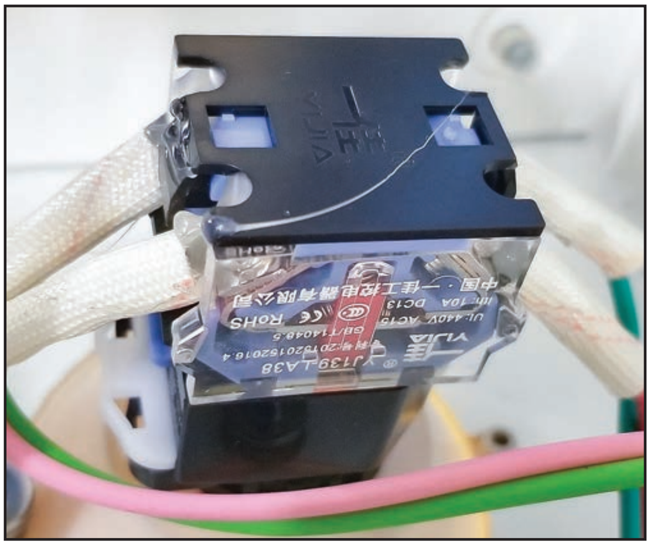


Figure 111. Emergency stop button.

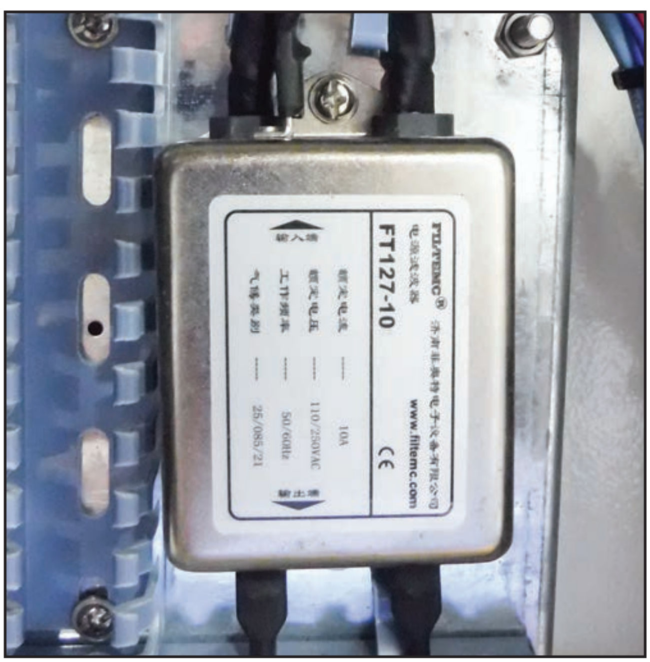


Figure 112. Power supply filter.

Figure 113. Digital readout (control panel).

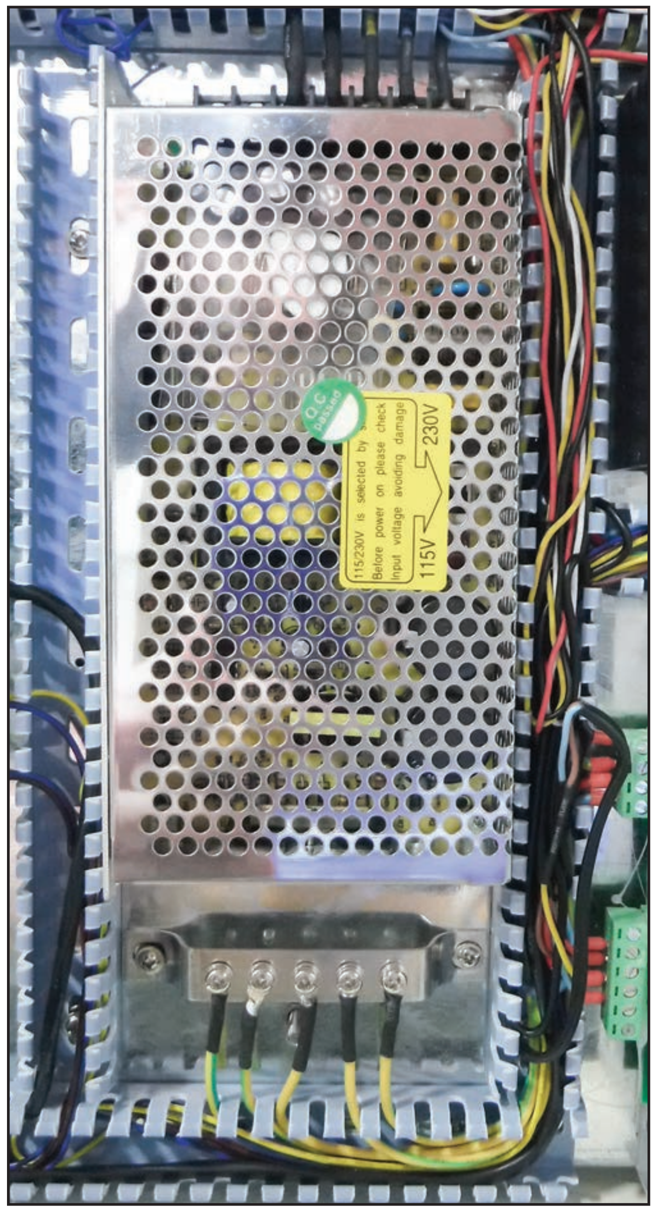
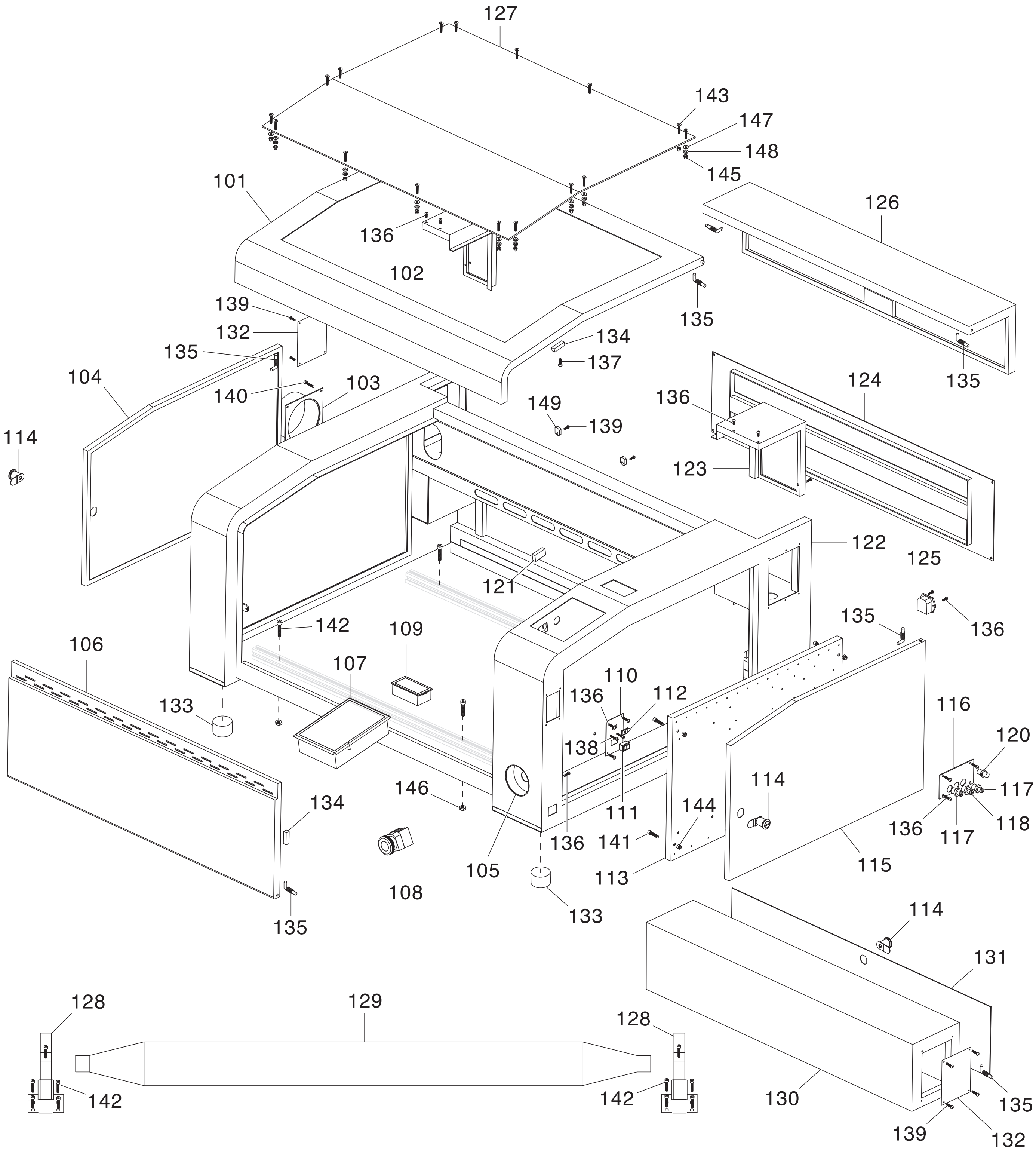


Figure 114. Transformer.

SECTION 9: PARTS

We do our best to stock replacement parts when possible, but we cannot guarantee that all parts shown are available for purchase. Call **(800) 523-4777** or visit **www.grizzly.com/parts** to check for availability.

Main

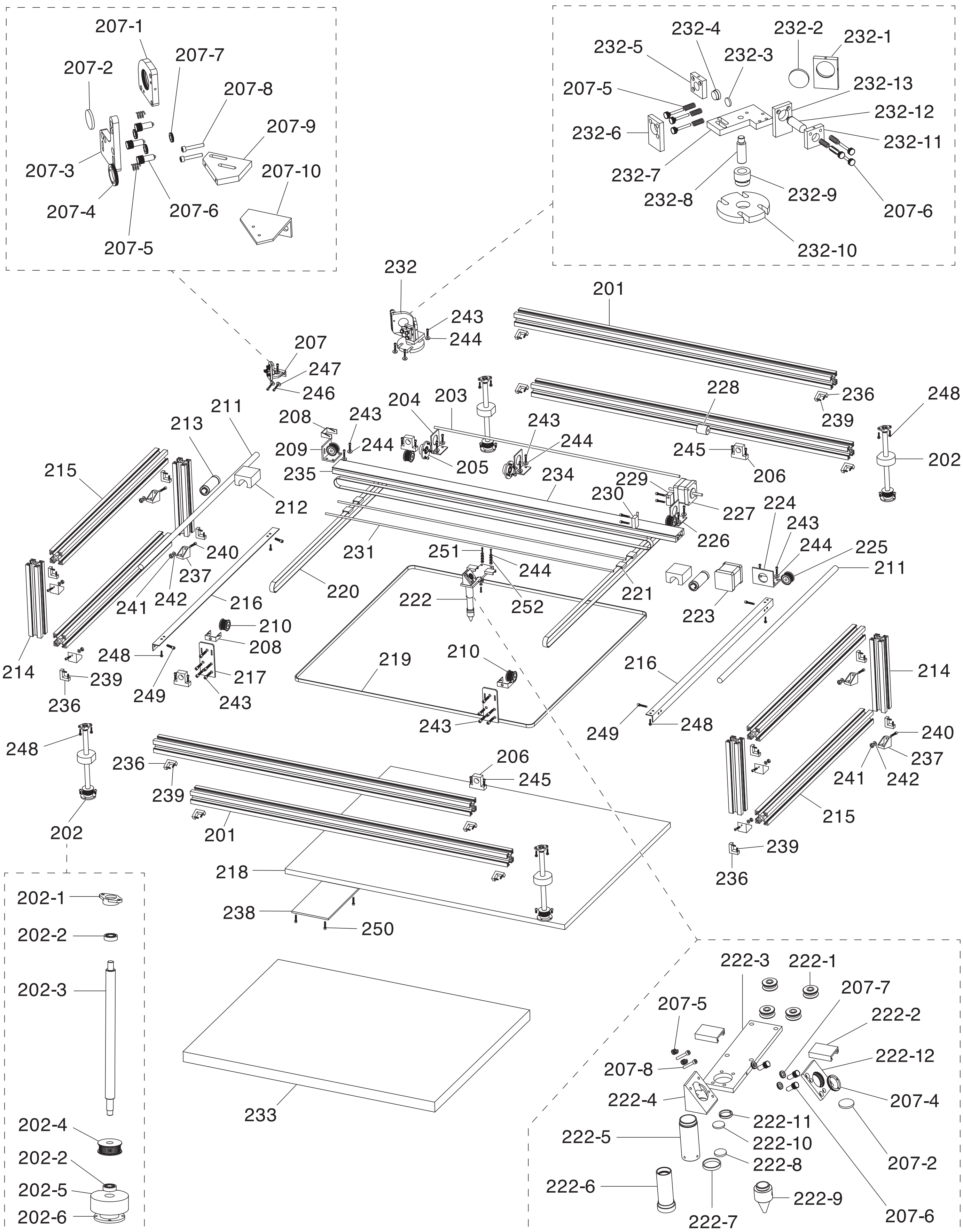


Main Parts List

REF	PART #	DESCRIPTION
101	P0872101	LOADING DOOR (TOP)
102	P0872102	LASER TUBE COVER (LEFT)
103	P0872103	EXHAUST PORT
104	P0872104	LOADING DOOR (SIDE)
105	P0872105	E-STOP MOUNTING BRACKET
106	P0872106	LOADING DOOR (BOTTOM)
107	P0872107	DIGITAL READOUT ST908
108	P0872108	E-STOP BUTTON YIJIA YJ139-LA38 22MM
109	P0872109	AMP METER ANALOG UXCELL 85C17 5A
110	P0872110	SWITCH MOUNTING PLATE
111	P0872111	ROCKER SWITCH DAIER KCD1-101
112	P0872112	USB 2.0 PORT
113	P0872113	MOUNTING BOARD
114	P0872114	KEYED CAM LOCK
115	P0872115	ELECTRICAL COMPARTMENT DOOR
116	P0872116	AUX SYSTEMS MOUNTING PLATE
117	P0872117	WATER FITTING 3/8 NPT STRAIGHT BR
118	P0872118	AIR FITTING 5/16 NPT STRAIGHT BR
120	P0872120	DIRECT GROUND POST
121	P0872121	PROXIMITY SENSOR PIC MS-324/325
122	P0872122	CABINET FRAME
123	P0872123	LASER TUBE COVER (RIGHT)
124	P0872124	ACCESS PANEL
125	P0872125	POWER CORD RECEPTACLE

REF	PART #	DESCRIPTION
126	P0872126	LASER TUBE ACCESS DOOR
127	P0872127	ACRYLIC WINDOW
128	P0872128	LASER TUBE SADDLE
129	P0872129	LASER TUBE 60W
130	P0872130	LASER TUBE EXTENSION CABINET
131	P0872131	EXTENSION CABINET DOOR
132	P0872132	END COVER
133	P0872133	RUBBER FOOT
134	P0872134	DOOR MAGNET
135	P0872135	DOOR HINGE
136	P0872136	BUTTON HD CAP SCR M4-.7 X 8
137	P0872137	FLAT HD CAP SCR M6-1 X 10
138	P0872138	CAP SCREW M3-.5 X 8
139	P0872139	BUTTON HD CAP SCR M3-.5 X 6
140	P0872140	CAP SCREW M4-.7 X 6
141	P0872141	CAP SCREW M5-.8 X 8
142	P0872142	CAP SCREW M5-.8 X 10
143	P0872143	CAP SCREW M6-1 X 10
144	P0872144	LOCK NUT M5-.8
145	P0872145	ACORN NUT M6-1
146	P0872146	HEX NUT M5-.8
147	P0872147	FLAT WASHER 6MM
148	P0872148	LOCK WASHER 6MM
149	P0872149	WIRE CLAMP

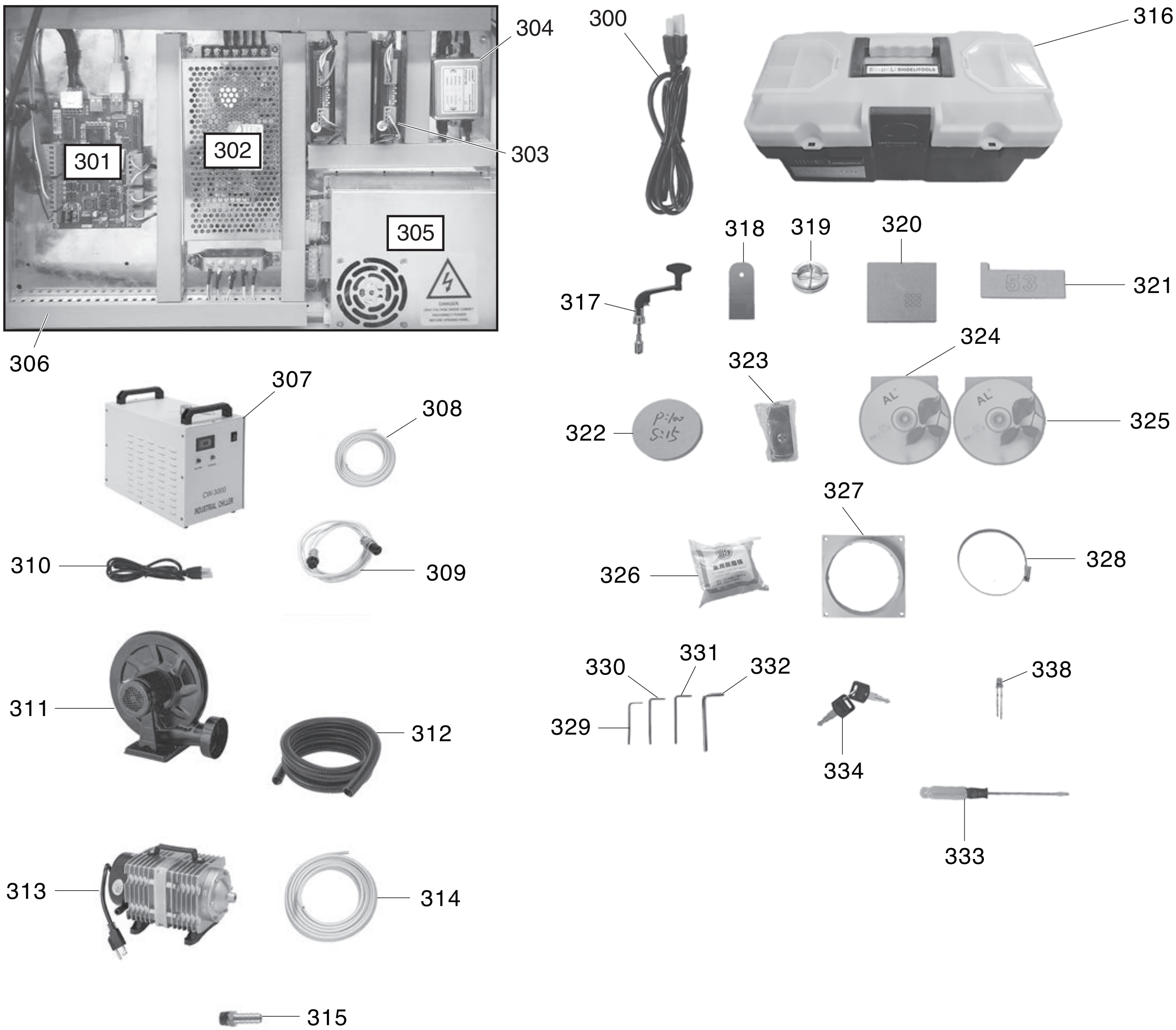
Gantry & Optics



Gantry & Optics Parts List

REF	PART #	DESCRIPTION
201	P0872201	CROSSBAR
202	P0872202	LEADSCREW ASSEMBLY
202-1	P0872202-1	LEADSCREW MOUNT (UPPER)
202-2	P0872202-2	BALL BEARING 6202-2RS
202-3	P0872202-3	LEADSCREW
202-4	P0872202-4	LEADSCREW PULLEY
202-5	P0872202-5	LEADSCREW LIFT BLOCK
202-6	P0872202-6	LEADSCREW MOUNT (LOWER)
203	P0872203	X-AXIS DRIVE SHAFT
204	P0872204	DRIVE SHAFT SUPPORT
205	P0872205	DRIVE SHAFT MOUNT
206	P0872206	STOP PAD
207	P0872207	MIRROR ASSEMBLY (FRONT)
207-1	P0872207-1	MIRROR HOUSING
207-2	P0872207-2	MIRROR
207-3	P0872207-3	VERTICAL ADJUSTMENT PLATE
207-4	P0872207-4	MIRROR COMPRESSION RING
207-5	P0872207-5	COMPRESSION SPRING 1 X 7.5 X 12
207-6	P0872207-6	KNURLED THUMB SCREW M3-.5 X 14, D8
207-7	P0872207-7	HEX THUMB NUT M3-.5
207-8	P0872207-8	CAP SCREW M4-.7 X 15
207-9	P0872207-9	ADJUSTMENT PLATE (UPPER)
207-10	P0872207-10	ADJUSTMENT PLATE (LOWER)
208	P0872208	WHEEL BRACKET
209	P0872209	X-AXIS PULLEY SUPPORT
210	P0872210	Y-AXIS PULLEY
211	P0872211	Y-AXIS DRIVE SHAFT
212	P0872212	BEARING MOUNT
213	P0872213	LINEAR BEARING LMF20LUU
214	P0872214	SHELF SUPPORT COLUMN
215	P0872215	SHELF SUPPORT RAIL
216	P0872216	SHELF SUPPORT BRACKET
217	P0872217	Y-AXIS PULLEY SUPPORT
218	P0872218	TABLE BASE
219	P0872219	Z-AXIS TIMING BELT 848MXL
220	P0872220	X/Y-AXIS TIMING BELT 37MXL
221	P0872221	Y-AXIS BELT LOCKING TAB
222	P0872222	LASER HEAD ASSEMBLY
222-1	P0872222-1	ROLLER
222-2	P0872222-2	X-AXIS BELT LOCKING TAB
222-3	P0872222-3	LASER HEAD MOUNTING PLATE
222-4	P0872222-4	45 DEG MIRROR HOUSING
222-5	P0872222-5	FIXED FOCUS BARREL
222-6	P0872222-6	TELESCOPING FOCUS BARREL
222-7	P0872222-7	BARREL RING
222-8	P0872222-8	FOCUS LENS (ENGRAVING)
222-9	P0872222-9	AIR NOZZLE
222-10	P0872222-10	FOCUS LENS (CUTTING)
222-11	P0872222-11	COMPRESSION RING
222-12	P0872222-12	45 DEG ADJUSTMENT PLATE
223	P0872223	X-AXIS STEP MOTOR NEMA 17
224	P0872224	X-AXIS MOTOR MOUNT
225	P0872225	X-AXIS PULLEY
226	P0872226	Y-AXIS MOTOR MOUNT
227	P0872227	Y-AXIS STEP MOTOR NEMA 17
228	P0872228	COUPLING AP16/23-04/05D
229	P0872229	LIMIT SWITCH UXCELL RV-163-1C25
230	P0872230	LIMIT SWITCH HIGHLY VS10N021C2
231	P0872231	X-AXIS TRACK
232	P0872232	MIRROR ASSEMBLY (REAR)
232-1	P0872232-1	MIRROR HOUSING
232-2	P0872232-2	MIRROR
232-3	P0872232-3	ANTI-REFLECTIVE LENS
232-4	P0872232-4	LENS COMPRESSION RING
232-5	P0872232-5	LENS MOUNTING PLATE
232-6	P0872232-6	LENS ADJUSTMENT PLATE
232-7	P0872232-7	MIRROR SUPPORT PLATE
232-8	P0872232-8	ADJUSTMENT ROD
232-9	P0872232-9	LOCKING DIAL
232-10	P0872232-10	MIRROR ASSEMBLY BASE PLATE
232-11	P0872232-11	LASER ADJUSTMENT PLATE
232-12	P0872232-12	REFERENCE LASER 30 X 40MM
232-13	P0872232-13	LASER MOUNTING PLATE
233	P0872233	HONEYCOMB TABLE 17" X 23"
234	P0872234	LASER HEAD SUPPORT RAIL
235	P0872235	LED WORKLIGHT
236	P0872236	L-BRACKET
237	P0872237	ANGLE BRACKET
238	P0872238	TABLE BRACE
239	P0872239	SET SCREW M4-.7 X 8
240	P0872240	CAP SCREW M6-1 X 10
241	P0872241	HEX NUT M6-1
242	P0872242	FLAT WASHER 6MM
243	P0872243	CAP SCREW M4-.7 X 8
244	P0872244	FLAT WASHER 4MM
245	P0872245	CAP SCREW M5-.8 X 6
246	P0872246	CAP SCREW M5-.8 X 10
247	P0872247	FLAT WASHER 5MM
248	P0872248	BUTTON HD CAP SCR M4-.7 X 6
249	P0872249	CAP SCREW M4-.7 X 6
250	P0872250	BUTTON HD CAP SCR M4-.7 X 8
251	P0872251	CAP SCREW M4-.7 X 12
252	P0872252	HEX NUT M4-.7

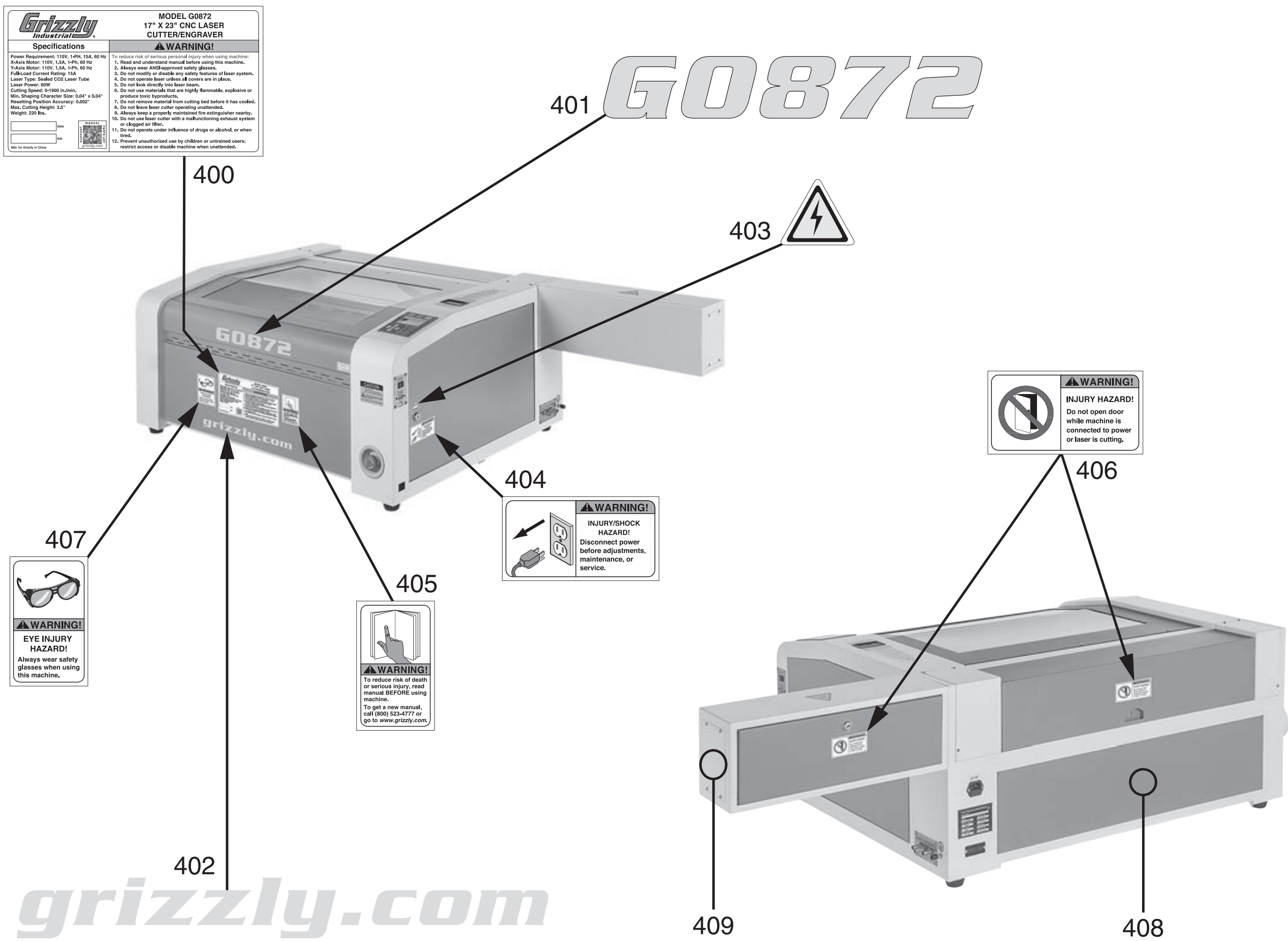
Electrical & Accessories



Electrical & Accessories Parts List

REF	PART #	DESCRIPTION
300	P0872300	POWER CORD 18G 3W 72" 5-15P
301	P0872301	CONTROLLER RUIDA ST908(EC)
302	P0872302	TRANSFORMER TL-POWER TL-150D-B
303	P0872303	STEPPER DRIVER JMCMOTION 2DM420
304	P0872304	POWER SUPPLY FILTER FT127-10
305	P0872305	HIGH VOLTAGE POWER SUPPLY WK60S
306	P0872306	WIRE LOOM
307	P0872307	WATER CHILLER
308	P0872308	WATER TUBING 3/8 ID 52"L
309	P0872309	SIGNAL CORD 20G 2W 78"L
310	P0872310	POWER CORD 18G 3W 72" 5-15P
311	P0872311	EXHAUST FAN
312	P0872312	EXHAUST DUCTING 4" ID 60"L
313	P0872313	AIR PUMP
314	P0872314	AIR TUBING 5/16 ID 56"L
315	P0872315	AIR FITTING 5/16 NPT STRAIGHT BR
316	P0872316	TOOLBOX
317	P0872317	Z-AXIS CRANK HANDLE
318	P0872318	OPTICS DISASSEMBLY TOOL
319	P0872319	LASER BEAM ALIGNMENT GAUGE
320	P0872320	MIRROR ALIGNMENT GAUGE
321	P0872321	FOCUS GAUGE
322	P0872322	LASER CUTTING DEPTH GAUGE
323	P0872323	USB FLASH DRIVE 8GB
324	P0872324	SOFTWARE INSTALLATION DISC
325	P0872325	USER MANUAL DISC
326	P0872326	ABSORBENT COTTON BAG
327	P0872327	EXHAUST PORT 3-3/4"
328	P0872328	HOSE CLAMP 4"
329	P0872329	HEX WRENCH 2MM
330	P0872330	HEX WRENCH 2.5MM
331	P0872331	HEX WRENCH 3MM
332	P0872332	HEX WRENCH 5MM
333	P0872333	SCREWDRIVER FLAT HEAD 1/4"
334	P0872334	LOCKING DOOR KEYS (2-PC SET)
338	P0872338	LIGHT EMITTING DIODE

Labels & Cosmetics



REF	PART #	DESCRIPTION
400	P0872400	MACHINE ID LABEL
401	P0872401	MODEL NUMBER LABEL
402	P0872402	GRIZZLY.COM LABEL
403	P0872403	ELECTRICITY LABEL
404	P0872404	DISCONNECT 110V LABEL

REF	PART #	DESCRIPTION
405	P0872405	READ MANUAL LABEL
406	P0872406	DON'T OPEN LASER LABEL
407	P0872407	SAFETY GLASSES LABEL
408	P0872408	TOUCH-UP PAINT, GRIZZLY GREEN
409	P0872409	TOUCH-UP PAINT, GRIZZLY BEIGE

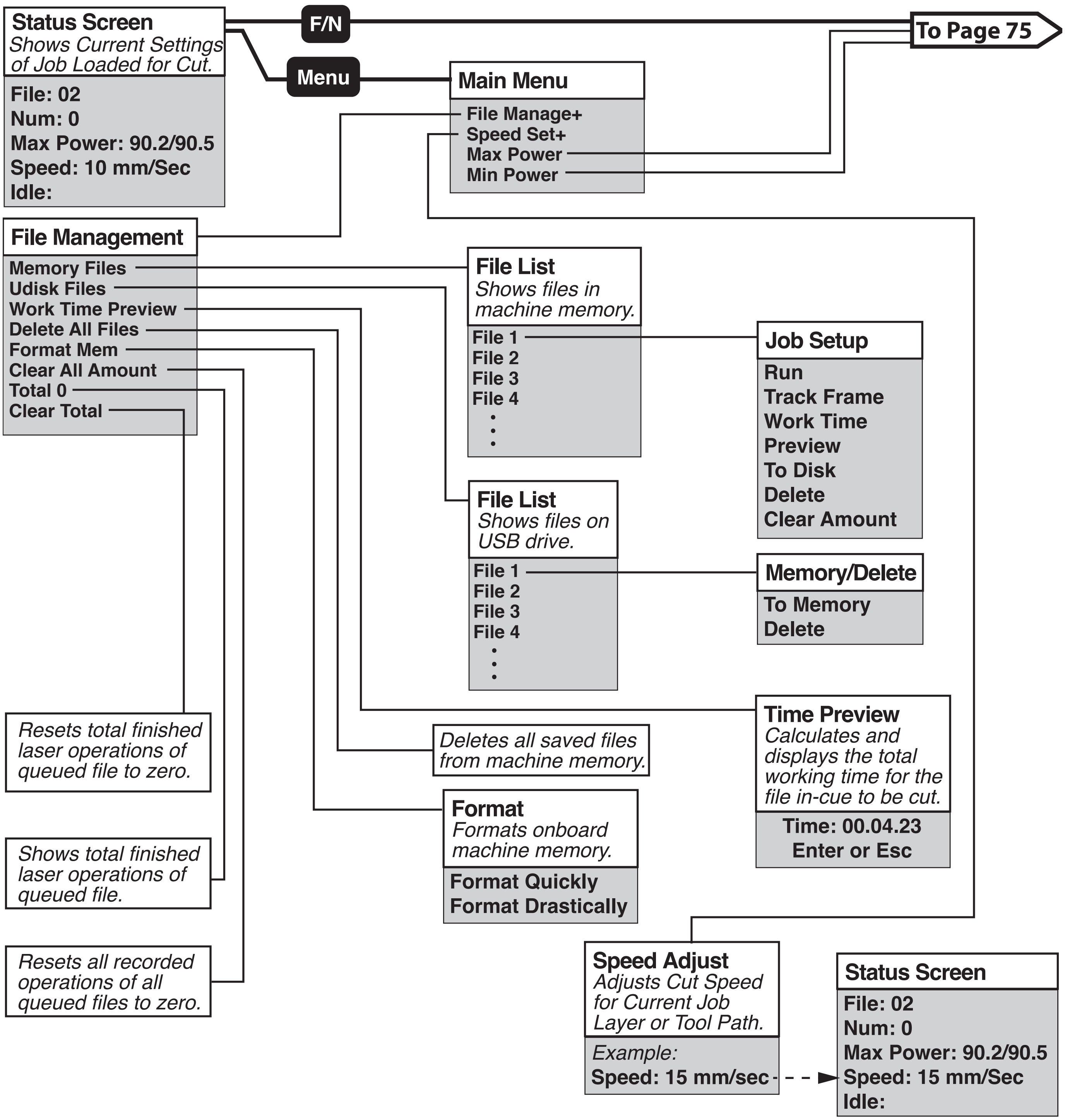
WARNING

Safety labels help reduce the risk of serious injury caused by machine hazards. If any label comes off or becomes unreadable, the owner of this machine MUST replace it in the original location before resuming operations. For replacements, contact (800) 523-4777 or www.grizzly.com.

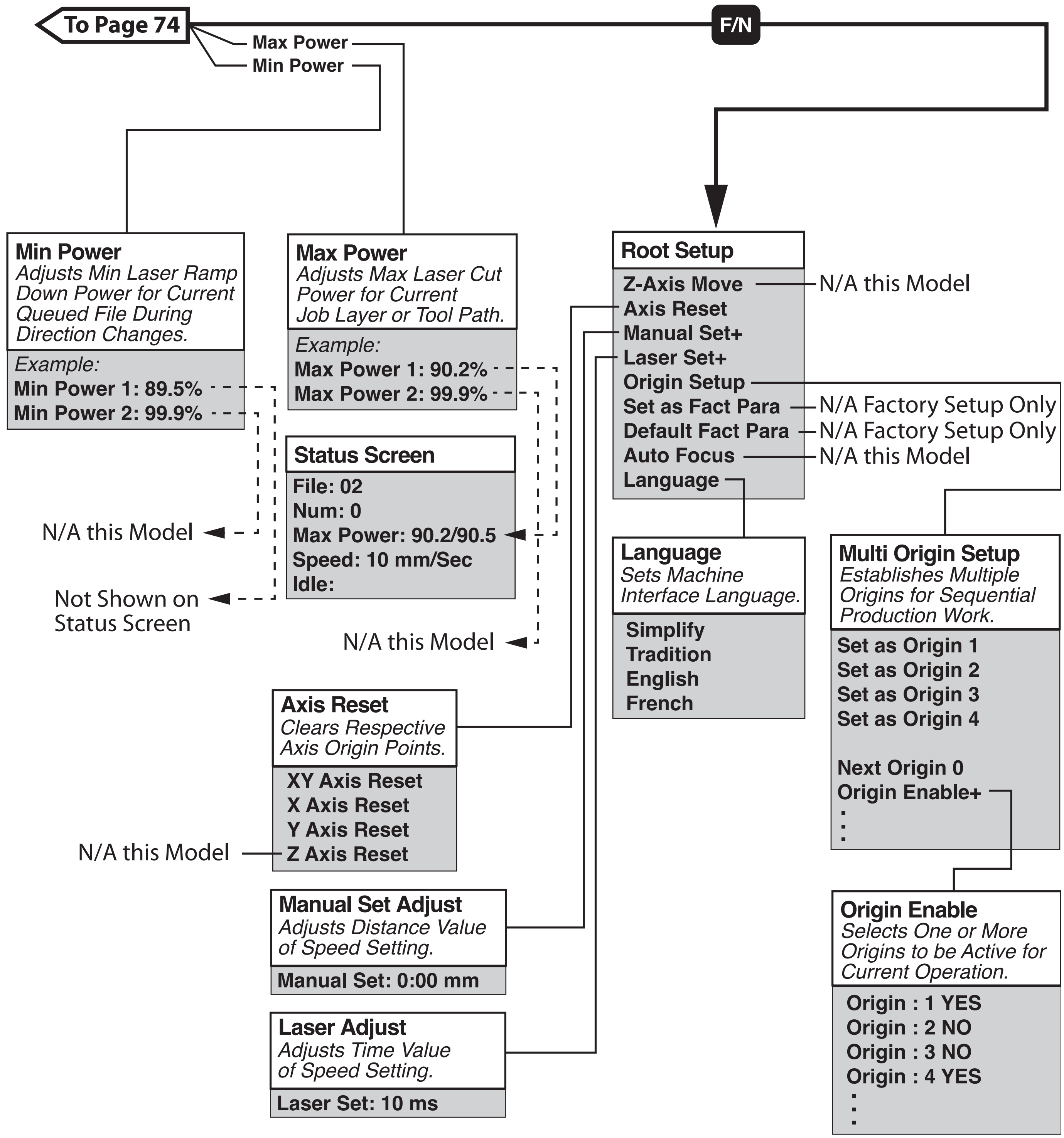
SECTION 10: APPENDIX

Command Tree

This section is an overview of the menus and features used to control this machine. For instructions on navigating the interface and the basics of operation, refer to **Controls & Components** on **Page 4**. Since software changes can affect the user interface, check **www.grizzly.com** for an up to date command tree if this one seems out of date.



Command Tree (Cont.)



WARRANTY & RETURNS

Grizzly Industrial, Inc. warrants every product it sells for a period of **1 year** to the original purchaser from the date of purchase. This warranty does not apply to defects due directly or indirectly to misuse, abuse, negligence, accidents, repairs or alterations or lack of maintenance. This is Grizzly's sole written warranty and any and all warranties that may be implied by law, including any merchantability or fitness, for any particular purpose, are hereby limited to the duration of this written warranty. We do not warrant or represent that the merchandise complies with the provisions of any law or acts unless the manufacturer so warrants. In no event shall Grizzly's liability under this warranty exceed the purchase price paid for the product and any legal actions brought against Grizzly shall be tried in the State of Washington, County of Whatcom.

We shall in no event be liable for death, injuries to persons or property or for incidental, contingent, special, or consequential damages arising from the use of our products.

The manufacturers reserve the right to change specifications at any time because they constantly strive to achieve better quality equipment. We make every effort to ensure that our products meet high quality and durability standards and we hope you never need to use this warranty.

In the event you need to use this warranty, contact us by mail or phone and give us all the details. We will then issue you a "Return Number,'' which must be clearly posted on the outside as well as the inside of the carton. We will not accept any item back without this number. Proof of purchase must accompany the merchandise.

Please feel free to write or call us if you have any questions about the machine or the manual.

Thank you again for your business and continued support. We hope to serve you again soon.

To take advantage of this warranty, you must register it at **https://www.grizzly.com/forms/warranty**, or you can scan the QR code below to be automatically directed to our warranty registration page. Enter all applicable information for the product.

WARRANTY

grizzly.com